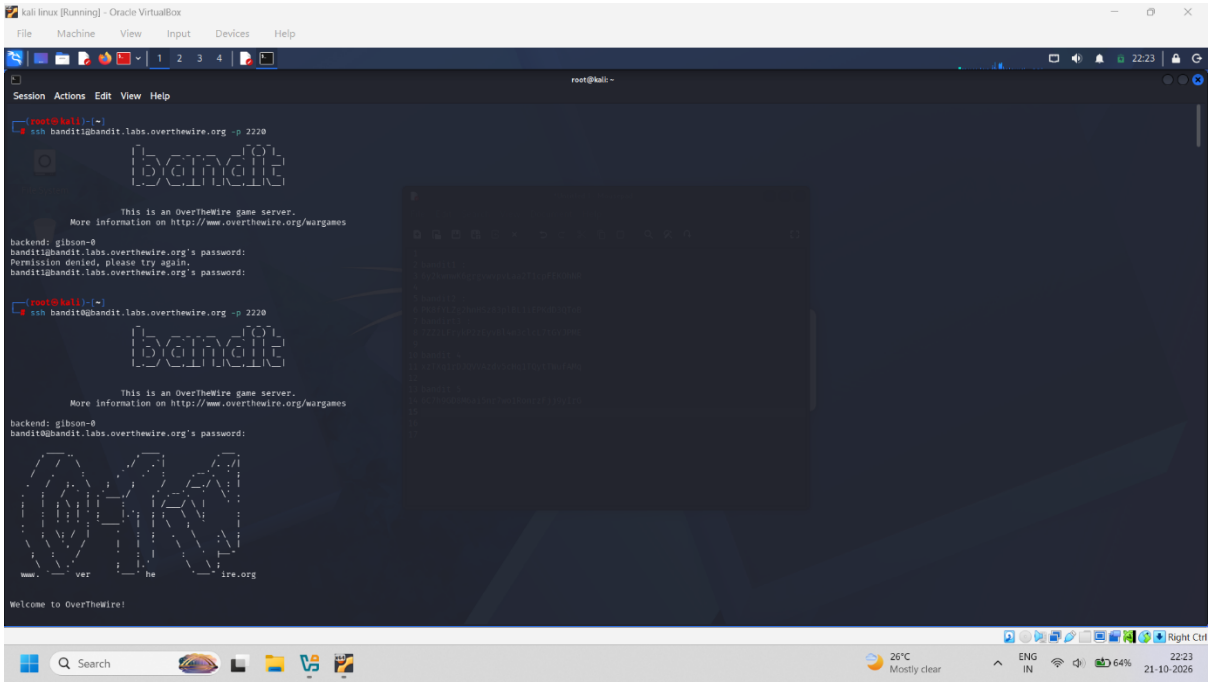
Topic – Over the Wire

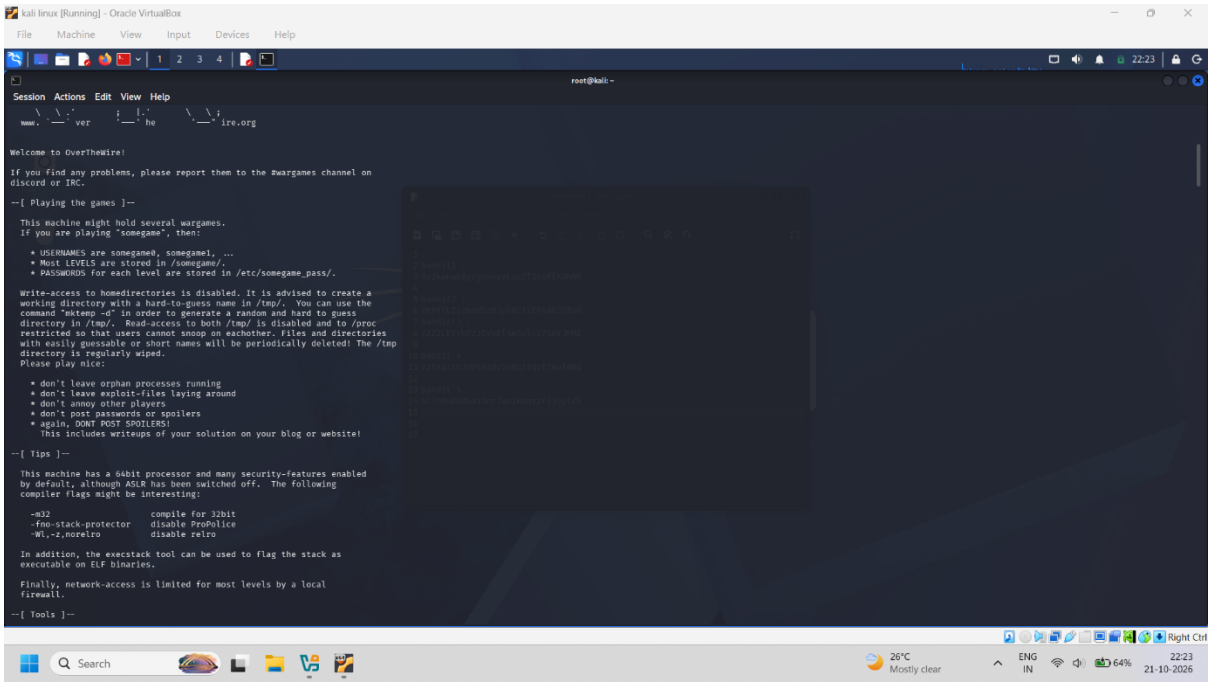
Level 0 –

Step 1 - SSH Command

ssh bandit0@bandit.labs.overthewire.org -p 2220

Explanation: Connects to the Bandit server using SSH on port 2220**.**





Step 2 - **ls**

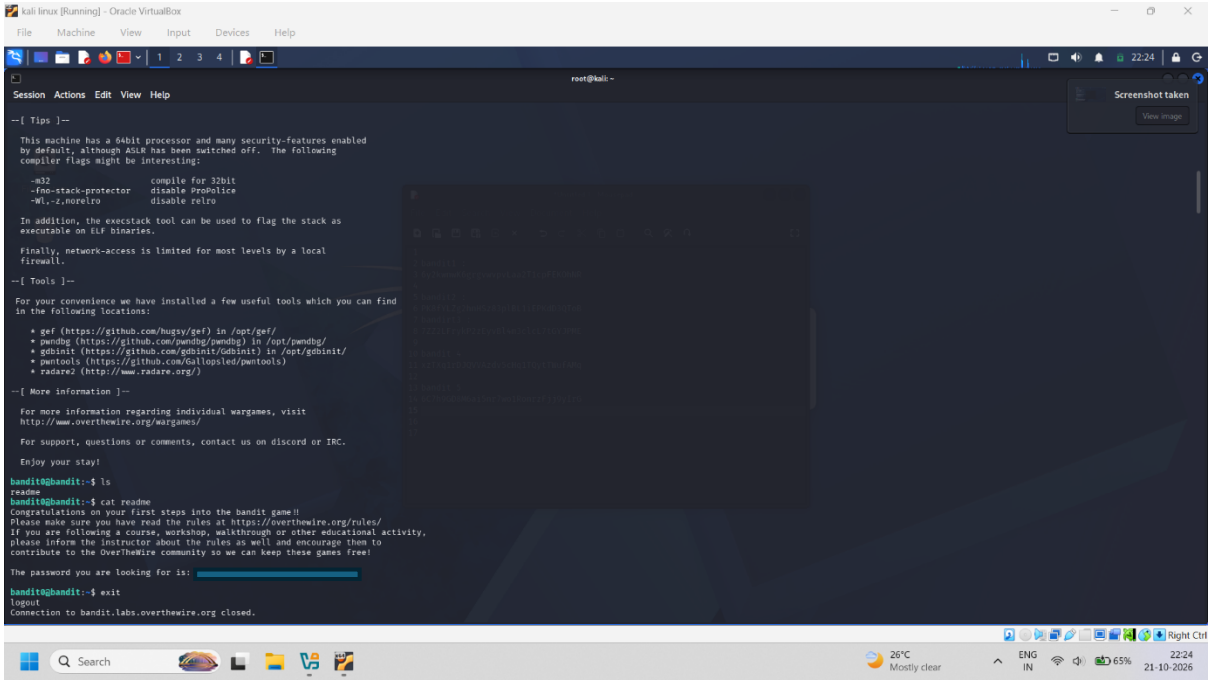
Explanation: Lists files and directories in the current directory.

Step 3 - **cat readme**

Explanation: Displays the contents of the readme file to find the password for Level 1.

Step 4 - **exit**

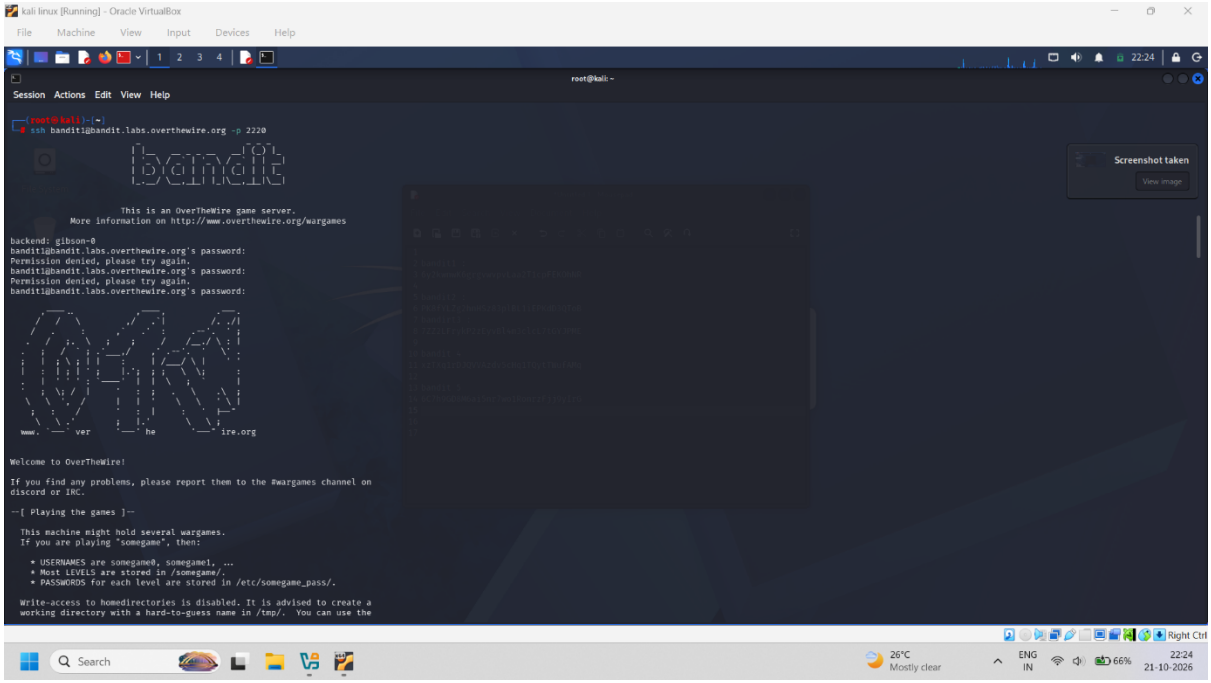
Explanation: Closes the current terminal session or remote connection.



LEVEL 1 – LEVEL 2

Step 1: Login to Level 1

ssh bandit1@bandit.labs.overthewire.org -p 2220



Step 2 – **ls**

Explanation: Lists the files and directories in the current directory. In Level 1, the target file is named -.

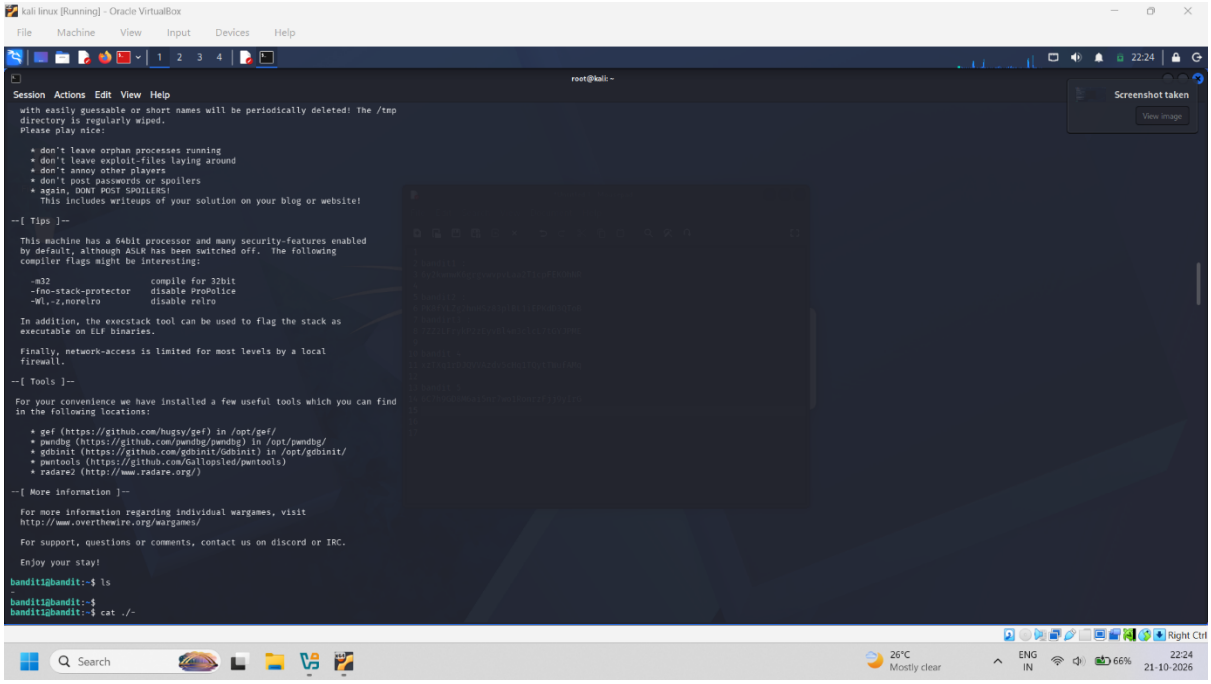
Step 3 - **cat ./-**

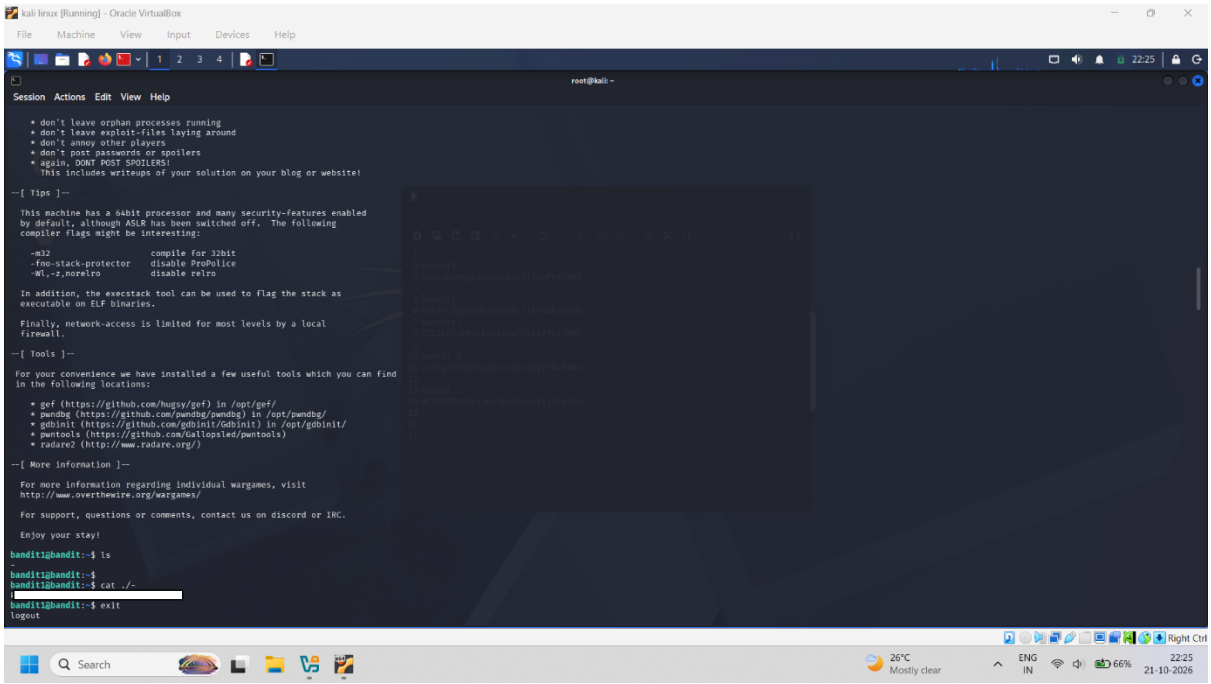
Explanation: Displays the contents of the file named -. The ./ specifies that the file is in the current directory.

Step 4 - **exit**

Explanation: Closes the SSH connection to the remote server.

Result: The password for Bandit Level 2 is obtained by reading the contents of the file named -.

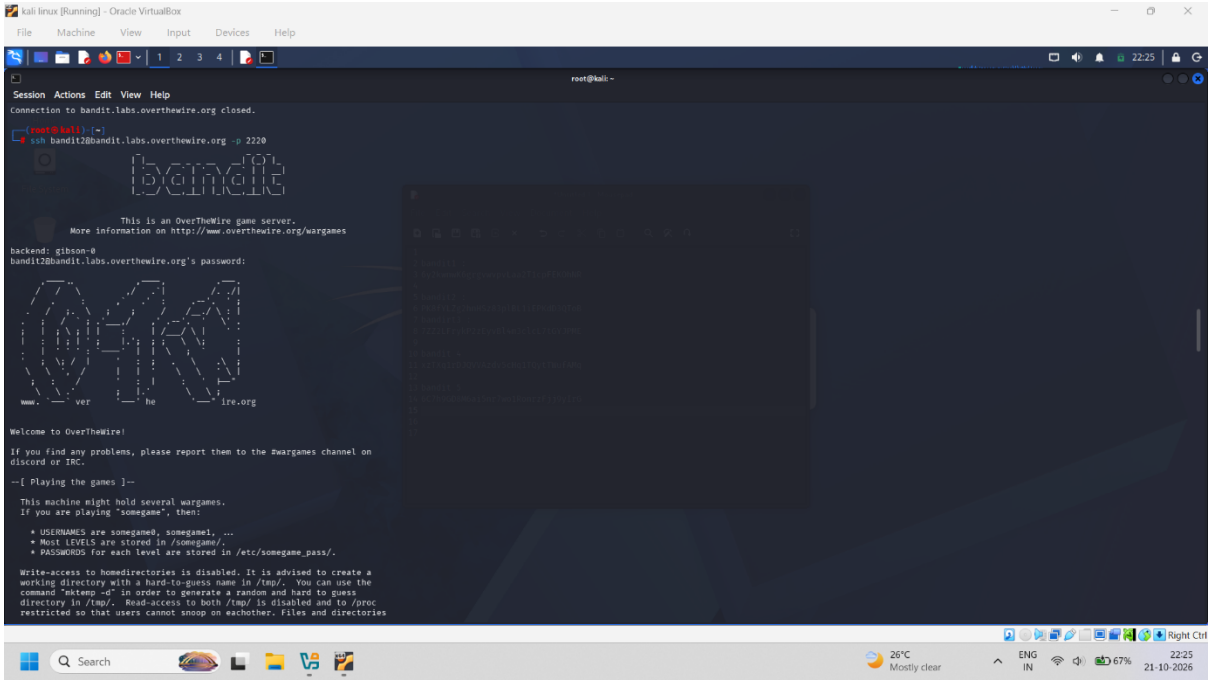

kali linux [Running] - Oracle VirtualBox
File Machine View Input Devices Help
root@kali: ~
Session Actions Edit View Help
with easily guessable or short names will be periodically deleted! The /tmp
directory is regularly wiped.
Please play nice:
* don't leave orphan processes running
* don't leave exploit-files laying around
* don't annoy other players
* don't post passwords or spoilers
* again, DONT POST SPOILERS!
This includes writeups of your solution on your blog or website!
--[Tips]--
This machine has a 64bit processor and many security-features enabled
by default, although ASLR has been switched off. The following
compiler flags might be interesting:
-m32 compile for 32bit
-fno-stack-protector disable ProPolice
-Wl,-z,norelro disable relro
In addition, the execstack tool can be used to flag the stack as
executable on ELF binaries.
Finally, network-access is limited for most levels by a local
firewall.
--[Tools]--
For your convenience we have installed a few useful tools which you can find
in the following locations:
* gef (https://github.com/hugsy/gef) in /opt/gef/
* pwndbg (https://github.com/pwndbg/pwndbg) in /opt/pwndbg/
* gdbinit (https://github.com/gdbinit/Gdbinit) in /opt/gdbinit/
* pwntools (https://github.com/Gallopsled/pwntools)
* radare2 (http://www.radare.org/)
--[More information]--
For more information regarding individual wargames, visit
http://www.overthewire.org/wargames/
For support, questions or comments, contact us on discord or IRC.
Enjoy your stay!
bandit1@bandit:~$ ls
-
bandit1@bandit:~$
bandit1@bandit:~$ cat ./-
Screenshot taken
View image
Search
26°C
Mostly clear
ENG
IN
66%
22:24
21-10-2026
Right Ctrl


kali linux [Running] - Oracle VirtualBox
File Machine View Input Devices Help
root@kali: ~
Session Actions Edit View Help
* don't leave orphan processes running
* don't leave exploit-files laying around
* don't annoy other players
* don't post passwords or spoilers
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Enjoy your stay!
bandit1@bandit:~$ ls
-
bandit1@bandit:~$
bandit1@bandit:~$ cat ./-
bandit1@bandit:~$ exit
logout
Search
26°C
Mostly clear
ENG
IN
66%
22:25
21-10-2026
Right Ctrl

LEVEL 2 – LEVEL 3

Step 1: Login to Level 2

ssh bandit2@bandit.labs.overthewire.org -p 2220



Step 2 **- ls**

Explanation: Lists files and directories in the current directory.

Step 3 - **cat**

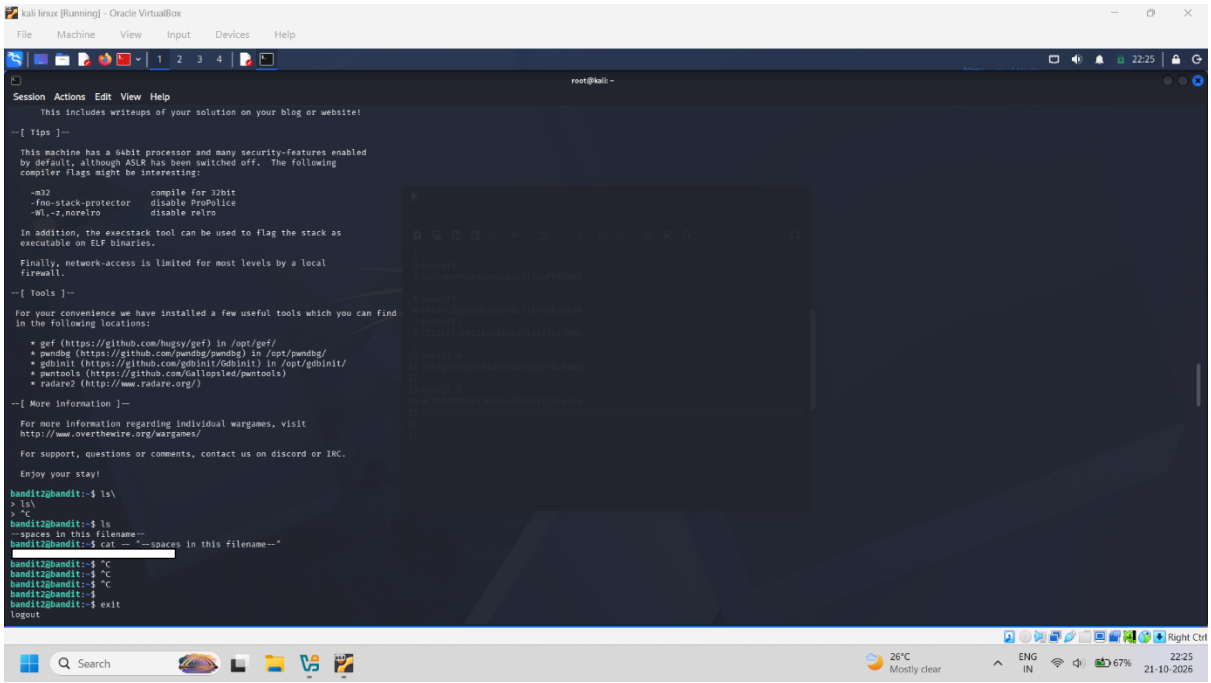
cat -- "--spaces in this filename--"

Explanation: Displays the contents of a file whose name contains spaces. The -- indicates that the following argument is a filename, not an option.

Step 4 - **exit**

Explanation: Closes the SSH connection to the remote server.

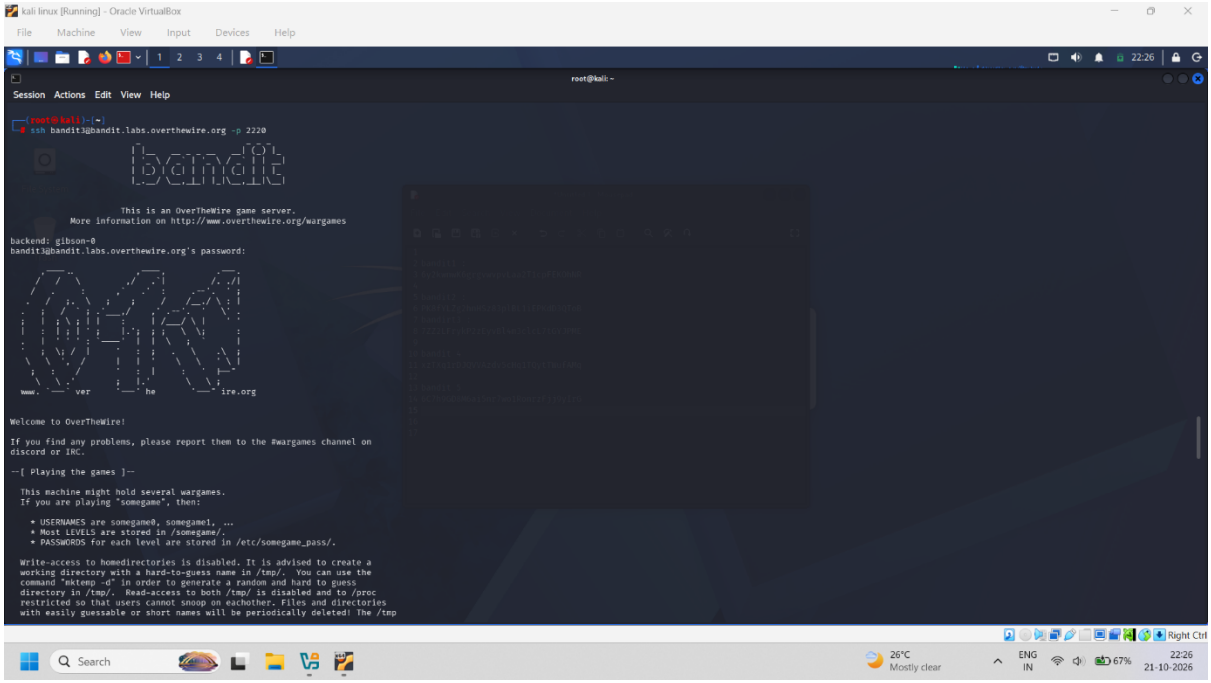
Result: The password for Bandit Level 3 is obtained by reading the contents of the file named --spaces in this filename--.


kali linux [Running] - Oracle VirtualBox
File Machine View Input Devices Help
root@kali: ~
Session Actions Edit View Help
This includes writeups of your solution on your blog or website!
--[Tips]--
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by default, although ASLR has been switched off. The following
compiler flags might be interesting:
-m32 compile for 32bit
-fno-stack-protector disable ProPolice
-Wl,-z,norelro disable relro
In addition, the execstack tool can be used to flag the stack as
executable on ELF binaries.
Finally, network-access is limited for most levels by a local
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--[Tools]--
For your convenience we have installed a few useful tools which you can find
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* pwndbg (https://github.com/pwndbg/pwndbg) in /opt/pwndbg/
* gdbinit (https://github.com/gdbinit/Gdbinit) in /opt/gdbinit/
* pwntools (https://github.com/Gallopsled/pwntools)
* radare2 (http://www.radare.org/)
--[More information]--
For more information regarding individual wargames, visit
http://www.overthewire.org/wargames/
For support, questions or comments, contact us on discord or IRC.
Enjoy your stay!
bandit2@bandit:~$ ls\
> ls\
> ^C
bandit2@bandit:~$ ls
--spaces in this filename--
bandit2@bandit:~$ cat -- "--spaces in this filename--"
bandit2@bandit:~$ ^C
bandit2@bandit:~$ ^C
bandit2@bandit:~$ ^C
bandit2@bandit:~$
bandit2@bandit:~$ exit
logout
Right Ctrl
Search
26°C
Mostly clear
ENG
IN
67%
22:25
21-10-2026

LEVEL 3 – LEVEL 4

Step 1: Login to Level 3

ssh bandit3@bandit.labs.overthewire.org -p 2220



Step 2 - **ls**

Explanation: Lists files and directories in the current directory.

Step 3 - **ls inhere**

Explanation: Lists the contents of the inhere directory.

Step 4 - **cd inhere**

Explanation: Changes the current directory to inhere.

Step 5 **- ls -la**

Explanation: Lists all files, including hidden files, with detailed information such as permissions, ownership, size, and modification date.

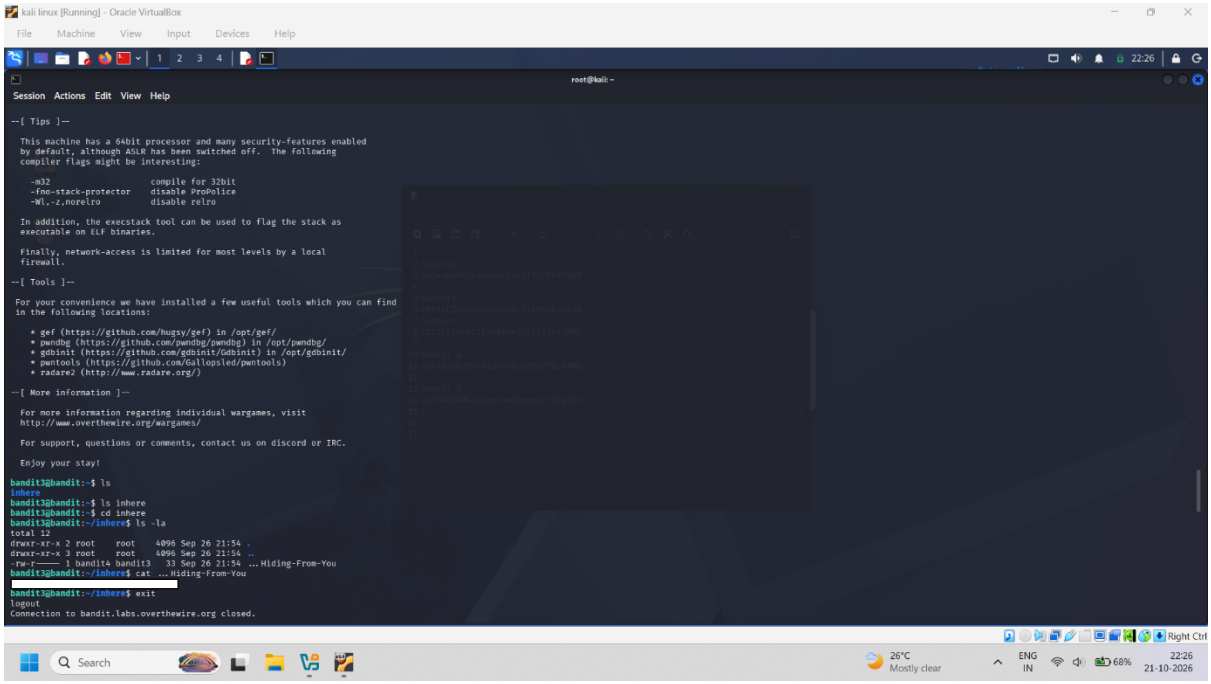
Step 6 - **cat ...Hiding-From-You**

Explanation: Displays the contents of the hidden file containing the password for Level 4.

Step 7 **- exit**

Explanation: Closes the SSH connection to the remote server.

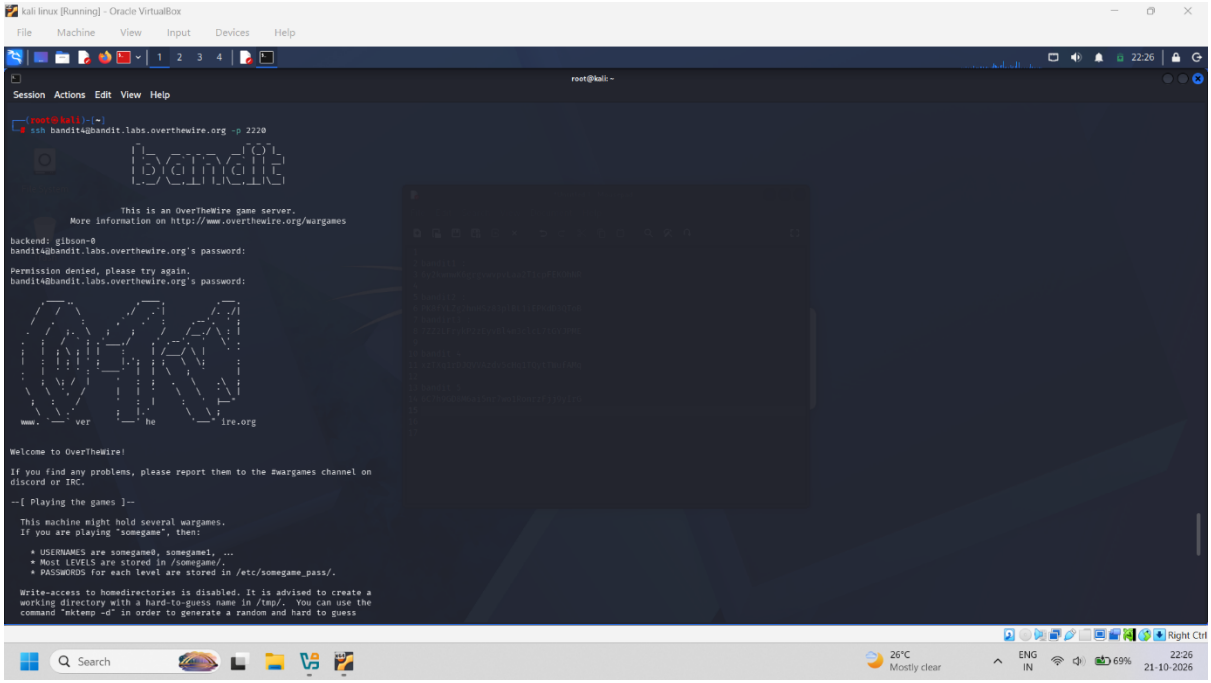
Result: The password for Bandit Level 4 was obtained by locating and reading the hidden file inside the inhere directory.



LEVEL 4 – LEVEL 5

Step 1: Login to Level 4

ssh bandit4@bandit.labs.overthewire.org -p 2220



Step 2 – **ls**

Explanation: Lists files and directories in the current directory.

Step 3 - **cd inhere**

Explanation: Changes the current directory to inhere.

Step 4 - **ls -la**

Explanation: Lists all files, including hidden files, with details such as permissions, ownership, and file size.

Step 5 - **ls -a**

Explanation: Displays all files, including hidden files, in the current directory.

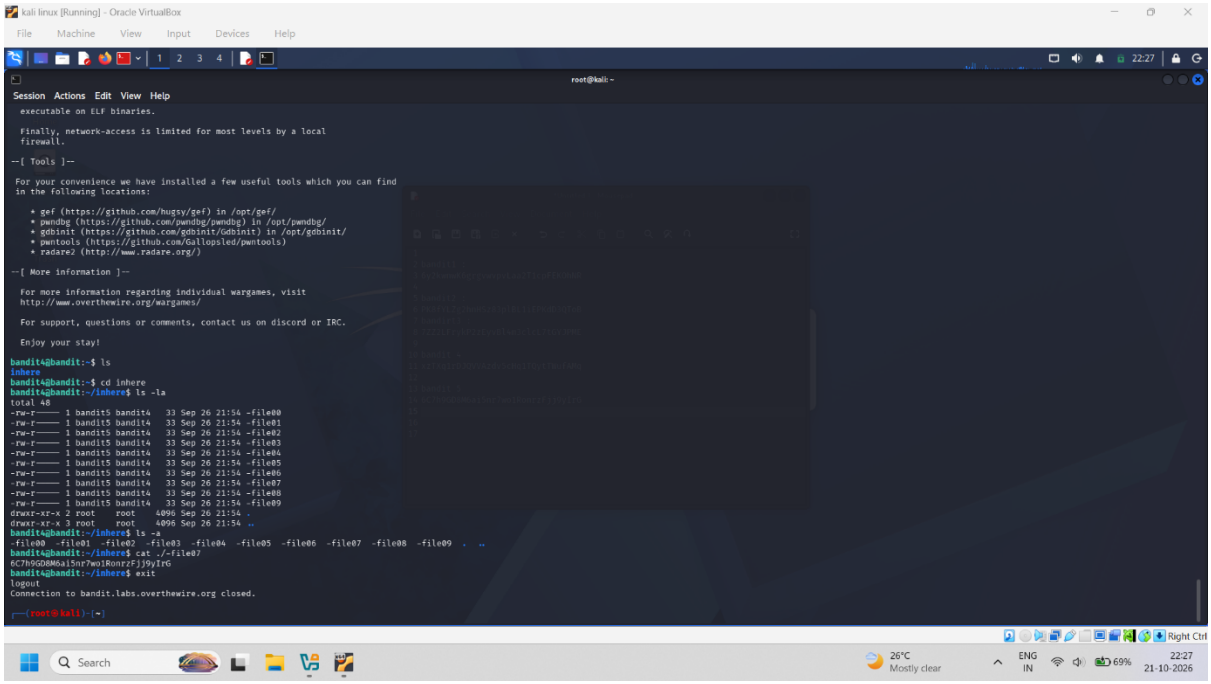
Step 6 - **cat ./-file07**

Explanation: Displays the contents of -file07. The ./ ensures the filename is treated as a file in the current directory.

Step 7 - **exit**

Explanation: Closes the SSH connection to the remote server.

Result: The password for Bandit Level 5 was obtained by reading the contents of -file07.

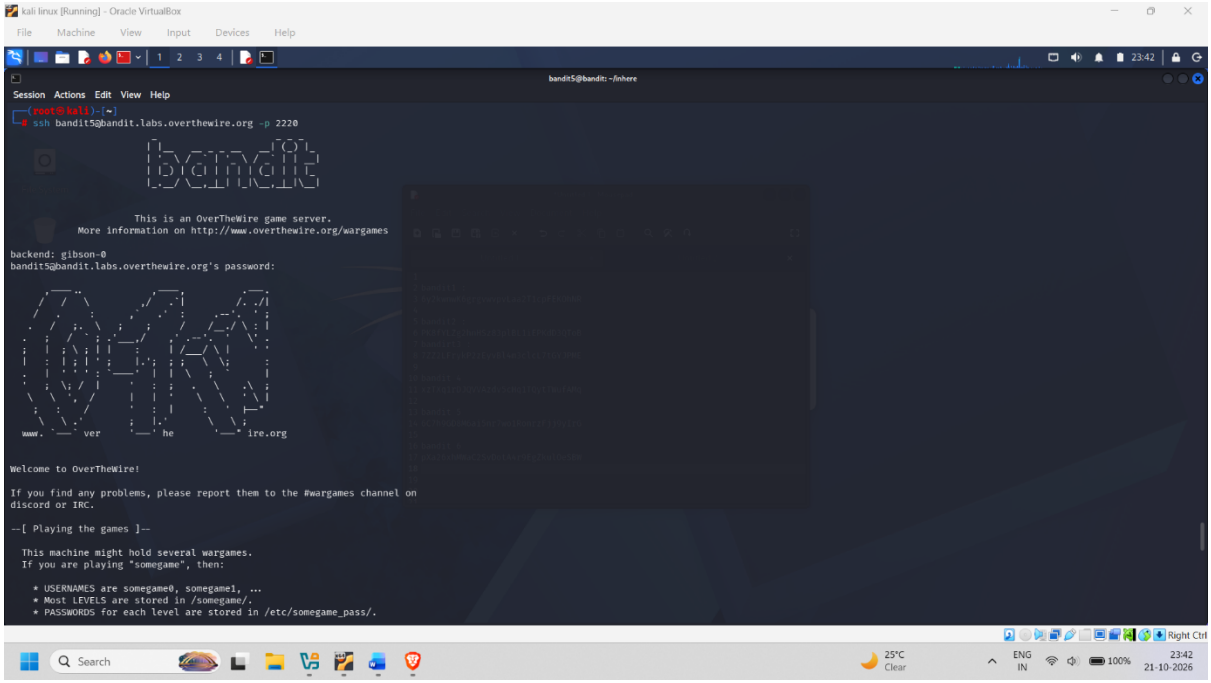


LEVEL 5 – LEVEL 6

Step 1: Login to Level 5

ssh bandit5@bandit.labs.overthewire.org -p 2220

Password – 6C7h9GD8M6ai5nr7wo1RonrzFjj9yIrG



Step 2 - **ls**

Explanation: Lists files and directories in the current directory.

Step 3 **- cd inhere**

Explanation: Moves into the inhere directory**.**

Step 4 - **ls -la**

Explanation: Lists all files and directories with detailed information, including permissions and file sizes.

Step 5 - **ls -a**

Explanation: Displays all files, including hidden files.

Step 6 – **find**

find . -type f -size 1033c

Explanation: Searches for a regular file that is exactly 1033 bytes in size. Here, . means the current directory, -type f specifies regular files, and -size 1033c specifies 1033 bytes.

Step 7 - **find with executable permission**

find . -type f -size 1033c -executable

Explanation: Searches for files that are exactly 1033 bytes and have executable permission.

Step 8 - **cat**

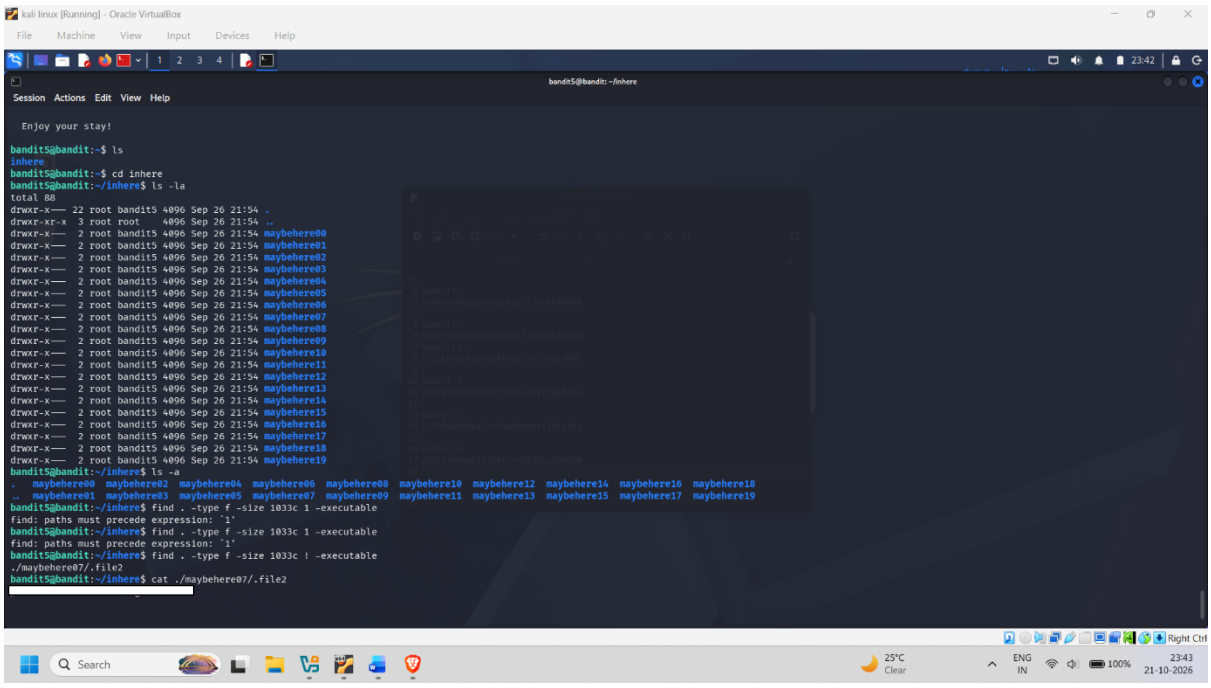
cat ./maybehere07/.file2

Explanation: Displays the contents of the file found in the maybehere07 directory. Use the exact path returned by your find command**.**

Step 9 - **exit**

Explanation: Closes the SSH connection.

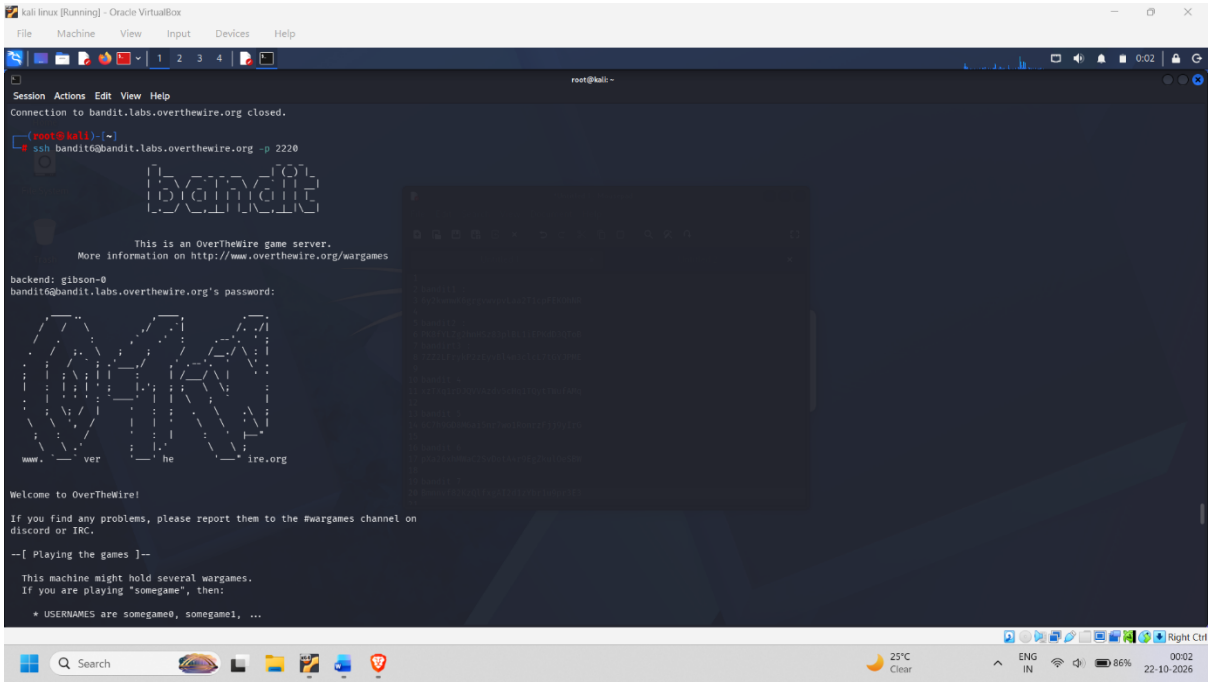
Result: The password for Bandit Level 6 was obtained by finding and reading the file that matched the required properties.



LEVEL 6 – LEVEL 7

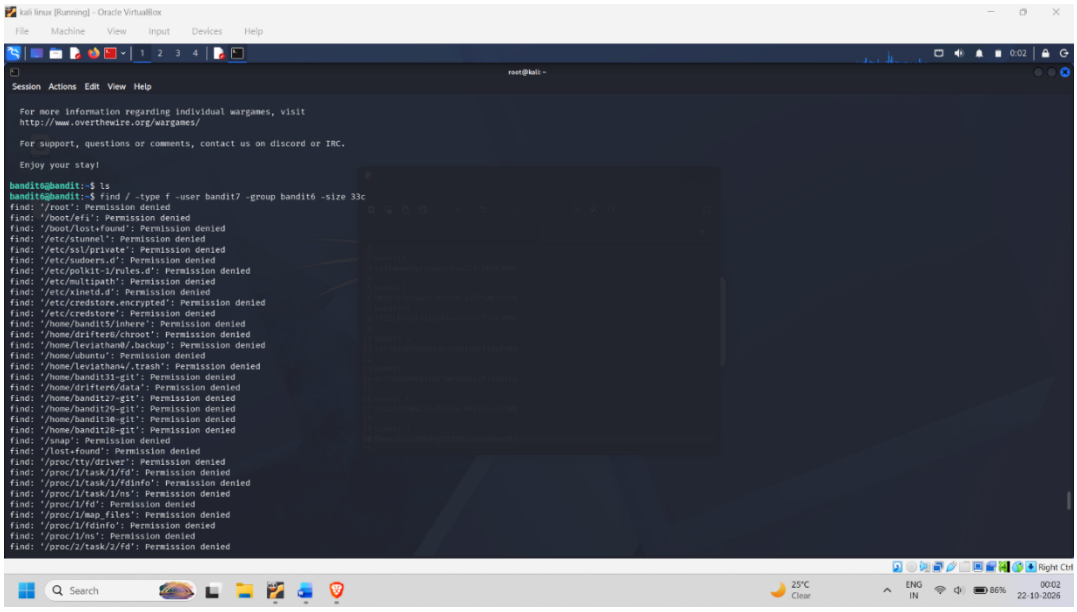
Step 1: Login to Level 6

ssh bandit6@bandit.labs.overthewire.org -p 2220



Step 2 **- ls**

Explanation: Lists files and directories in the current directory.



Step 3 - **find**

find / -type f -user bandit7 -group bandit6 -size 33c

Explanation: Searches the entire filesystem for a file owned by user bandit7, belonging to group bandit6, and exactly 33 bytes in size.

Step 4 - **cat**

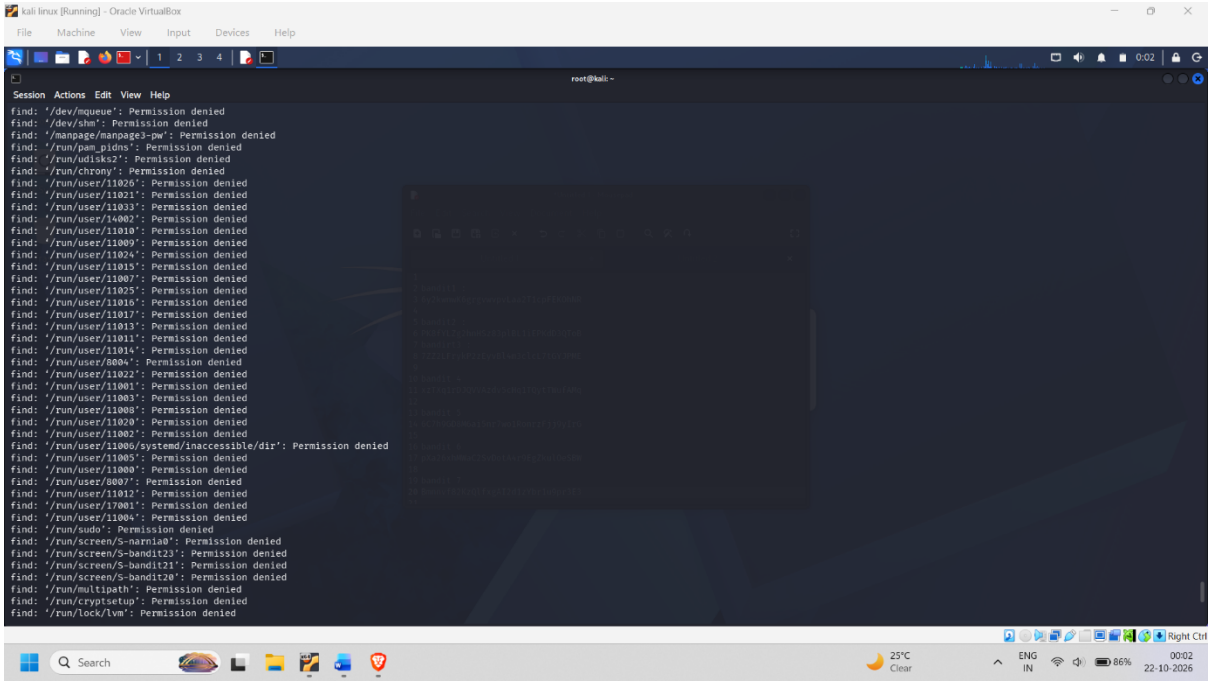
cat /var/lib/dpkg/info/bandit7.password

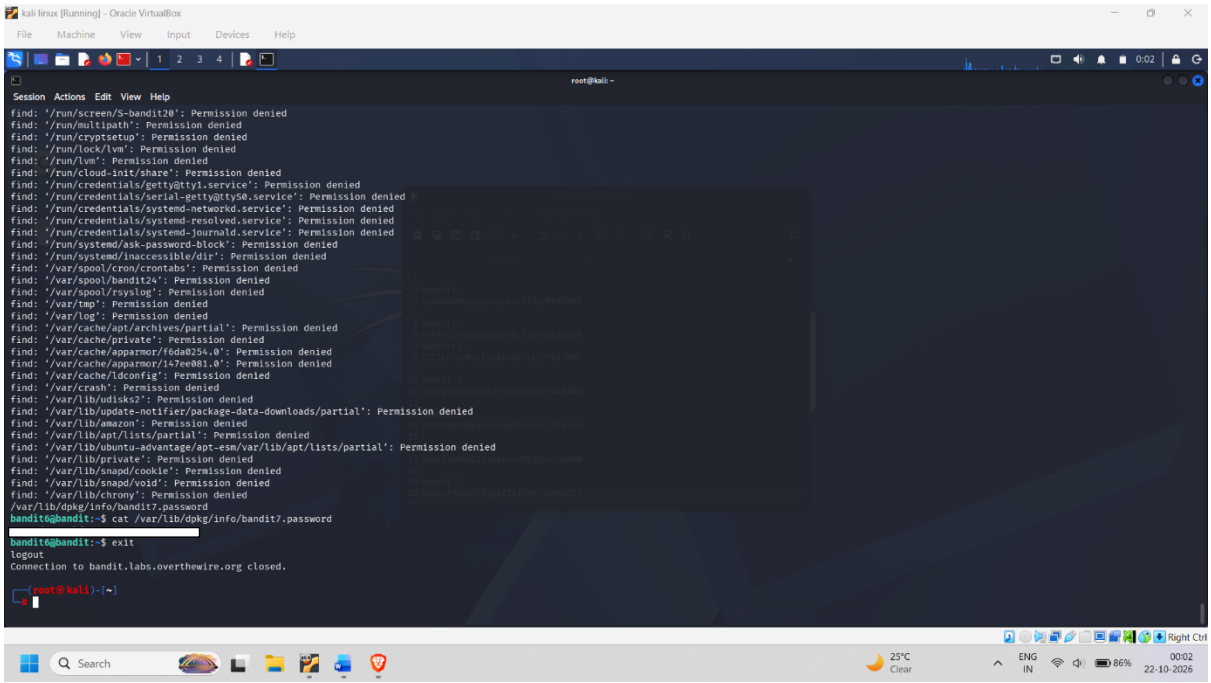
Explanation: Displays the contents of the file containing the password for Bandit Level 7.

Step 5 - **exit**

Explanation: Closes the SSH connection to the remote server.

Result: The password for Bandit Level 7 was obtained by locating and reading the file that matched the specified conditions.

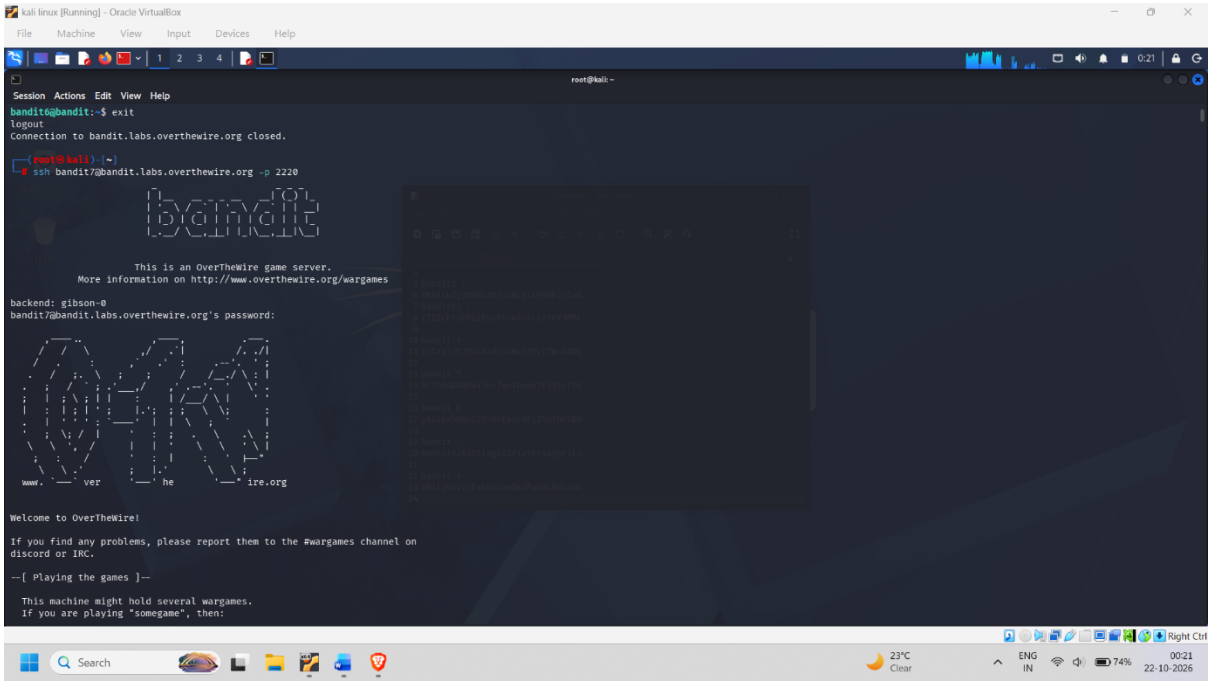



kali linux [Running] - Oracle VirtualBox
File Machine View Input Devices Help
root@kali: ~
Session Actions Edit View Help
find: '/run/screen/S-bandit20': Permission denied
find: '/run/multipath': Permission denied
find: '/run/cryptsetup': Permission denied
find: '/run/lock/lvm': Permission denied
find: '/run/lvm': Permission denied
find: '/run/cloud-init/share': Permission denied
find: '/run/credentials/getty@tty1.service': Permission denied
find: '/run/credentials/serial-getty@ttyS0.service': Permission denied
find: '/run/credentials/systemd-networkd.service': Permission denied
find: '/run/credentials/systemd-resolved.service': Permission denied
find: '/run/credentials/systemd-journald.service': Permission denied
find: '/run/systemd/ask-password-block': Permission denied
find: '/run/systemd/inaccessible/dir': Permission denied
find: '/var/spool/cron/crontabs': Permission denied
find: '/var/spool/bandit24': Permission denied
find: '/var/spool/rsyslog': Permission denied
find: '/var/tmp': Permission denied
find: '/var/log': Permission denied
find: '/var/cache/apt/archives/partial': Permission denied
find: '/var/cache/private': Permission denied
find: '/var/cache/apparmor/f6da0256.0': Permission denied
find: '/var/cache/apparmor/147ee081.0': Permission denied
find: '/var/cache/ldconfig': Permission denied
find: '/var/crash': Permission denied
find: '/var/lib/udisks2': Permission denied
find: '/var/lib/update-notifier/package-data-downloads/partial': Permission denied
find: '/var/lib/amazon': Permission denied
find: '/var/lib/apt/lists/partial': Permission denied
find: '/var/lib/ubuntu-advantage/apt-esm/var/lib/apt/lists/partial': Permission denied
find: '/var/lib/private': Permission denied
find: '/var/lib/snapd/cookie': Permission denied
find: '/var/lib/snapd/void': Permission denied
find: '/var/lib/chrony': Permission denied
/var/lib/dpkg/info/bandit7.password
bandit6@bandit:~$ cat /var/lib/dpkg/info/bandit7.password
bandit6@bandit:~$ exit
logout
Connection to bandit.labs.overthewire.org closed.
(root㉿kali)-[~]
Right Ctrl
Search
25°C
Clear
ENG
IN
86%
00:02
22-10-2026

LEVEL 7 – LEVEL 8

Step 1: Login to Level 7

ssh bandit7@bandit.labs.overthewire.org -p 2220

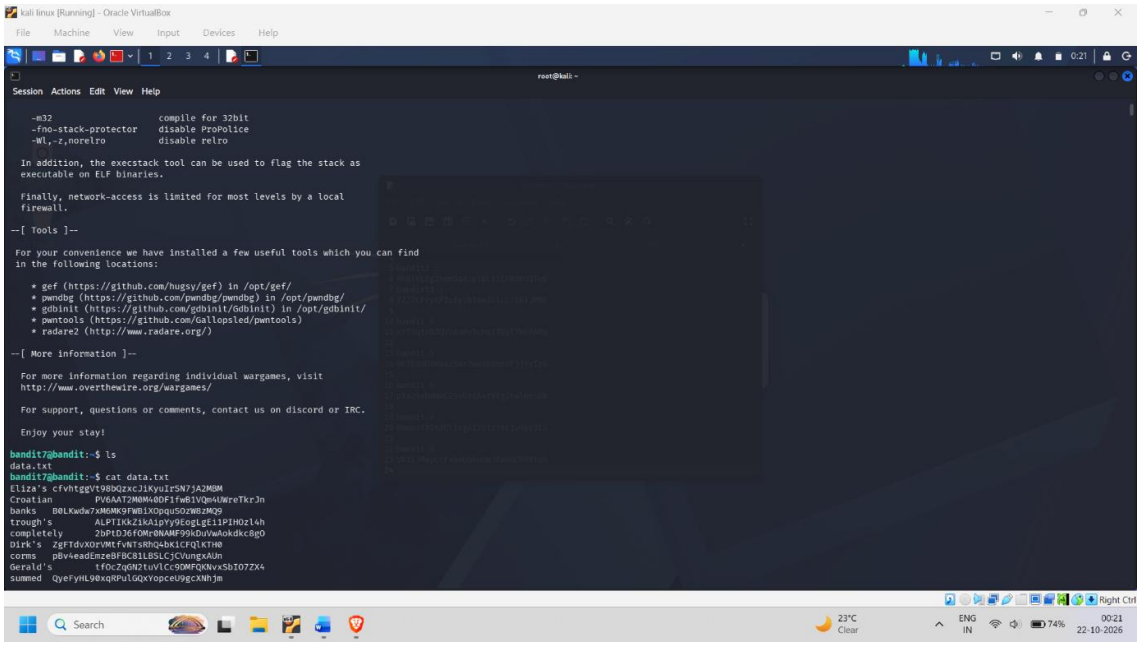


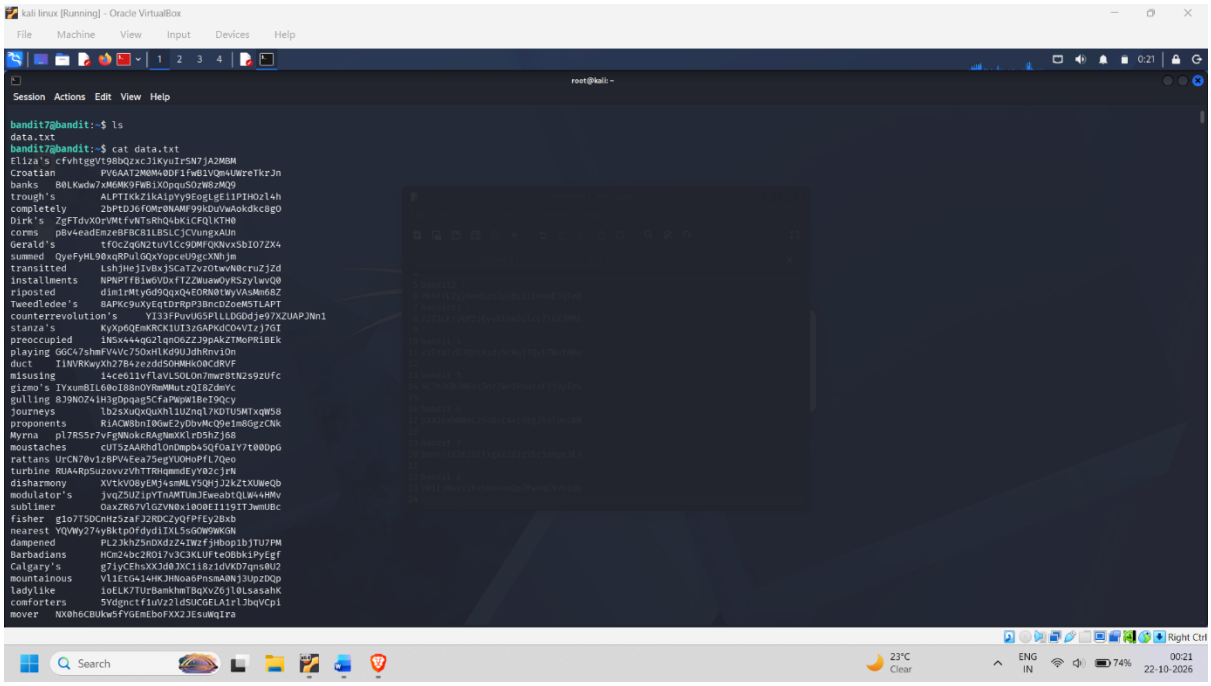
Step 2 - **ls**

Explanation: Lists files and directories in the current directory.

Step 3 - **cat data.txt**

Explanation: Displays the contents of data.txt in the terminal.





Step 4 - **grep**

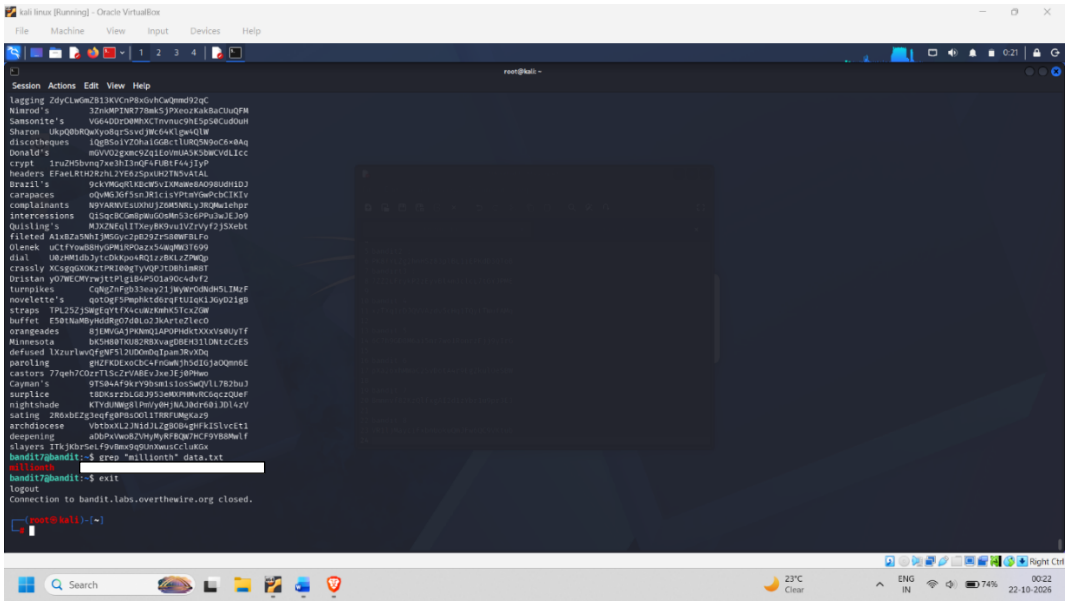
grep "millionth" data.txt

Explanation: Searches data.txt for the word millionth and displays the matching line containing the password for Level 8.

Step 5 – **exit**

Explanation: Closes the SSH connection to the remote server.

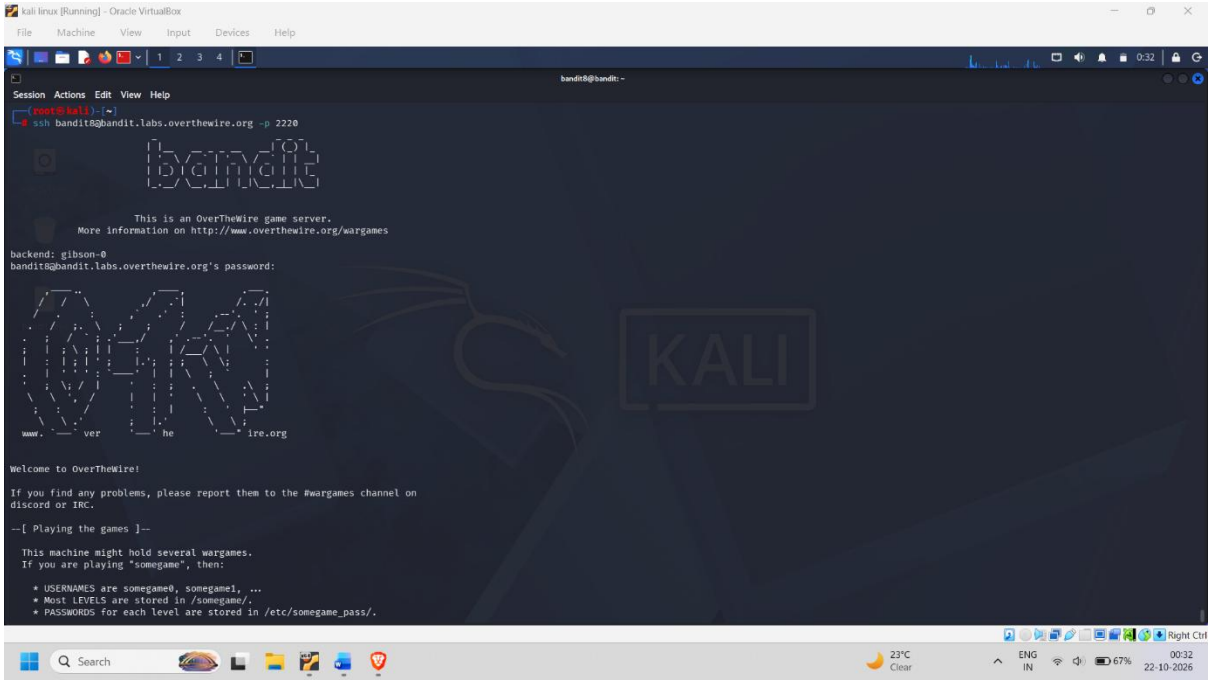
Result: The password for Bandit Level 8 was found by searching for the word millionth in data.txt.



LEVEL 8 – LEVEL 9

Step 1: Login to Level 8

ssh bandit8@bandit.labs.overthewire.org -p 2220



Step 2 **- sort data.txt**

Explanation: Sorts the lines in data.txt alphabetically.

Step 3 - **(Pipe operator)**

Explanation: Passes the output of one command as input to another command.

Step 4 - **uniq -u**

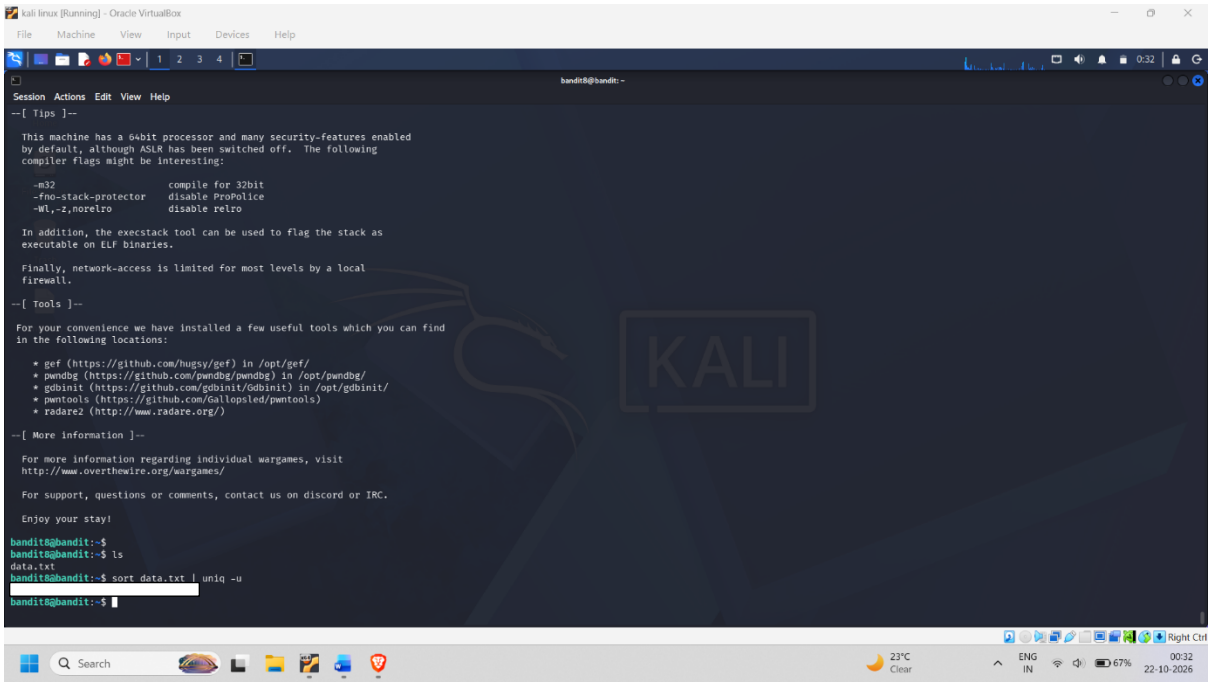
Explanation: Displays only lines that occur exactly once in the input.

Step 5 - **Complete command**

sort data.txt | uniq -u

Explanation: Sorts the file and displays the line that appears only once, revealing the password for Level 9.

Result: The unique line in data.txt provides the password for Bandit Level 9.

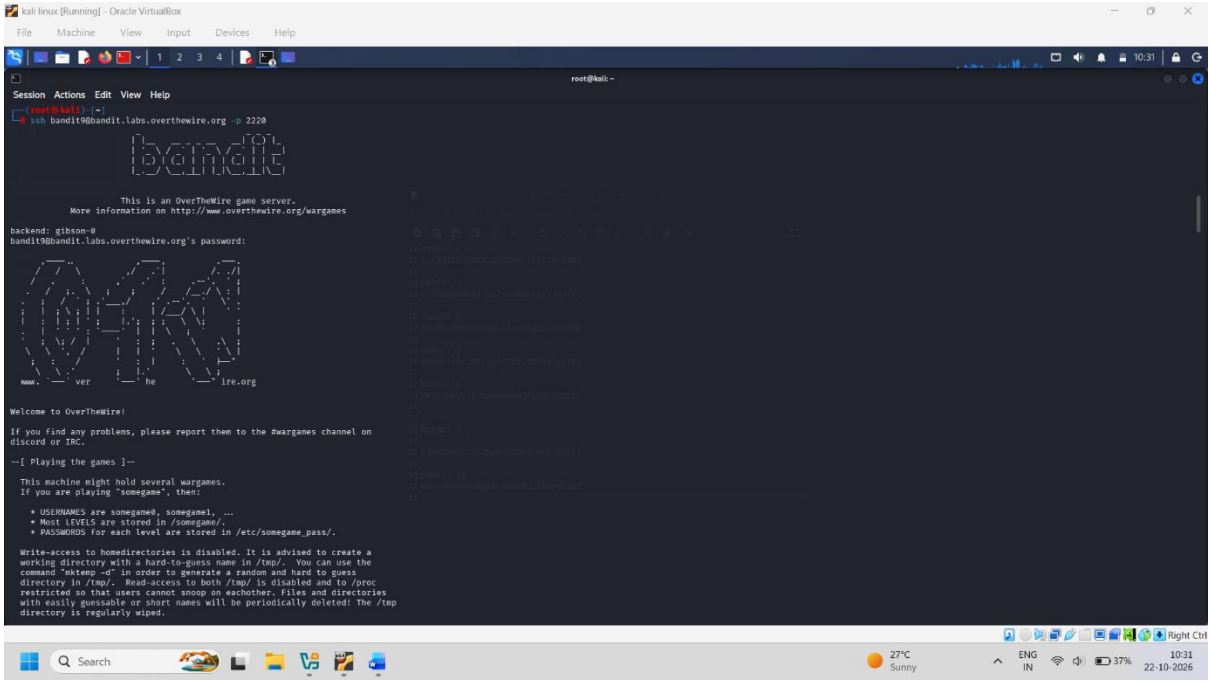

kali linux [Running] - Oracle VirtualBox
File Machine View Input Devices Help
bandit8@bandit: ~
Session Actions Edit View Help
--[Tips]--
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by default, although ASLR has been switched off. The following
compiler flags might be interesting:
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-Wl,-z,norelro disable relro
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executable on ELF binaries.
Finally, network-access is limited for most levels by a local
firewall.
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* gdbinit (https://github.com/gdbinit/Gdbinit) in /opt/gdbinit/
* pwntools (https://github.com/Gallopsled/pwntools)
* radare2 (http://www.radare.org/)
--[More information]--
For more information regarding individual wargames, visit
http://www.overthewire.org/wargames/
For support, questions or comments, contact us on discord or IRC.
Enjoy your stay!
bandit8@bandit:~$
bandit8@bandit:~$ ls
data.txt
bandit8@bandit:~$ sort data.txt | uniq -u
bandit8@bandit:~$
Right Ctrl
Search
23°C
Clear
ENG
IN
67%
00:32
22-10-2026

LEVEL 9 – LEVEL 10

Step 1: Login to Level 9

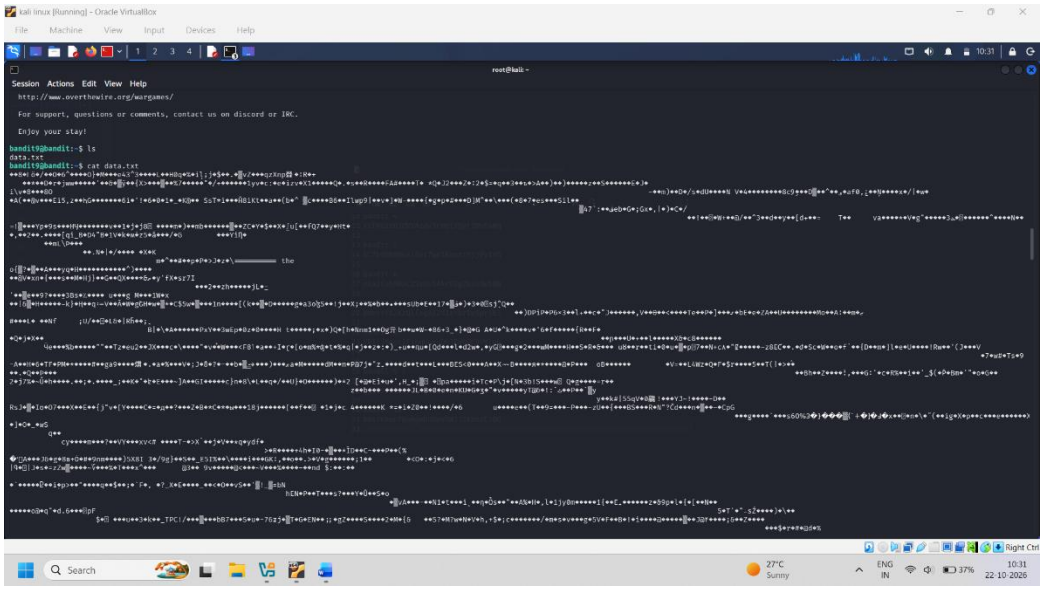
ssh bandit9@bandit.labs.overthewire.org -p 2220

Password –



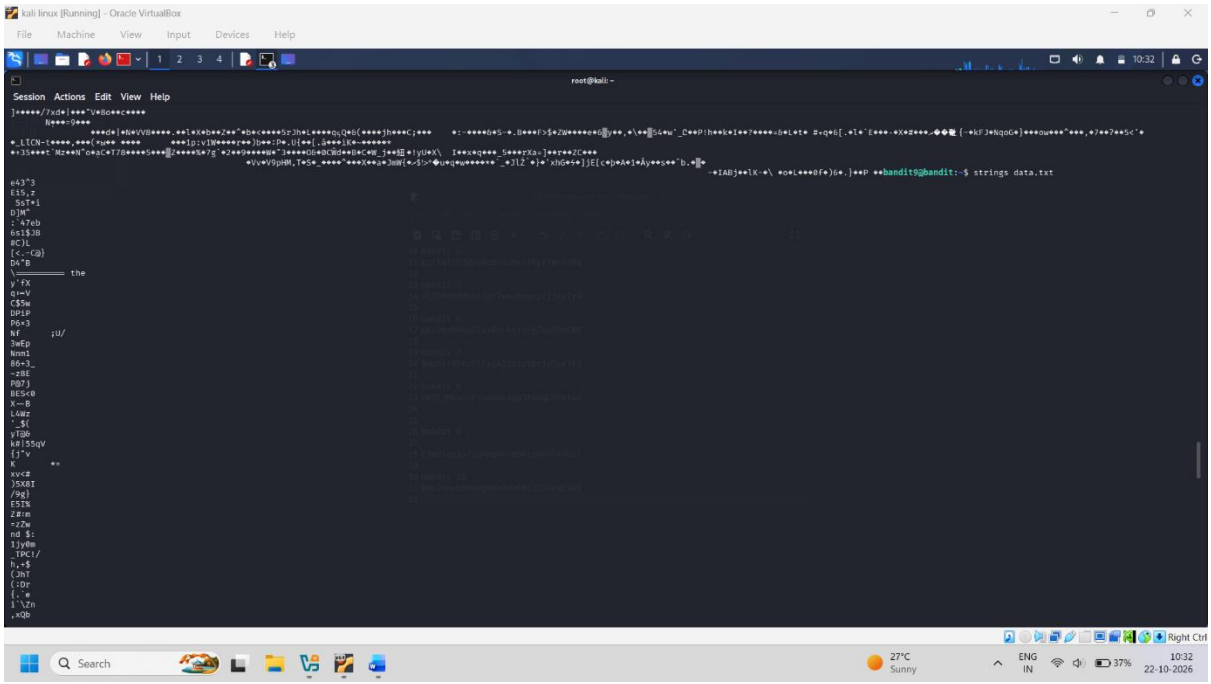
Step 2 - **cat data.txt**

Explanation: Displays the file contents, but binary data may appear unreadable.



Step 3 - **strings data.txt**

Explanation: Extracts readable text sequences from a binary file.



Step 4 - **grep -E**

grep -E '={2,}'

Explanation: Searches for lines containing two or more consecutive = characters using an extended regular expression.

Step 5 **- Pipe operator (|)**

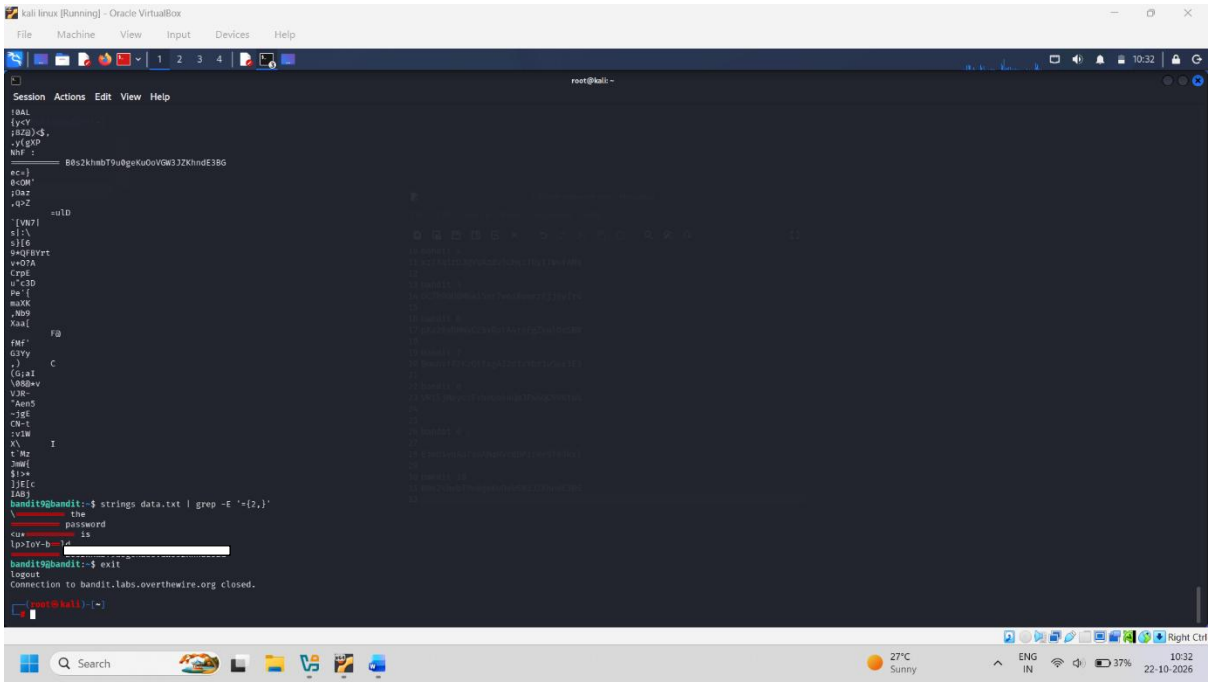
Explanation: Passes the output of one command as input to another command.

Step 6 - **Complete command**

strings data.txt | grep -E '={2,}'

Explanation: Extracts readable strings from data.txt and filters for lines containing at least two consecutive = characters to help locate the password for Level 10.

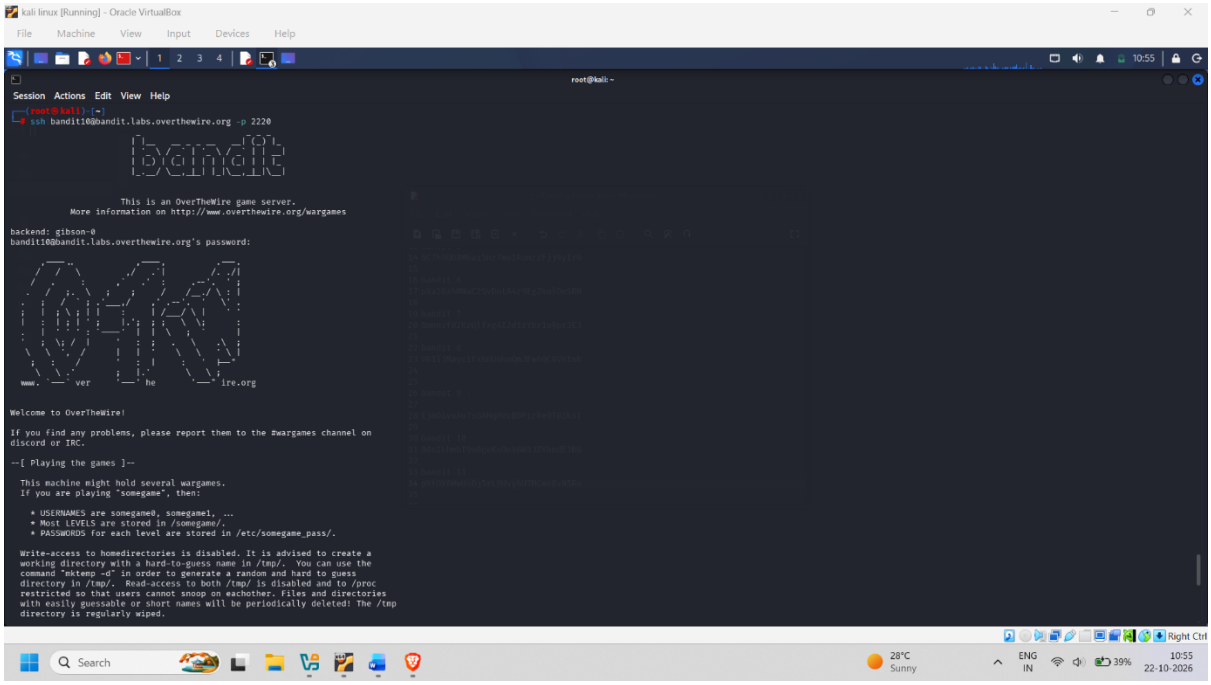
Result: The password for Bandit Level 10 is identified by examining the matching output.


kali linux [Running] - Oracle VirtualBox
File Machine View Input Devices Help
root@kali: ~
Session Actions Edit View Help
B0s2khmbT9u@goKuOoVGW3JZKhndE3BG
bandit9@bandit:~$ strings data.txt | grep -E '=[2,}'
the
password
is
bandit9@bandit:~$ exit
logout
Connection to bandit.labs.overthewire.org closed.
(root㉿kali)-[~]
Right Ctrl
Search
27°C
Sunny
ENG
IN
37%
10:32
22-10-2026

LEVEL 10 – LEVEL 11

Step 1: Login to Level 10

ssh bandit10@bandit.labs.overthewire.org -p 2220



Step 2 - **cat data.txt**

Explanation: Displays the contents of data.txt, revealing a Base64-encoded string.

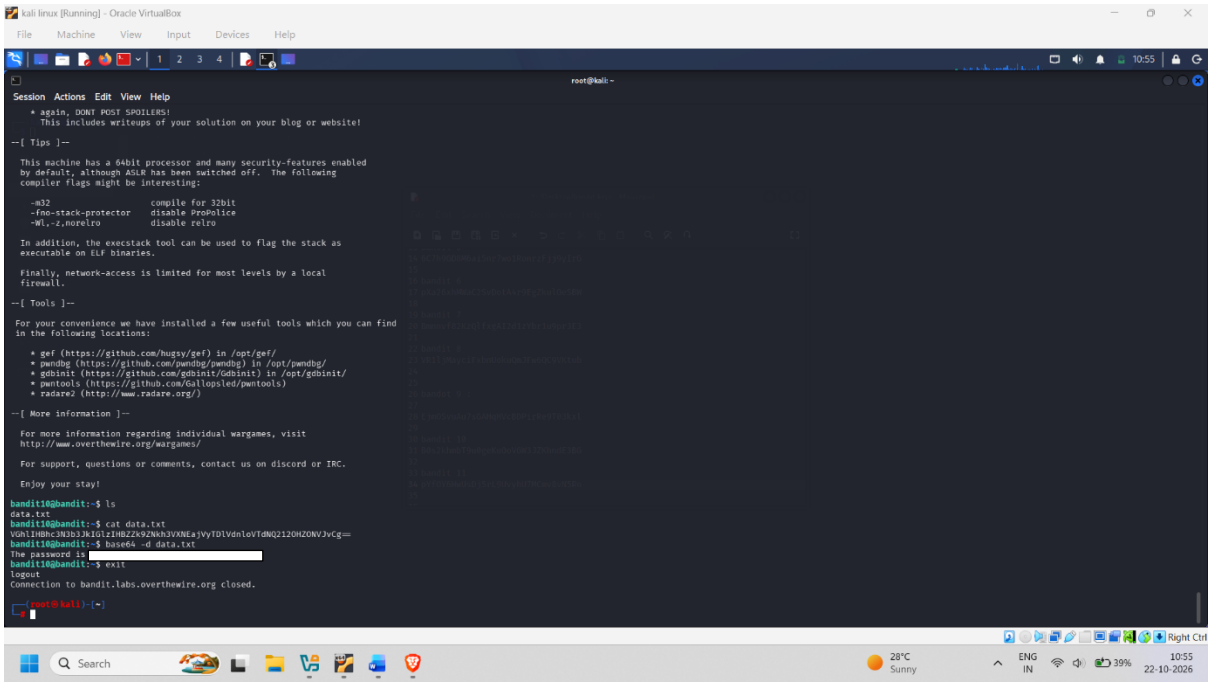
Step 3 - **base64 -d data.txt**

Explanation: Decodes the Base64-encoded content of data.txt into readable text.

Step 4 – **-d option**

Explanation: Tells the base64 command to decode the input instead of encoding it.

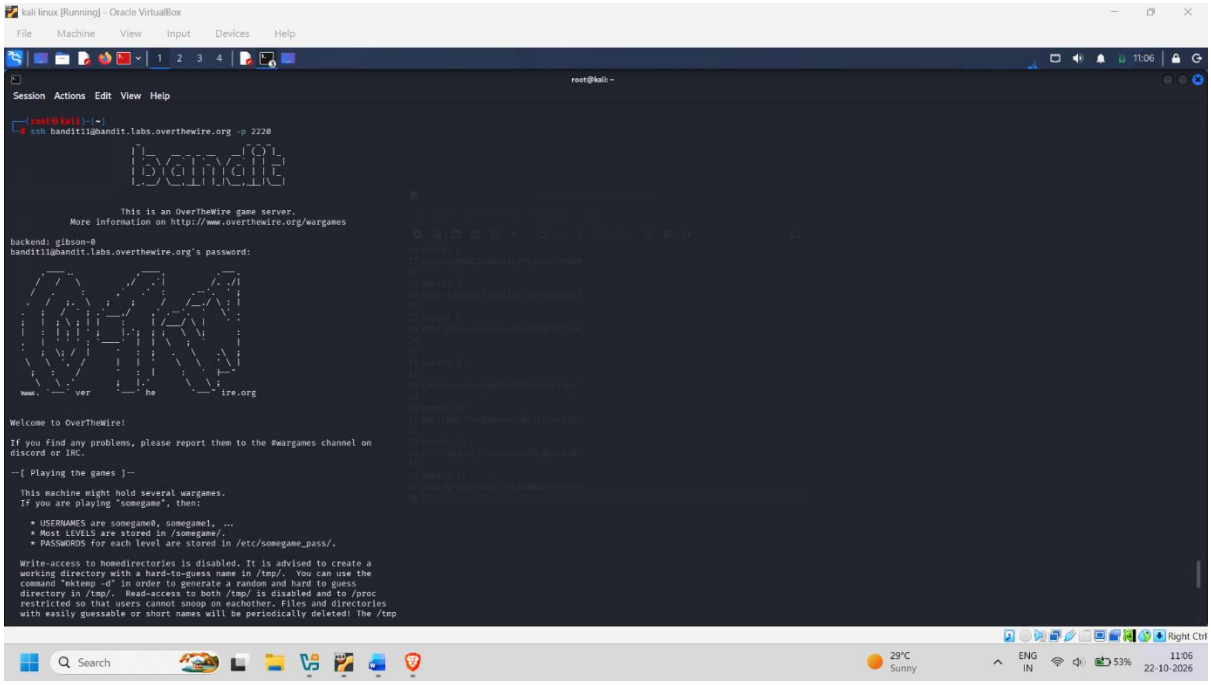
Result: The decoded text reveals the password for Bandit Level 11.


kali linux [Running] - Oracle VirtualBox
File Machine View Input Devices Help
root@kali: ~
Session Actions Edit View Help
* again, DONT POST SPOILERS!
This includes writeups of your solution on your blog or website!
--[Tips]--
This machine has a 64bit processor and many security-features enabled
by default, although ASLR has been switched off. The following
compiler flags might be interesting:
-m32 compile for 32bit
-fno-stack-protector disable ProPolice
-Wl,-z,norelro disable relro
In addition, the execstack tool can be used to flag the stack as
executable on ELF binaries.
Finally, network-access is limited for most levels by a local
firewall.
--[Tools]--
For your convenience we have installed a few useful tools which you can find
in the following locations:
* gef (https://github.com/hugsy/gef) in /opt/gef/
* pwndbg (https://github.com/pwndbg/pwndbg) in /opt/pwndbg/
* gdbinit (https://github.com/gdbinit/Gdbinit) in /opt/gdbinit/
* pwntools (https://github.com/Gallopsled/pwntools)
* radare2 (http://www.radare.org/)
--[More information]--
For more information regarding individual wargames, visit
http://www.overthewire.org/wargames/
For support, questions or comments, contact us on discord or IRC.
Enjoy your stay!
bandit10@bandit:~$ ls
data.txt
bandit10@bandit:~$ cat data.txt
VGhlIHBhc3N3b3JkIGlzIHBZk9ZNkh3VXNEajVyTDlVdnloVTdNQ2l2OHZONVJvCg==
bandit10@bandit:~$ base64 -d data.txt
The password is
bandit10@bandit:~$ exit
logout
Connection to bandit.labs.overthewire.org closed.
(root㉿kali)-[~]
Right Ctrl
Search
28°C
Sunny
ENG
IN
39%
10:55
22-10-2026

LEVEL 11 – LEVEL 12

Step 1: Login to Level 11

ssh bandit11@bandit.labs.overthewire.org -p 2220



Step 2 - **cat data.txt**

Explanation: Displays the contents of data.txt, which contains text encoded using ROT13.

Step 3 - **tr command**

tr 'A-Za-z' 'N-ZA-Mn-za-m'

Explanation: Translates each letter to its ROT13 equivalent to decode the encrypted text.

Step 4 - **Pipe operator (|)**

Explanation: Passes the output of one command as input to another command.

Step 5 - **Complete command**

cat data.txt | tr 'A-Za-z' 'N-ZA-Mn-za-m'

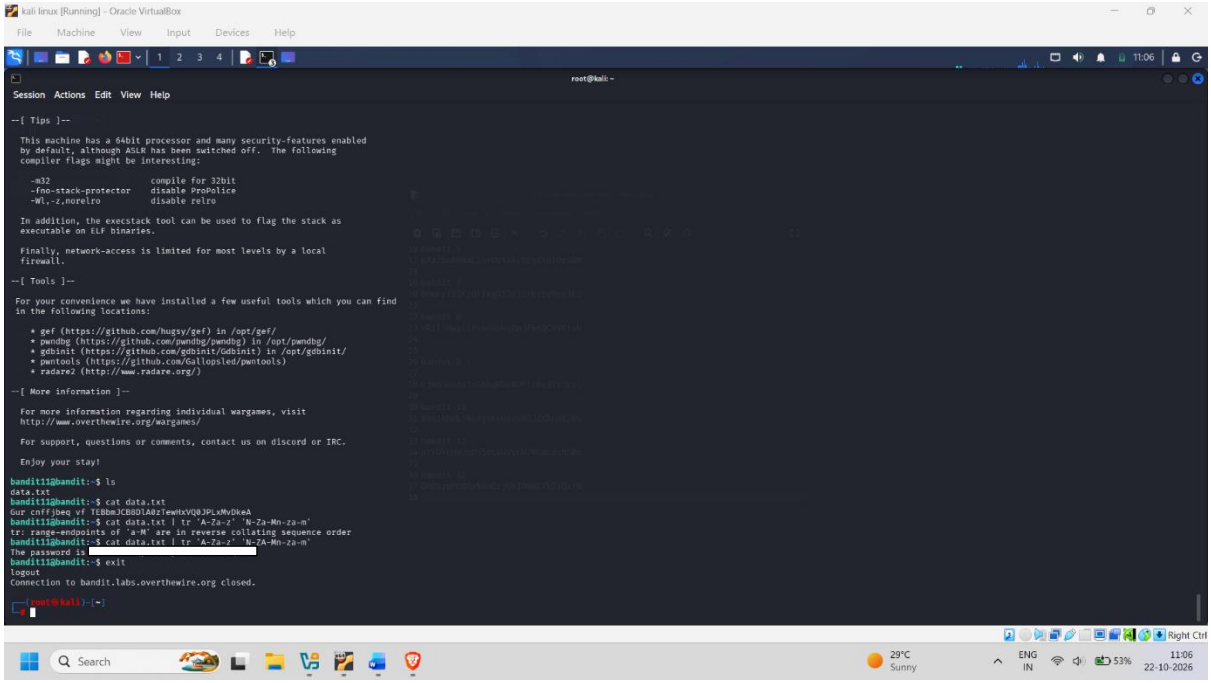
Explanation: Reads the file and applies ROT13 decoding to reveal the original text.

Step 6 - **Alternative command**

tr 'A-Za-z' 'N-ZA-Mn-za-m' < data.txt

Explanation: Uses input redirection (<) to provide data.txt directly to the tr command.

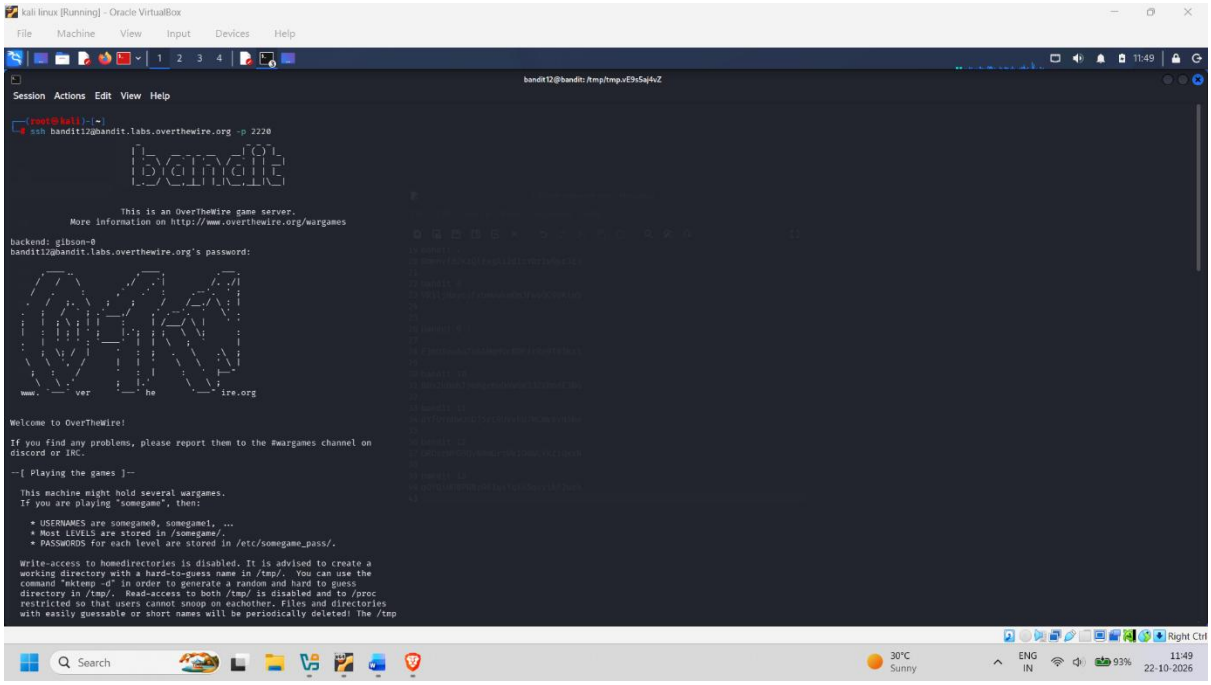
Result: The decoded text reveals the password for Bandit Level 12.



LEVEL 12 - LEVEL 13

Step 1: Login to Level 12

ssh bandit12@bandit.labs.overthewire.org -p 2220



Step 2 - **ls**

Explanation: Lists files and directories in the current directory.

Step 3 - **mktemp -d**

Explanation: Creates a temporary directory for safely working with files**.**

Step 4 - **cd**

cd /tmp/tmp.vE9s5aj4vZ

Explanation: Changes the current directory to the temporary directory**.**

Step 5 - **cp**

cp ~/data.txt .

Explanation: Copies data.txt from the home directory to the current directory.

Step 6 - **xxd -r**

xxd -r data.txt file1

Explanation: Converts the hexadecimal dump back into binary data.

Step 7 - **file**

file file1

Explanation: Identifies the file type to determine the appropriate decompression method.

Step 8 - **mv**

mv file1 file1.gz

Explanation: Renames the file with a .gz extension**.**

Step 9 - **gzip -d**

gzip -d file1.gz

Explanation: Decompresses a gzip-compressed file.

Step 10 - **bzip2 -d**

bzip2 -d file.bz2

Explanation: Decompresses a bzip2-compressed file.

Step 11 - **tar -xf**

tar -xf file.tar

Explanation: Extracts files from a TAR archive**.**

Step 12 - **ls -la**

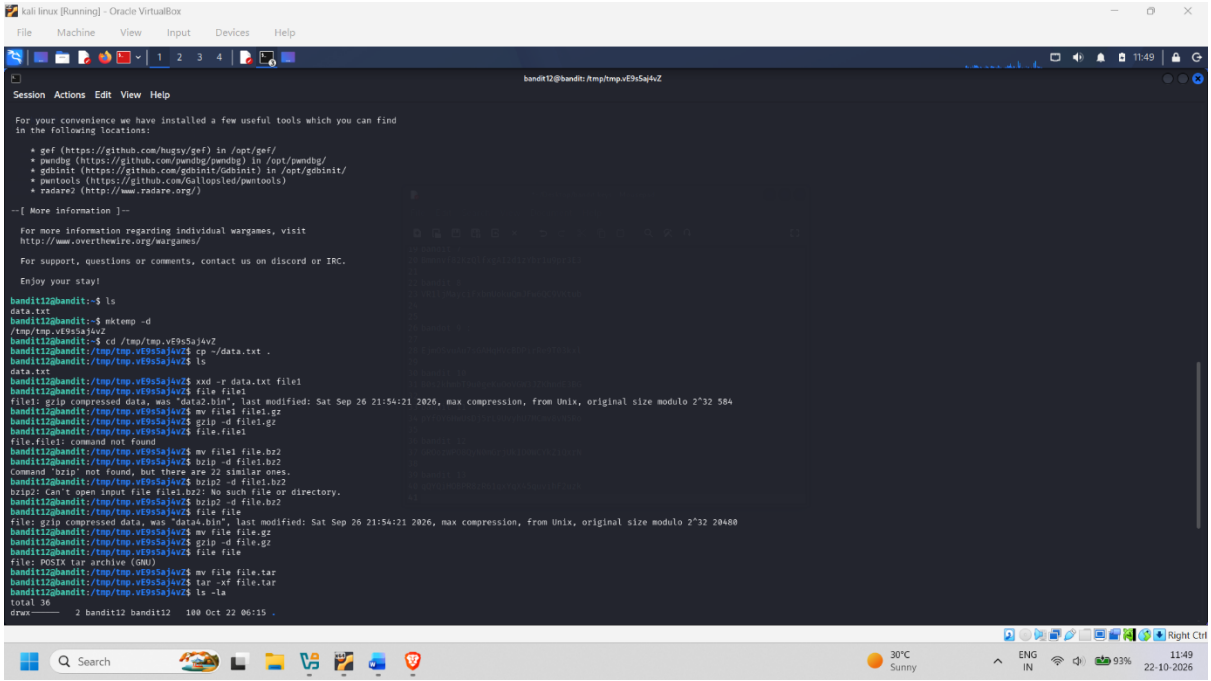
Explanation: Lists all files, including hidden files, with detailed information.

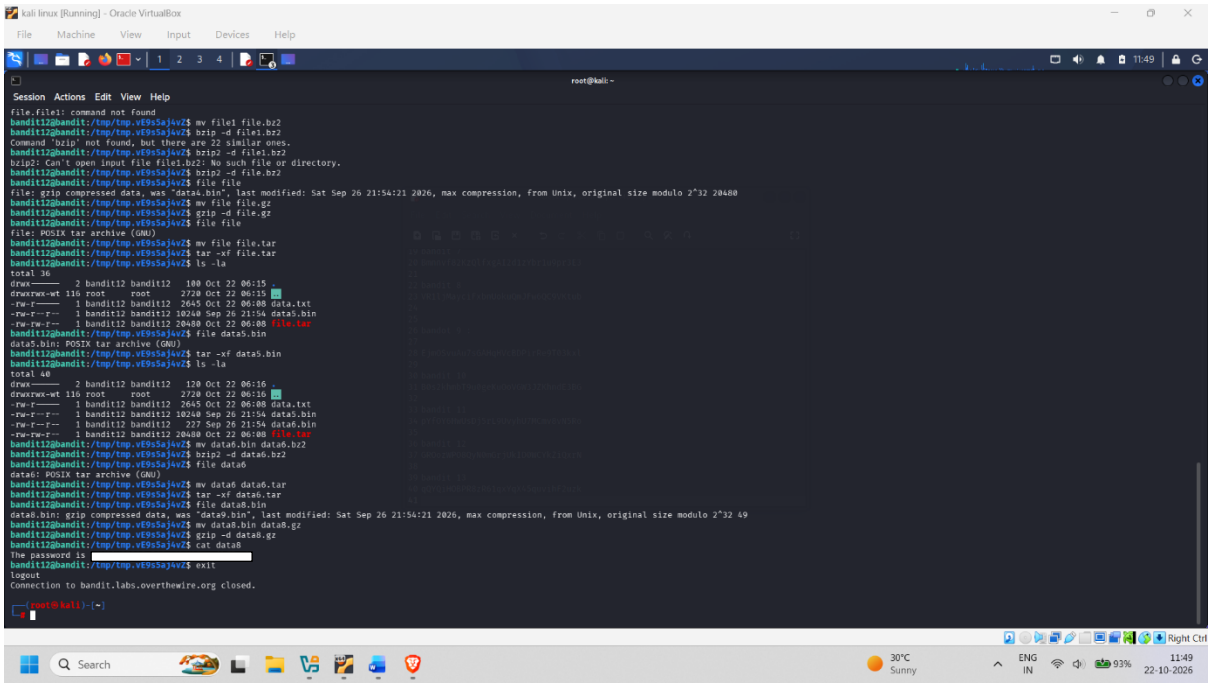
Step 13 - **cat**

cat data8

Explanation: Displays the final text containing the password for Bandit Level 13.

Result: The hexadecimal dump was reversed, and the nested compressed files and archives were extracted until the password for Bandit Level 13 was obtained.

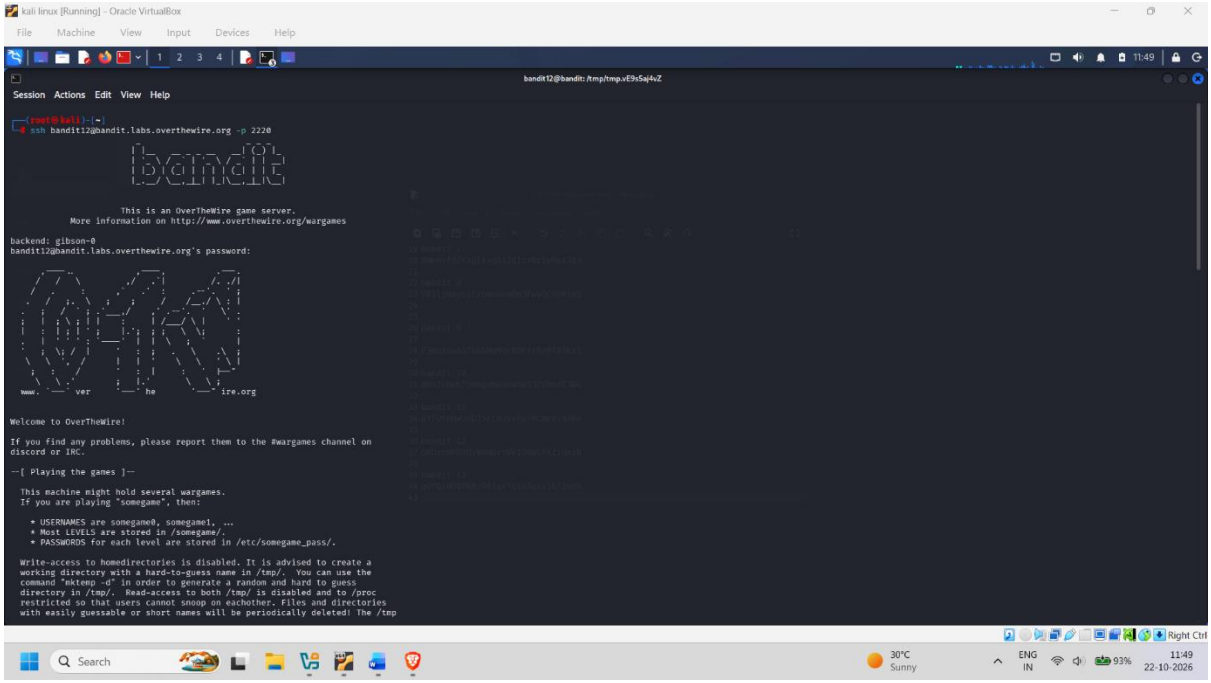

kali linux [Running] - Oracle VirtualBox
File Machine View Input Devices Help
bandit12@bandit: /tmp/tmp.vE9s5aj4vZ
Session Actions Edit View Help
For your convenience we have installed a few useful tools which you can find
in the following locations:
* gef (https://github.com/hugsy/gef) in /opt/gef/
* pwndbg (https://github.com/pwndbg/pwndbg) in /opt/pwndbg/
* gdbinit (https://github.com/gdbinit/Gdbinit) in /opt/gdbinit/
* pwntools (https://github.com/Gallopsled/pwntools)
* radare2 (http://www.radare.org/)
--[More information]--
For more information regarding individual wargames, visit
http://www.overthewire.org/wargames/
For support, questions or comments, contact us on discord or IRC.
Enjoy your stay!
bandit12@bandit:~$ ls
data.txt
bandit12@bandit:~$ mktemp -d
/tmp/tmp.vE9s5aj4vZ
bandit12@bandit:~$ cd /tmp/tmp.vE9s5aj4vZ
bandit12@bandit:/tmp/tmp.vE9s5aj4vZ$ cp ~/data.txt .
bandit12@bandit:/tmp/tmp.vE9s5aj4vZ$ ls
data.txt
bandit12@bandit:/tmp/tmp.vE9s5aj4vZ$ xxd -r data.txt file1
bandit12@bandit:/tmp/tmp.vE9s5aj4vZ$ file file1
file1: gzip compressed data, was "data2.bin", last modified: Sat Sep 26 21:54:21 2026, max compression, from Unix, original size modulo 2^32 584
bandit12@bandit:/tmp/tmp.vE9s5aj4vZ$ mv file1 file1.gz
bandit12@bandit:/tmp/tmp.vE9s5aj4vZ$ gzip -d file1.gz
bandit12@bandit:/tmp/tmp.vE9s5aj4vZ$ file.file1
file.file1: command not found
bandit12@bandit:/tmp/tmp.vE9s5aj4vZ$ mv file1 file.bz2
bandit12@bandit:/tmp/tmp.vE9s5aj4vZ$ bzip -d file1.bz2
Command 'bzip' not found, but there are 22 similar ones.
bandit12@bandit:/tmp/tmp.vE9s5aj4vZ$ bzip2 -d file1.bz2
bzip2: Can't open input file file1.bz2: No such file or directory.
bandit12@bandit:/tmp/tmp.vE9s5aj4vZ$ bzip2 -d file.bz2
bandit12@bandit:/tmp/tmp.vE9s5aj4vZ$ file file
file: gzip compressed data, was "data4.bin", last modified: Sat Sep 26 21:54:21 2026, max compression, from Unix, original size modulo 2^32 20480
bandit12@bandit:/tmp/tmp.vE9s5aj4vZ$ mv file file.gz
bandit12@bandit:/tmp/tmp.vE9s5aj4vZ$ gzip -d file.gz
bandit12@bandit:/tmp/tmp.vE9s5aj4vZ$ file file
file: POSIX tar archive (GNU)
bandit12@bandit:/tmp/tmp.vE9s5aj4vZ$ mv file file.tar
bandit12@bandit:/tmp/tmp.vE9s5aj4vZ$ tar -xf file.tar
bandit12@bandit:/tmp/tmp.vE9s5aj4vZ$ ls -la
total 36
drwx------ 2 bandit12 bandit12 100 Oct 22 06:15 .
Right Ctrl
Search
30°C
Sunny
ENG
IN
93%
11:49
22-10-2026


kali linux [Running] - Oracle VirtualBox
File Machine View Input Devices Help
root@kali: ~
Session Actions Edit View Help
file.file1: command not found
bandit12@bandit:/tmp/tmp.vE9s5aj4vZ$ mv file1 file.bz2
bandit12@bandit:/tmp/tmp.vE9s5aj4vZ$ bzip -d file1.bz2
Command 'bzip' not found, but there are 22 similar ones.
bandit12@bandit:/tmp/tmp.vE9s5aj4vZ$ bzip2 -d file1.bz2
bzip2: Can't open input file file1.bz2: No such file or directory.
bandit12@bandit:/tmp/tmp.vE9s5aj4vZ$ bzip2 -d file.bz2
bandit12@bandit:/tmp/tmp.vE9s5aj4vZ$ file file
file: gzip compressed data, was "data4.bin", last modified: Sat Sep 26 21:54:21 2026, max compression, from Unix, original size modulo 2^32 20480
bandit12@bandit:/tmp/tmp.vE9s5aj4vZ$ mv file file.gz
bandit12@bandit:/tmp/tmp.vE9s5aj4vZ$ gzip -d file.gz
bandit12@bandit:/tmp/tmp.vE9s5aj4vZ$ file file
file: POSIX tar archive (GNU)
bandit12@bandit:/tmp/tmp.vE9s5aj4vZ$ mv file file.tar
bandit12@bandit:/tmp/tmp.vE9s5aj4vZ$ tar -xf file.tar
bandit12@bandit:/tmp/tmp.vE9s5aj4vZ$ ls -la
total 36
drwx------ 2 bandit12 bandit12 100 Oct 22 06:15 .
drwxrwxrwt 116 root root 2720 Oct 22 06:15 ..
-rw-r----- 1 bandit12 bandit12 2645 Oct 22 06:08 data.txt
-rw-r--r-- 1 bandit12 bandit12 10240 Sep 26 21:54 data5.bin
-rw-rw-r-- 1 bandit12 bandit12 20480 Oct 22 06:08 file.tar
bandit12@bandit:/tmp/tmp.vE9s5aj4vZ$ file data5.bin
data5.bin: POSIX tar archive (GNU)
bandit12@bandit:/tmp/tmp.vE9s5aj4vZ$ tar -xf data5.bin
bandit12@bandit:/tmp/tmp.vE9s5aj4vZ$ ls -la
total 40
drwx------ 2 bandit12 bandit12 120 Oct 22 06:16 .
drwxrwxrwt 116 root root 2720 Oct 22 06:16 ..
-rw-r----- 1 bandit12 bandit12 2645 Oct 22 06:08 data.txt
-rw-r--r-- 1 bandit12 bandit12 10240 Sep 26 21:54 data5.bin
-rw-r--r-- 1 bandit12 bandit12 227 Sep 26 21:54 data6.bin
-rw-rw-r-- 1 bandit12 bandit12 20480 Oct 22 06:08 file.tar
bandit12@bandit:/tmp/tmp.vE9s5aj4vZ$ mv data6.bin data6.bz2
bandit12@bandit:/tmp/tmp.vE9s5aj4vZ$ bzip2 -d data6.bz2
bandit12@bandit:/tmp/tmp.vE9s5aj4vZ$ file data6
data6: POSIX tar archive (GNU)
bandit12@bandit:/tmp/tmp.vE9s5aj4vZ$ mv data6 data6.tar
bandit12@bandit:/tmp/tmp.vE9s5aj4vZ$ tar -xf data6.tar
bandit12@bandit:/tmp/tmp.vE9s5aj4vZ$ file data8.bin
data8.bin: gzip compressed data, was "data9.bin", last modified: Sat Sep 26 21:54:21 2026, max compression, from Unix, original size modulo 2^32 49
bandit12@bandit:/tmp/tmp.vE9s5aj4vZ$ mv data8.bin data8.gz
bandit12@bandit:/tmp/tmp.vE9s5aj4vZ$ gzip -d data8.gz
bandit12@bandit:/tmp/tmp.vE9s5aj4vZ$ cat data8
The password is
bandit12@bandit:/tmp/tmp.vE9s5aj4vZ$ exit
logout
Connection to bandit.labs.overthewire.org closed.
┌──(root㉿kali)-[~]
└─#
Right Ctrl
Search
30°C
Sunny
ENG
IN
93%
11:49
22-10-2026

LEVEL 13 – LEVEL 14

Step 1: Login to Level 13

ssh bandit13@bandit.labs.overthewire.org -p 2220



Step 2 - **ls**

Explanation: Lists files and directories, including sshkey.private.

Step 3 - **cat sshkey.private**

Explanation: Displays the contents of the SSH private key file.

Step 4 – **scp**

scp -P 2220 bandit13@bandit.labs.overthewire.org:~/sshkey.private ./bandit14.key

Explanation: Copies the SSH private key from the remote server to the local Kali Linux machine.

Step 5 - **nano bandit14.key**

Explanation: Opens a text editor to create or edit the private key file manually.

Step 6 - **chmod 600**

chmod 600 bandit14.key

Explanation: Restricts the file permissions so only the owner can read and modify the private key.

Step 7 - **ssh -i**

ssh -i bandit14.key bandit14@bandit.labs.overthewire.org -p 2220

Explanation: Connects to the remote server as bandit14 using the specified private key for authentication.

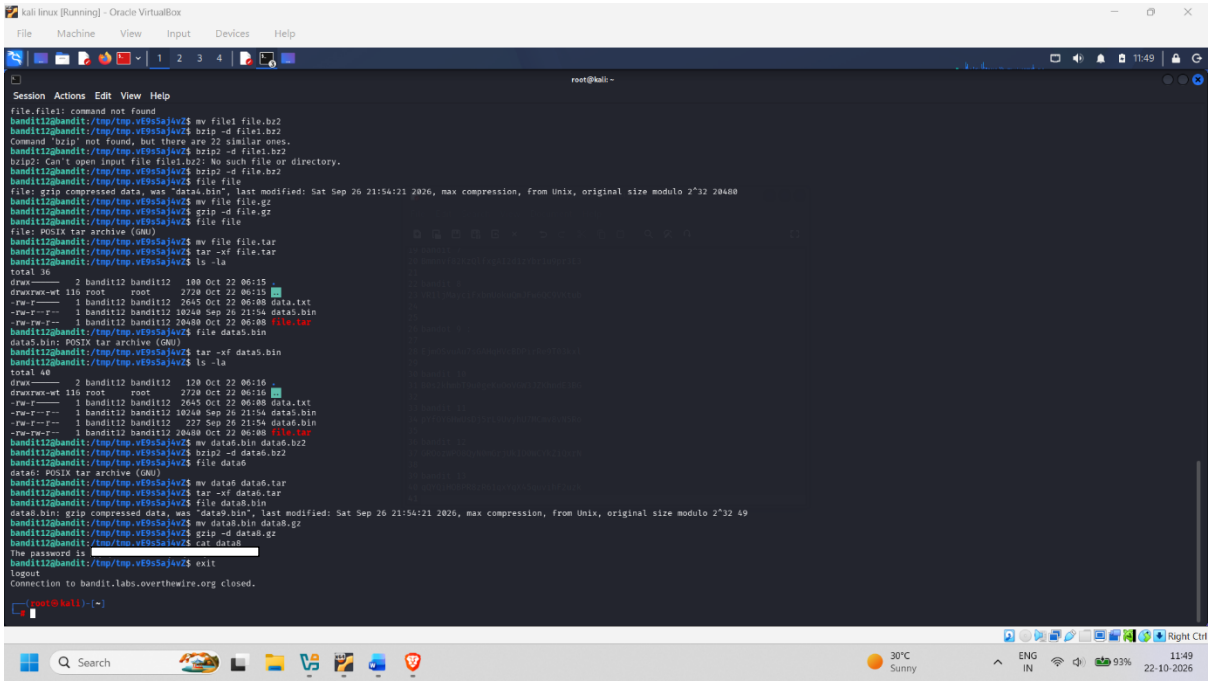
Step 8 - **cat /etc/bandit_pass/bandit14**

Explanation: Displays the password for Bandit Level 14.

Step 8 - **exit**

Explanation: Closes the current SSH session**.**

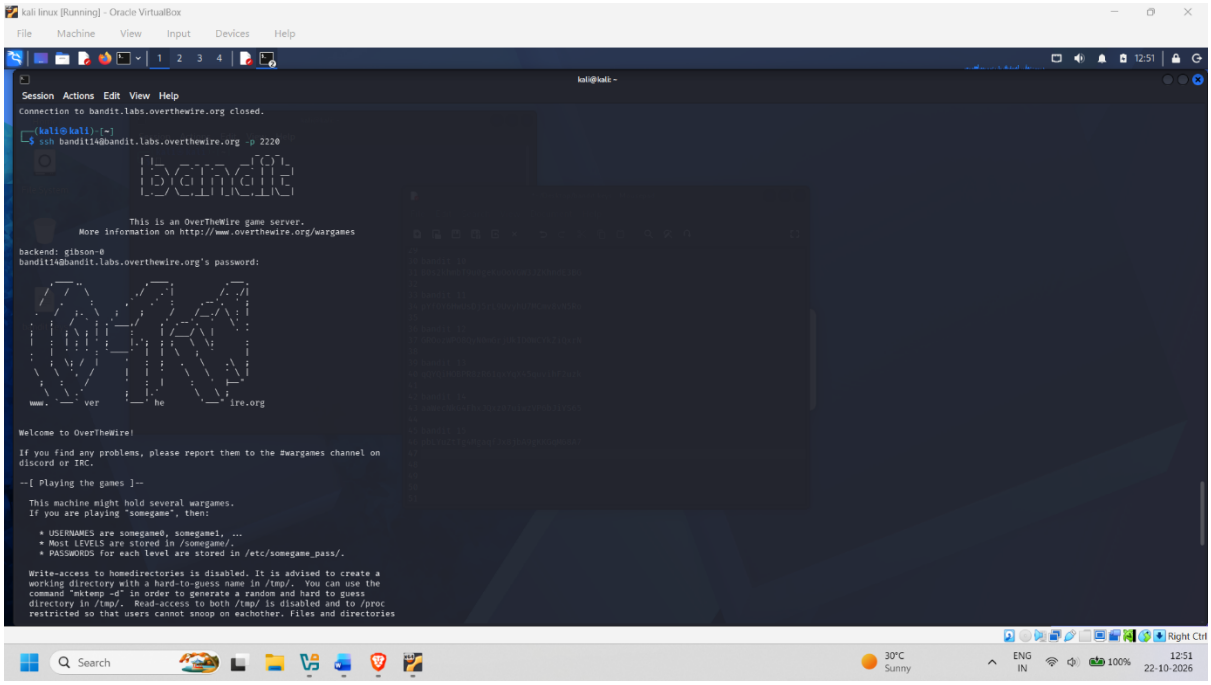
Result: Successfully authenticated to Bandit Level 14 using an SSH private key and retrieved the level's password.



LEVEL 14 – LEVEL 15

Step 1: Login to Level 14

ssh bandit14@bandit.labs.overthewire.org -p 2220



Step 2 - **ls**

Explanation - Lists files and directories in the current directory**.**

Step 3 - **nc localhost 30000**

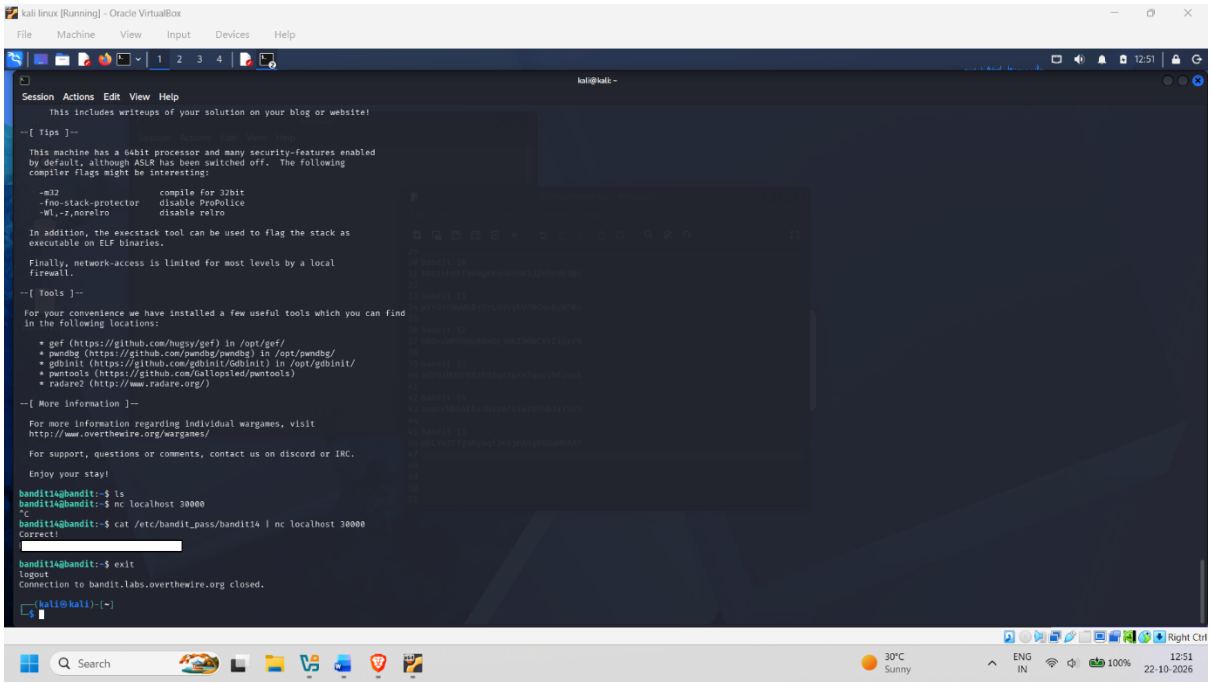
Explanation -Connects to port 30000 on the local machine using Netcat.

Step 4 - **cat /etc/bandit_pass/bandit14 | nc localhost 30000**

Explanation - Reads the Bandit 14 password and sends it to the local service on port 30000.

Step 5 - **exit**

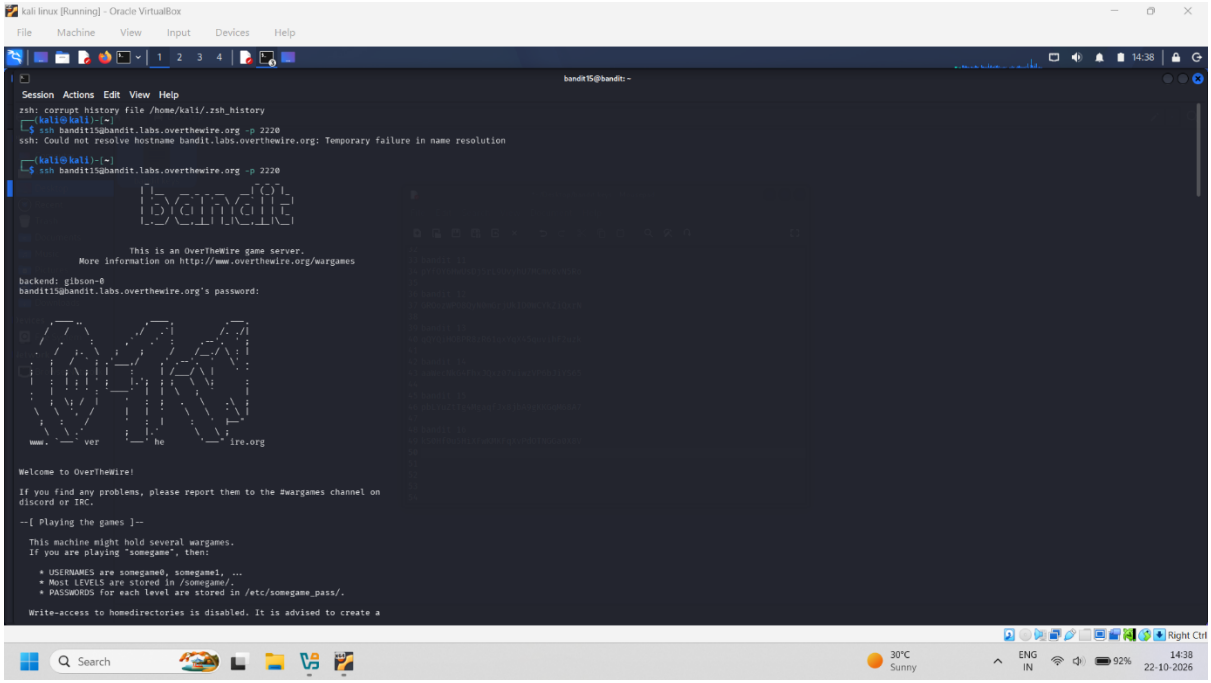
Explanation - Closes the current SSH session**.**


kali linux [Running] - Oracle VirtualBox
File Machine View Input Devices Help
kali@kali: ~
Session Actions Edit View Help
This includes writeups of your solution on your blog or website!
--[Tips]--
This machine has a 64bit processor and many security-features enabled
by default, although ASLR has been switched off. The following
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-fno-stack-protector disable ProPolice
-Wl,-z,norelro disable relro
In addition, the execstack tool can be used to flag the stack as
executable on ELF binaries.
Finally, network-access is limited for most levels by a local
firewall.
--[Tools]--
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in the following locations:
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* pwndbg (https://github.com/pwndbg/pwndbg) in /opt/pwndbg/
* gdbinit (https://github.com/gdbinit/Gdbinit) in /opt/gdbinit/
* pwntools (https://github.com/Gallopsled/pwntools)
* radare2 (http://www.radare.org/)
--[More information]--
For more information regarding individual wargames, visit
http://www.overthewire.org/wargames/
For support, questions or comments, contact us on discord or IRC.
Enjoy your stay!
bandit14@bandit:~$ ls
bandit14@bandit:~$ nc localhost 30000
^C
bandit14@bandit:~$ cat /etc/bandit_pass/bandit14 | nc localhost 30000
Correct!
bandit14@bandit:~$ exit
logout
Connection to bandit.labs.overthewire.org closed.
┌──(kali㉿kali)-[~]
└─$
Right Ctrl
Search
30°C
Sunny
ENG
IN
100%
12:51
22-10-2026

LEVEL 15 – LEVEL 16

Step 1: Login to Level 15

ssh bandit15@bandit.labs.overthewire.org -p 2220



Step 2 - Password Submission Using OpenSSL

cat /etc/bandit_pass/bandit15 | openssl s_client -connect localhost:30001 -quiet

Explanation:

- cat — Reads the password file for Bandit Level 15.

- /etc/bandit_pass/bandit15 — Path to the password file.

- | — Sends the output of the first command to the next command.

- openssl s_client — Establishes an SSL/TLS client connection.

- -connect localhost:30001 — Connects to port 30001 on the local machine.

- -quiet — Reduces connection-related output.

Step 3 - Exit Command

exit

Explanation: Closes the current terminal shell or SSH session.

Step 4 -Next Level SSH Command

ssh bandit16@bandit.labs.overthewire.org -p 2220

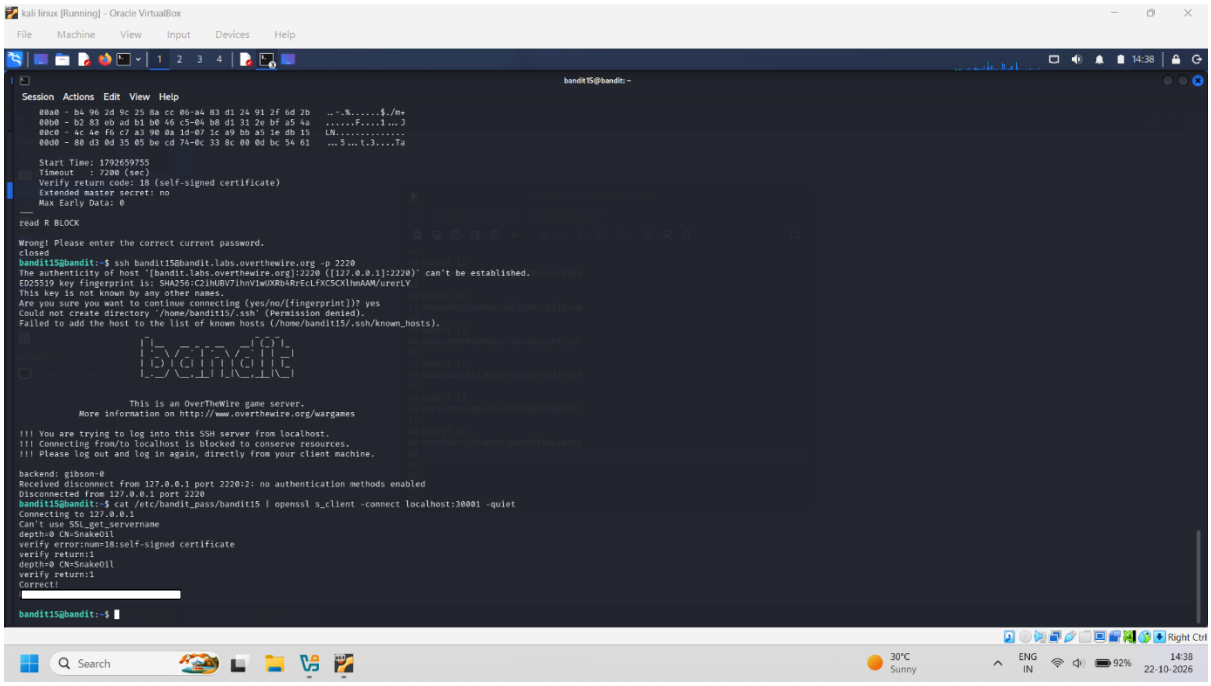
Explanation:

- ssh — Starts a secure remote connection.

- bandit16 — Username for Level 16.

- bandit.labs.overthewire.org — Bandit challenge server.

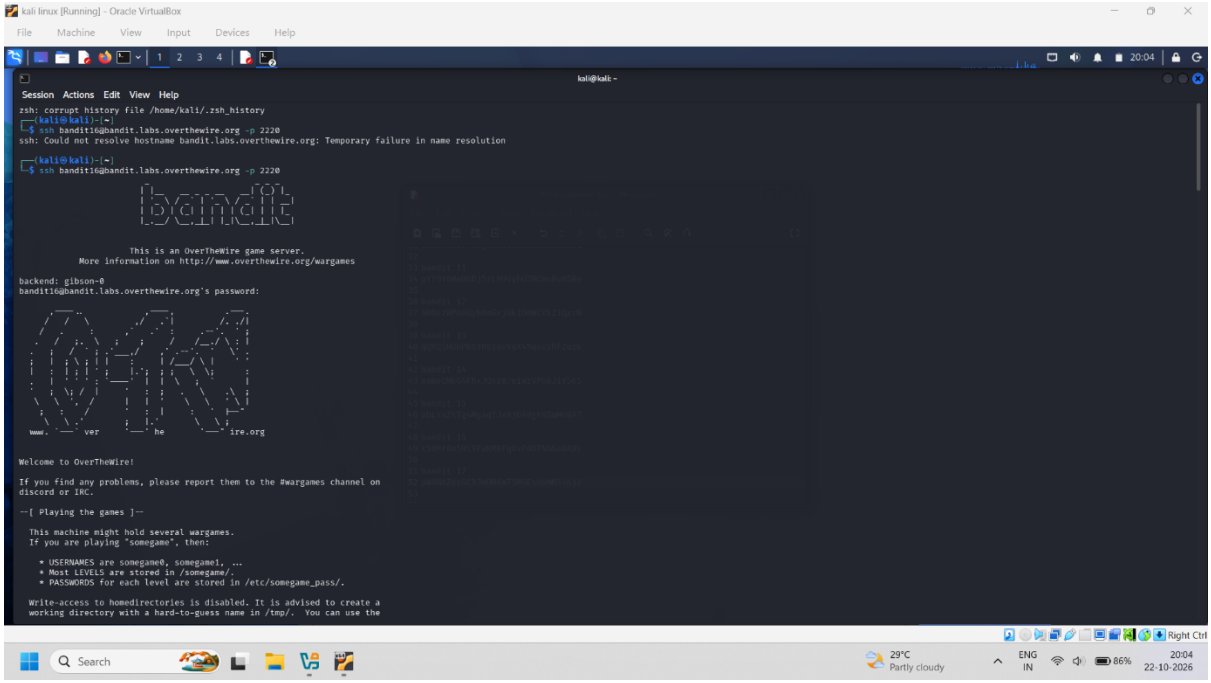
- -p 2220 — Specifies port 2220.



LEVEL 16 – LEVEL 17

Step 1: Login to Level 16

ssh bandit16@bandit.labs.overthewire.org -p 2220



Step 2 - **cat /etc/bandit_pass/bandit16 | openssl s_client -connect localhost:31790 -quiet > bandit17.key**

Explanation: Reads the Level 16 password, sends it to the SSL/TLS service, and saves the output in bandit17.key. It fails if the current directory is not writable.

Step 3 - **mkdir -p /tmp/mykey && cd /tmp/mykey**

Explanation: Creates a writable directory and changes into it.

Step 4 - **cat /etc/bandit_pass/bandit16 | openssl s_client -connect localhost:31790 -quiet > bandit17.key**

Explanation: Connects to the service and saves the returned private key in the writable directory.

Step 5 - **head -n 5 bandit17.key**

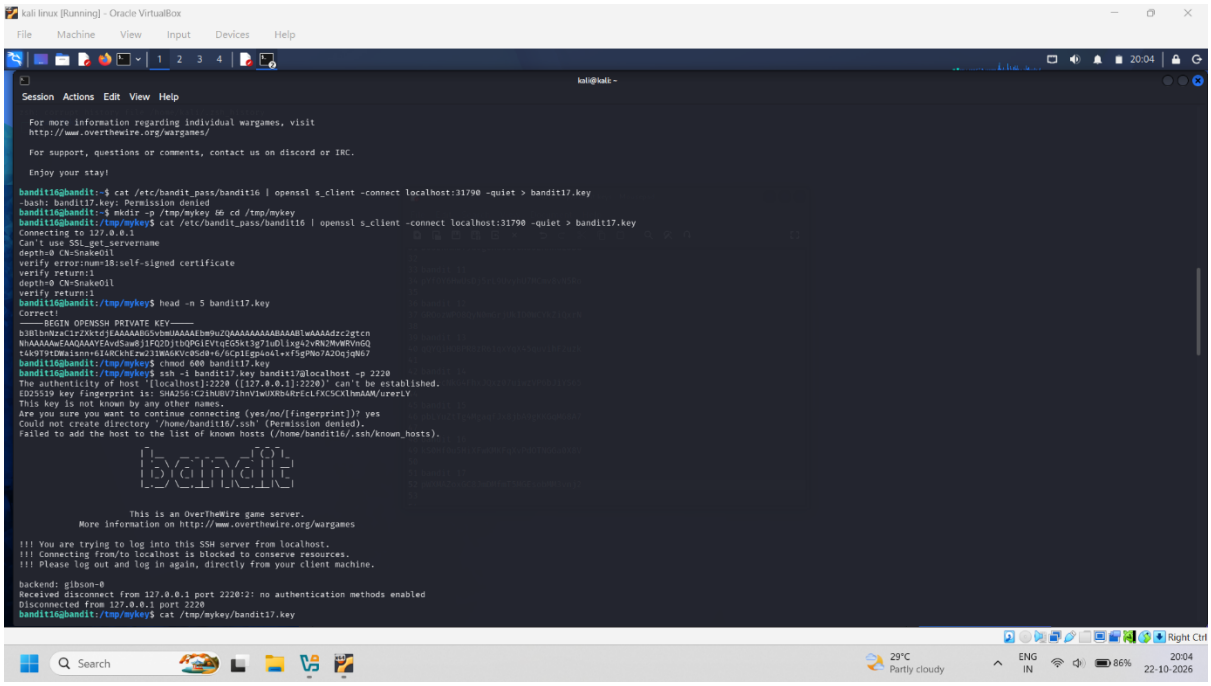
Explanation: Displays the first five lines to check the file contents.

Step 6 - **chmod 600 bandit17.key**

Explanation: Restricts file permissions so only the owner can read and write the key.

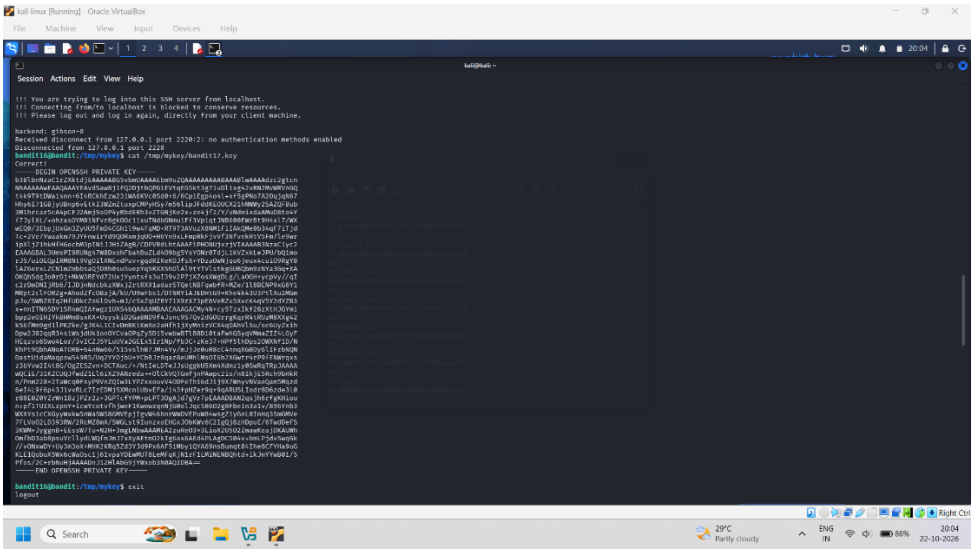
Step 7 - **ssh -i bandit17.key bandit17@localhost -p 2220**

Explanation: Attempts to log in to Level 17 using the private key through localhost.



Step 8 - **cat /tmp/mykey/bandit17.key**

Explanation: Displays the complete output so the private key can be copied.



Step 9 - **exit**

Explanation: Closes the current remote session and returns to the previous terminal.

Step 10 - **nano bandit17.key**

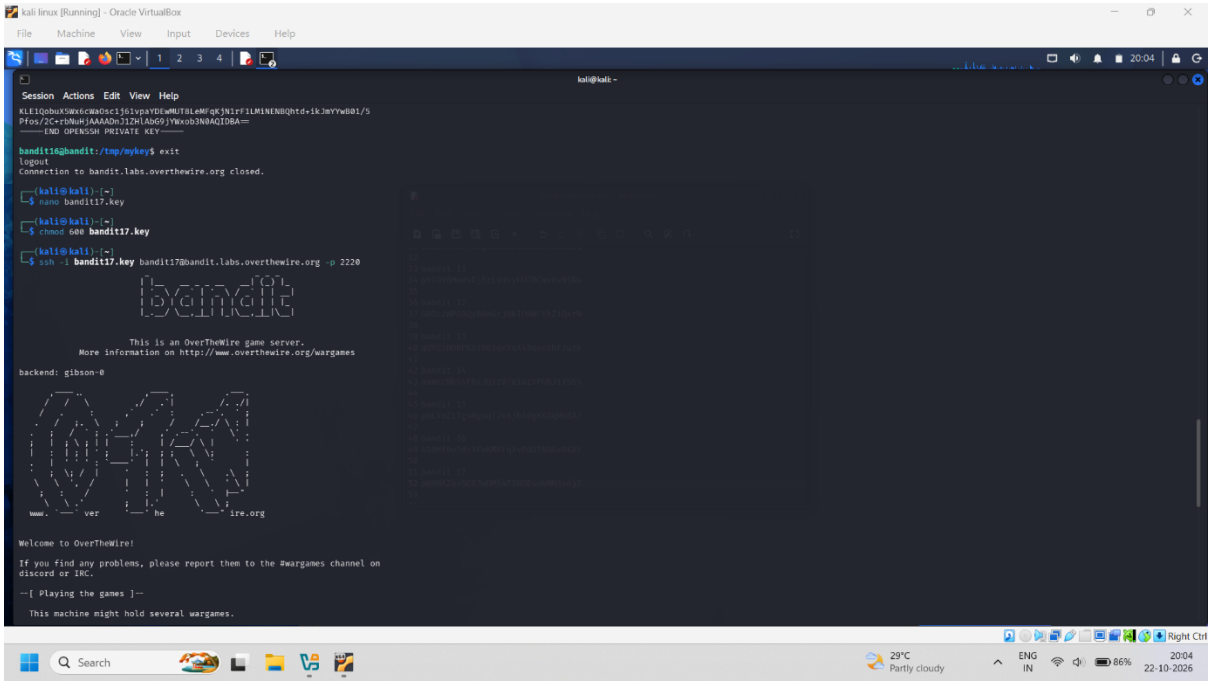
Explanation: Opens Nano to paste and save the clean private key on the local Kali machine.

Step 11 - **chmod 600 bandit17.key**

Explanation: Restricts the local private key's permissions for secure SSH authentication.

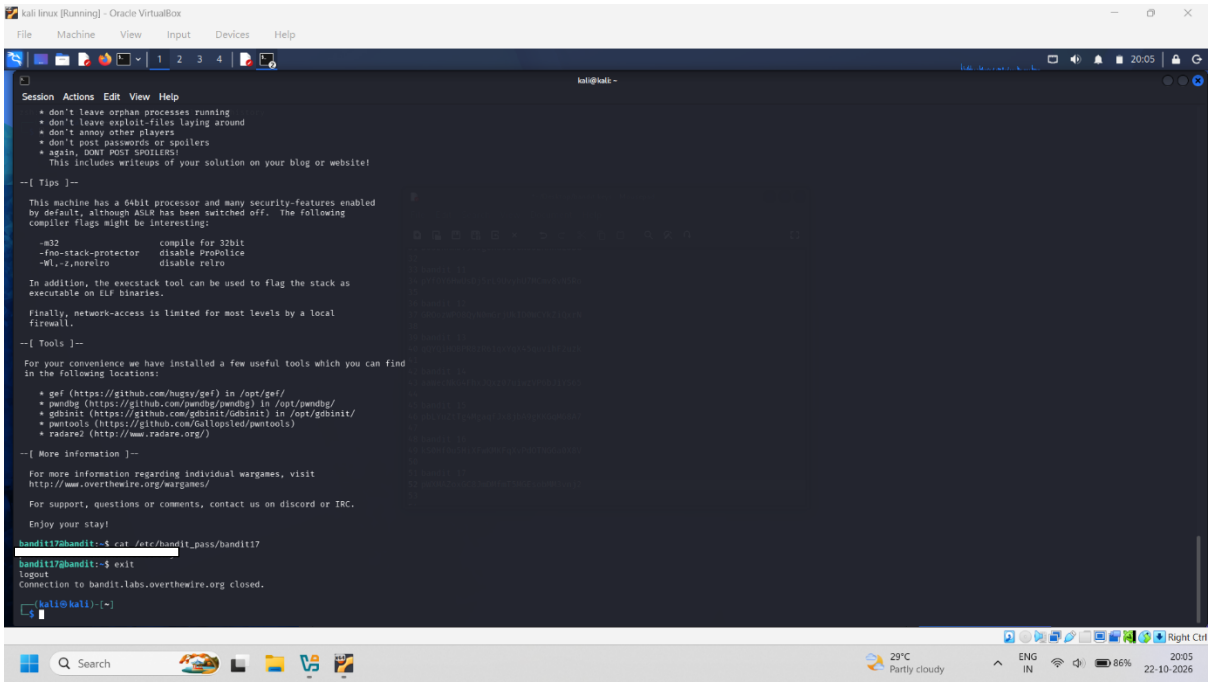
Step 12 - **ssh -i bandit17.key bandit17@bandit.labs.overthewire.org -p 2220**

Explanation: Connects to the Bandit server as bandit17 using the private key.



Step 13 - **cat /etc/bandit_pass/bandit17**

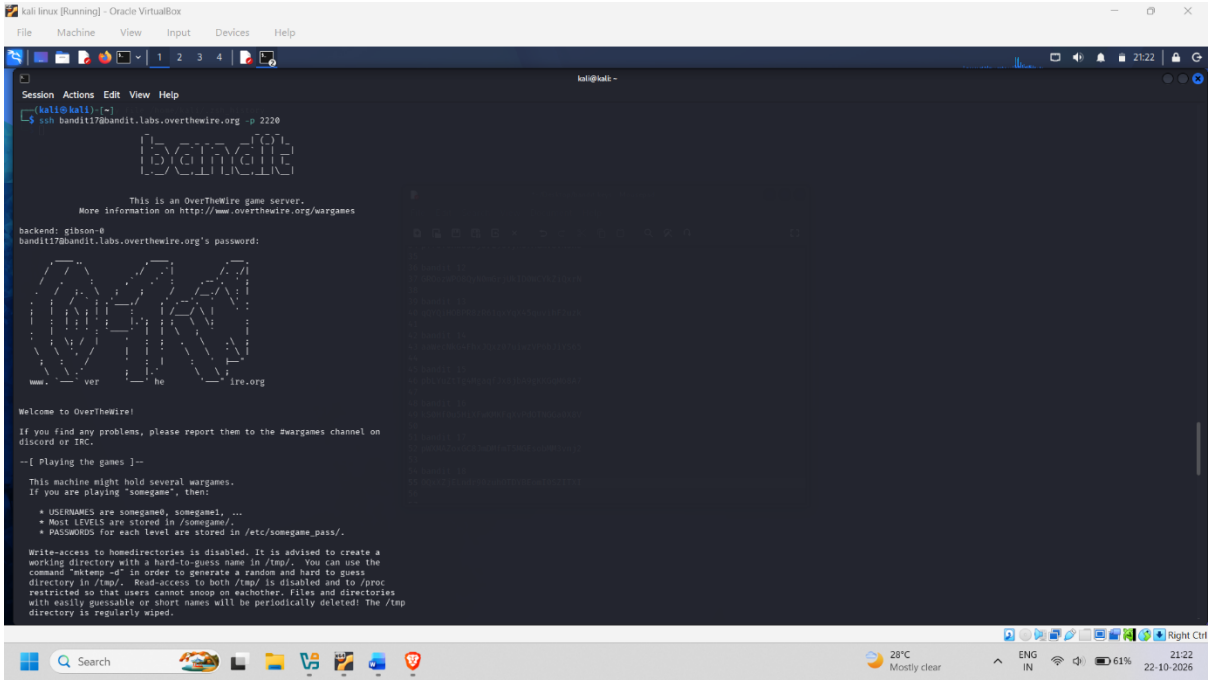
Explanation: Displays the Level 17 password if the current account has permission to read it.


kali linux [Running] - Oracle VirtualBox
File Machine View Input Devices Help
kali@kali: ~
Session Actions Edit View Help
* don't leave orphan processes running
* don't leave exploit-files laying around
* don't annoy other players
* don't post passwords or spoilers
* again, DONT POST SPOILERS!
This includes writeups of your solution on your blog or website!
--[Tips]--
This machine has a 64bit processor and many security-features enabled
by default, although ASLR has been switched off. The following
compiler flags might be interesting:
-m32 compile for 32bit
-fno-stack-protector disable ProPolice
-Wl,-z,norelro disable relro
In addition, the execstack tool can be used to flag the stack as
executable on ELF binaries.
Finally, network-access is limited for most levels by a local
firewall.
--[Tools]--
For your convenience we have installed a few useful tools which you can find
in the following locations:
* gef (https://github.com/hugsy/gef) in /opt/gef/
* pwndbg (https://github.com/pwndbg/pwndbg) in /opt/pwndbg/
* gdbinit (https://github.com/gdbinit/Gdbinit) in /opt/gdbinit/
* pwntools (https://github.com/Gallopsled/pwntools)
* radare2 (http://www.radare.org/)
--[More information]--
For more information regarding individual wargames, visit
http://www.overthewire.org/wargames/
For support, questions or comments, contact us on discord or IRC.
Enjoy your stay!
bandit17@bandit:~$ cat /etc/bandit_pass/bandit17
bandit17@bandit:~$ exit
logout
Connection to bandit.labs.overthewire.org closed.
Right Ctrl
Search
29°C
Partly cloudy
ENG
IN
86%
20:05
22-10-2026

LEVEL 17 – LEVEL 18

Step 1: Login to Level 17

ssh bandit17@bandit.labs.overthewire.org -p 2220



Step 2 - **diff passwords.old passwords.new**

Explanation: Compares two files and displays the difference to identify the Level 18 password.

Step 3 - **ssh bandit18@bandit.labs.overthewire.org -p 2220**

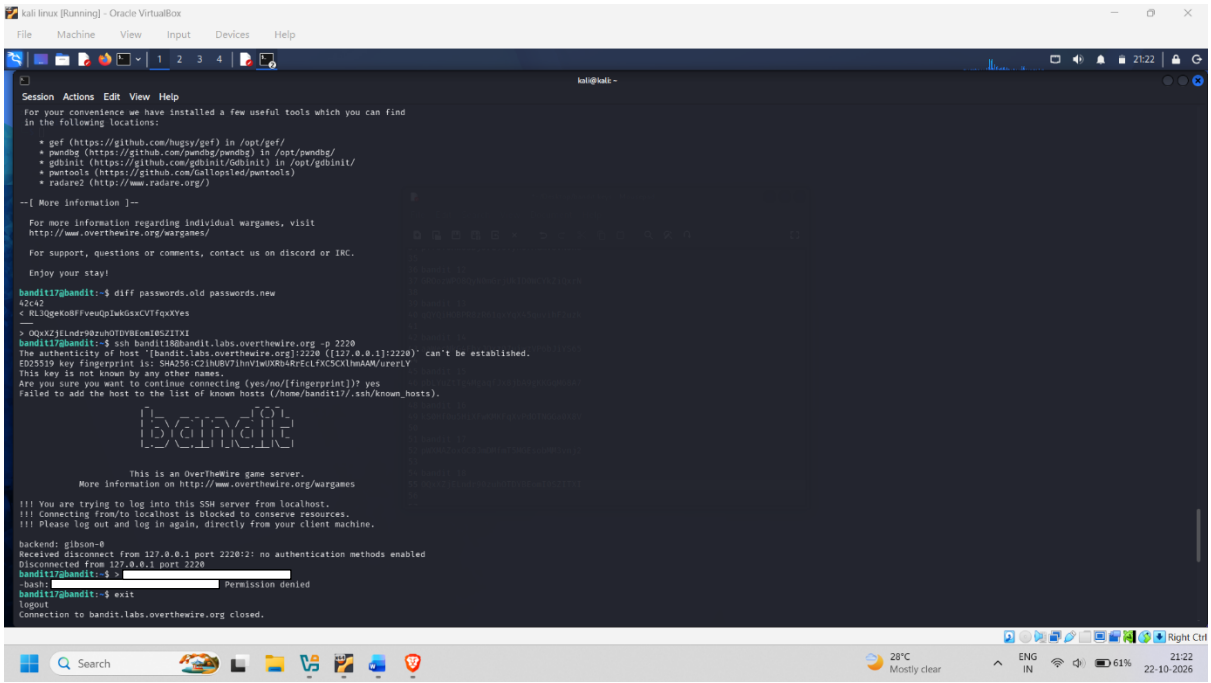
Explanation: Attempts to log in to Bandit 18 but fails because localhost SSH connections are blocked.

Step 4 - **> 0QxXZjELndr90zuhOTDYBEomI0SZITXI**

Explanation: Attempts to redirect output to a file named after the password, but fails due to insufficient write permissions.

Step 5 - **exit**

Explanation: Closes the SSH session and returns to the local terminal.

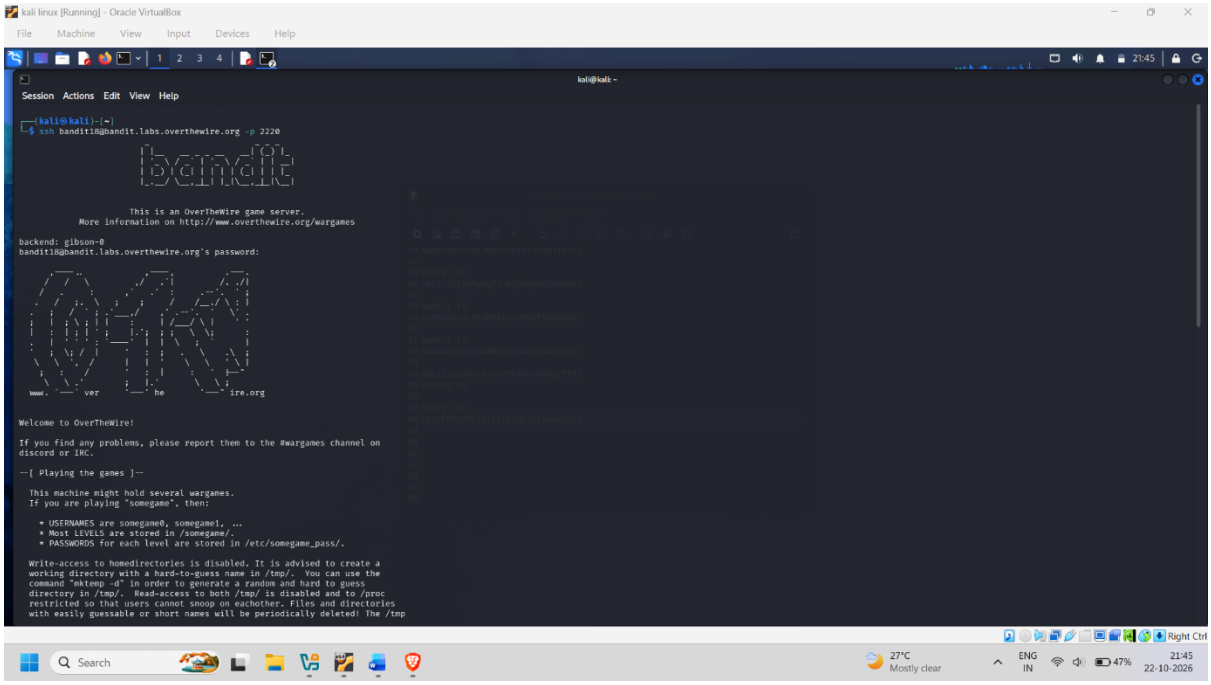

kali linux [Running] - Oracle VirtualBox
File Machine View Input Devices Help
kali@kali: ~
Session Actions Edit View Help
For your convenience we have installed a few useful tools which you can find
in the following locations:
* gef (https://github.com/hugsy/gef) in /opt/gef/
* pwndbg (https://github.com/pwndbg/pwndbg) in /opt/pwndbg/
* gdbinit (https://github.com/gdbinit/Gdbinit) in /opt/gdbinit/
* pwntools (https://github.com/Gallopsled/pwntools)
* radare2 (http://www.radare.org/)
--[More information]--
For more information regarding individual wargames, visit
http://www.overthewire.org/wargames/
For support, questions or comments, contact us on discord or IRC.
Enjoy your stay!
bandit17@bandit:~$ diff passwords.old passwords.new
42c42
< RL3QgeKoBFFveuQpIwkGsxCVTfqxXYes

> OQxXZjELndr90zuhOTDYBEomI0SZITXI
bandit17@bandit:~$ ssh bandit18@bandit.labs.overthewire.org -p 2220
The authenticity of host '[bandit.labs.overthewire.org]:2220 ([127.0.0.1]:2220)' can't be established.
ED25519 key fingerprint is: SHA256:C2ihUBV7ihnV1wUXRb4RrEcLfXC5CXlhmAAM/urerLY
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Failed to add the host to the list of known hosts (/home/bandit17/.ssh/known_hosts).
This is an OverTheWire game server.
More information on http://www.overthewire.org/wargames
!!! You are trying to log into this SSH server from localhost.
!!! Connecting from/to localhost is blocked to conserve resources.
!!! Please log out and log in again, directly from your client machine.
backend: gibson-0
Received disconnect from 127.0.0.1 port 2220:2: no authentication methods enabled
Disconnected from 127.0.0.1 port 2220
bandit17@bandit:~$ >
-bash: Permission denied
bandit17@bandit:~$ exit
logout
Connection to bandit.labs.overthewire.org closed.
Right Ctrl
Search
28°C
Mostly clear
ENG
IN
61%
21:22
22-10-2026

LEVEL 18 – LEVEL 19

Step 1: Login to Level 18

ssh bandit18@bandit.labs.overthewire.org -p 2220



Step 2 - **ssh bandit18@bandit.labs.overthewire.org -p 2220**

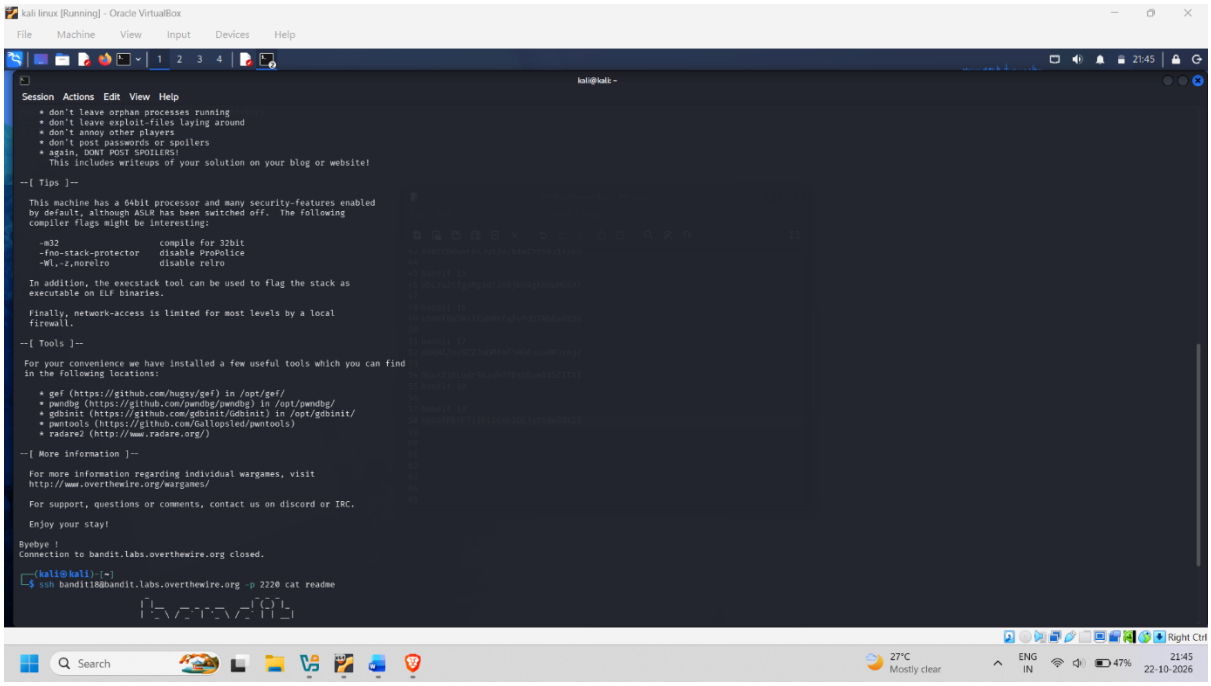
Explanation: Connects to the Bandit server as bandit18 on port 2220. The session closes immediately because the account's shell configuration terminates the session.

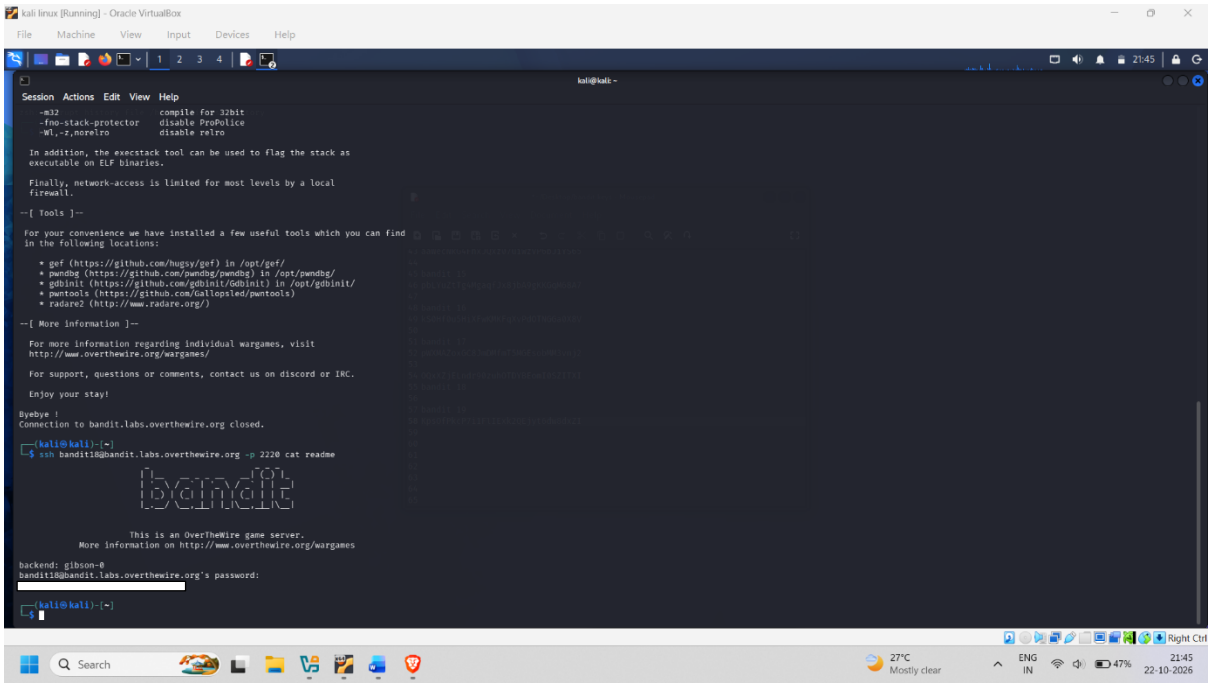
Step 3 - **ssh bandit18@bandit.labs.overthewire.org -p 2220 cat readme**

Explanation: Executes cat readme directly on the remote server to display the contents of the readme file without opening an interactive shell.

Step 5 - **exit**

Explanation: Closes the session and returns to the local terminal.

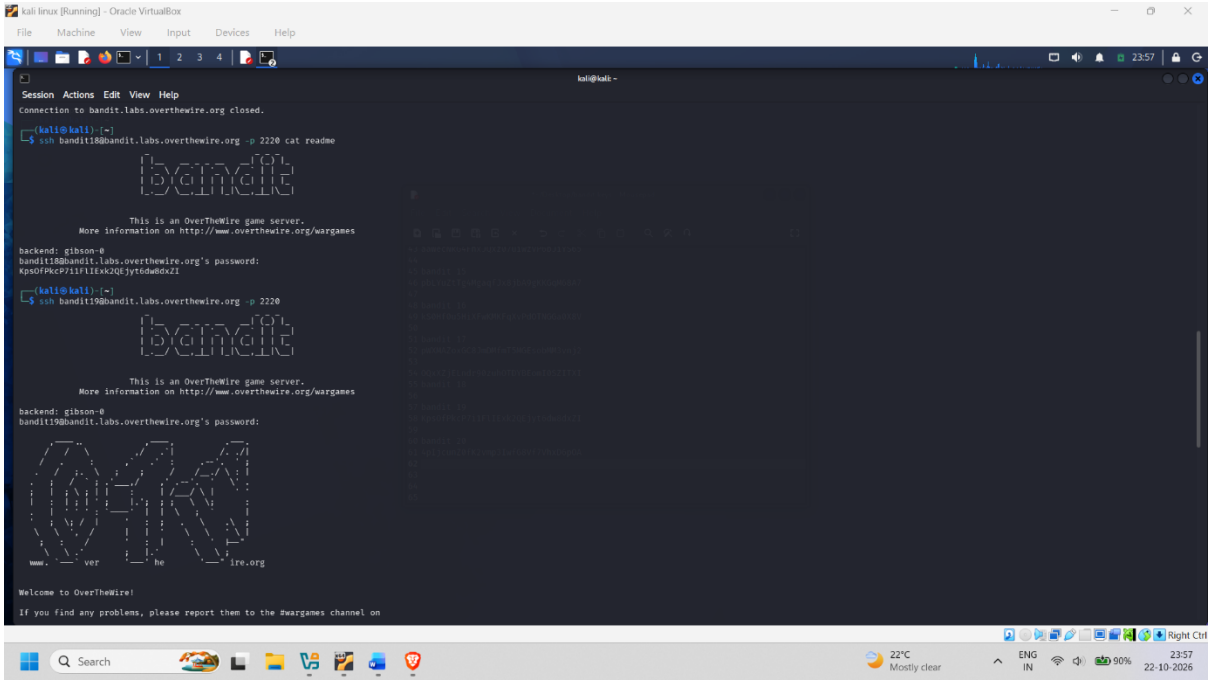

kali linux [Running] - Oracle VirtualBox
File Machine View Input Devices Help
kali@kali: ~
Session Actions Edit View Help
* don't leave orphan processes running
* don't leave exploit-files laying around
* don't annoy other players
* don't post passwords or spoilers
* again, DONT POST SPOILERS!
This includes writeups of your solution on your blog or website!
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This machine has a 64bit processor and many security-features enabled
by default, although ASLR has been switched off. The following
compiler flags might be interesting:
-m32 compile for 32bit
-fno-stack-protector disable ProPolice
-Wl,-z,norelro disable relro
In addition, the execstack tool can be used to flag the stack as
executable on ELF binaries.
Finally, network-access is limited for most levels by a local
firewall.
--[Tools]--
For your convenience we have installed a few useful tools which you can find
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* pwntools (https://github.com/Gallopsled/pwntools)
* radare2 (http://www.radare.org/)
--[More information]--
For more information regarding individual wargames, visit
http://www.overthewire.org/wargames/
For support, questions or comments, contact us on discord or IRC.
Enjoy your stay!
Byebye !
Connection to bandit.labs.overthewire.org closed.
(kali㉿kali)-[~]
$ ssh bandit18@bandit.labs.overthewire.org -p 2220 cat readme
Right Ctrl
Search
27°C
Mostly clear
ENG
IN
47%
21:45
22-10-2026


kali linux [Running] - Oracle VirtualBox
File Machine View Input Devices Help
kali@kali: ~
Session Actions Edit View Help
-m32 compile for 32bit
-fno-stack-protector disable ProPolice
-Wl,-z,norelro disable relro
In addition, the execstack tool can be used to flag the stack as
executable on ELF binaries.
Finally, network-access is limited for most levels by a local
firewall.
--[Tools]--
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* gdbinit (https://github.com/gdbinit/Gdbinit) in /opt/gdbinit/
* pwntools (https://github.com/Gallopsled/pwntools)
* radare2 (http://www.radare.org/)
--[More information]--
For more information regarding individual wargames, visit
http://www.overthewire.org/wargames/
For support, questions or comments, contact us on discord or IRC.
Enjoy your stay!
Byebye !
Connection to bandit.labs.overthewire.org closed.
(kali㉿kali)-[~]
$ ssh bandit18@bandit.labs.overthewire.org -p 2220 cat readme
This is an OverTheWire game server.
More information on http://www.overthewire.org/wargames
backend: gibson-0
bandit18@bandit.labs.overthewire.org's password:
(kali㉿kali)-[~]
$
Right Ctrl
Search
27°C
Mostly clear
ENG
IN
47%
21:45
22-10-2026

LEVEL 19 – LEVEL 20

Step 1: Login to Level 19

ssh bandit19@bandit.labs.overthewire.org -p 2220



Step 2 - **ls**

Explanation: Lists the files and directories in the current directory.

Step 3 - **./bandit20-do id**

Explanation: Runs the id command using the privileges of the bandit20 user.

Step 4 - **id**

Explanation: Displays the current user's UID, GID, and group information.

Step 5 - **./bandit20-do cat /etc/bandit_pass/bandit20**

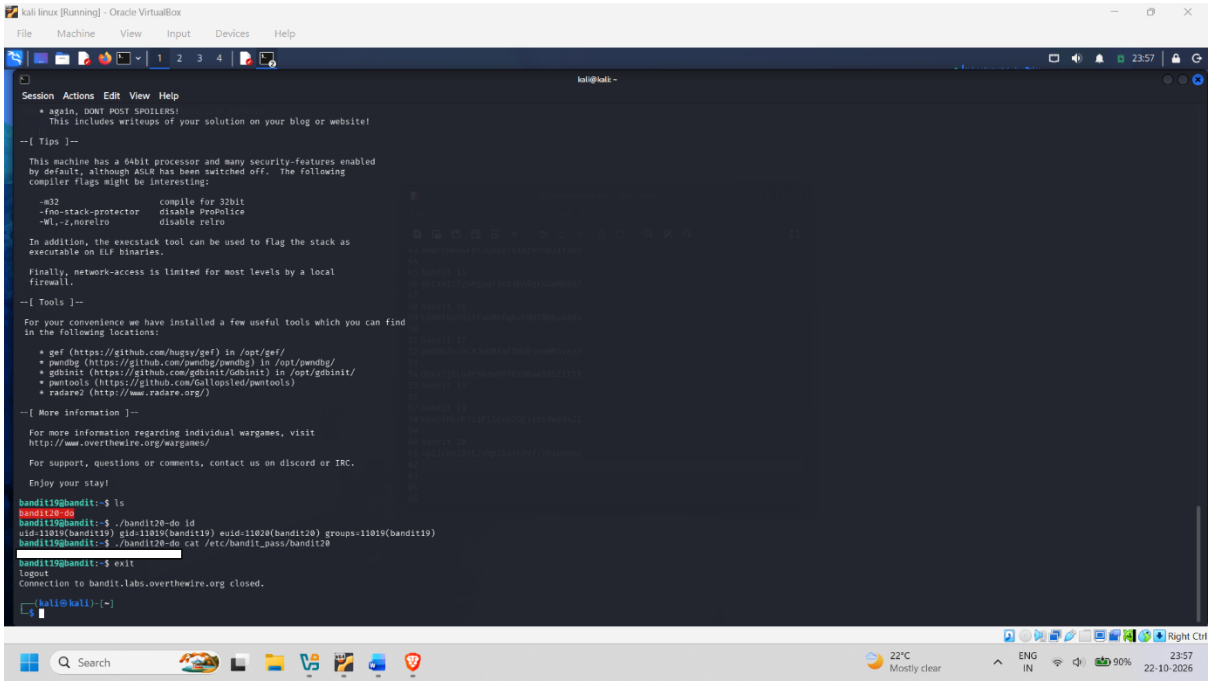
Explanation: Uses bandit20 privileges to read the Bandit Level 20 password file.

Step 6 - **cat /etc/bandit_pass/bandit20**

Explanation: Displays the contents of the Bandit 20 password file.

Step 7 - exit

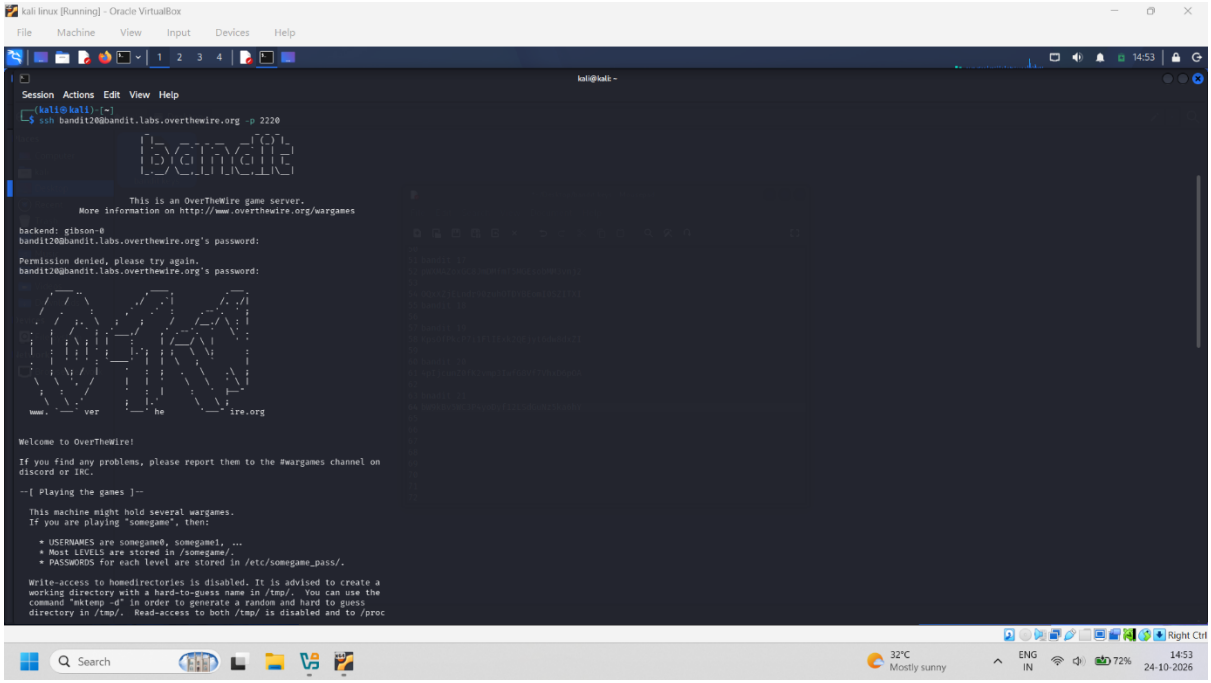
Explanation: Closes the current Bandit SSH session and returns to the Kali Linux terminal



LEVEL 20 -LEVEL 21

Step 1: Login to Level 20

ssh bandit20@bandit.labs.overthewire.org -p 2220



Step 2 - **ls -la**

Explanation - Lists all files, including hidden files, along with their permissions and ownership.

Step 3 - **cat /etc/bandit_pass/bandit20 | nc -lvnp 45454 &**

Explanation -

- cat: Reads the Bandit 20 password.

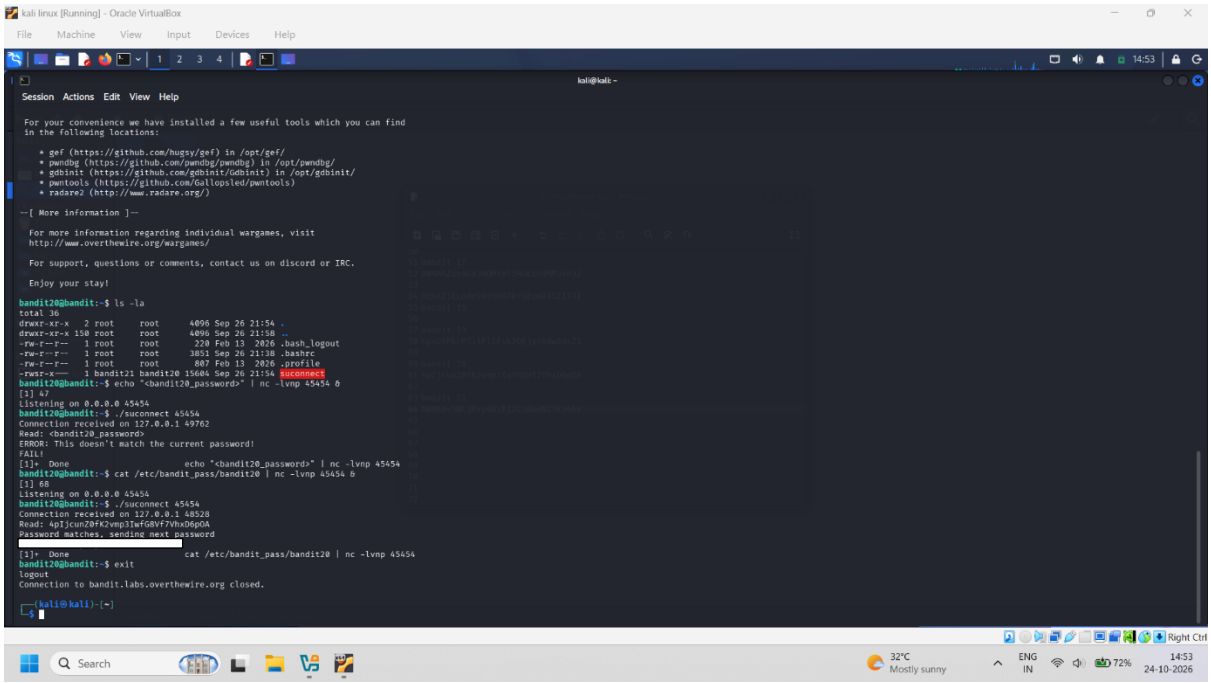
- |: Sends the password to Netcat.

- nc -lvnp 45454: Listens for a connection on port 45454.

- &: Runs the process in the background**.**

Step 4 - **./suconnect 45454**

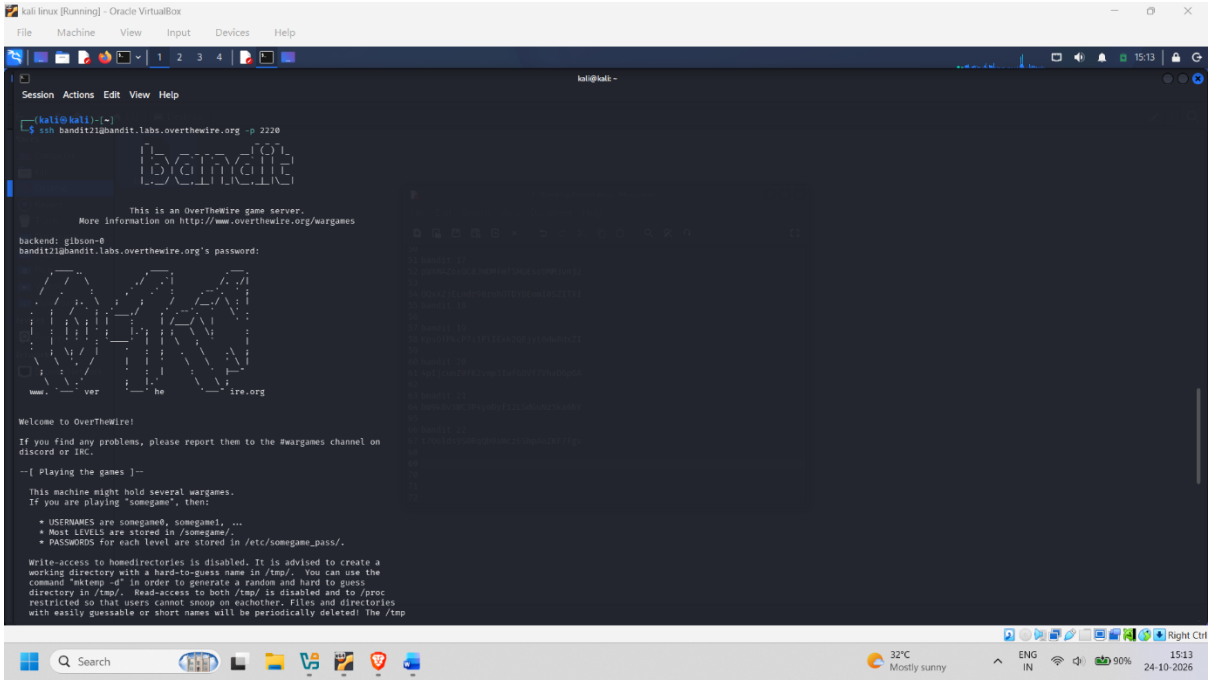
Explanation - Runs the suconnect program, which connects to port 45454 and checks the received password. If correct, it sends back the Bandit 21 password.


kali linux [Running] - Oracle VirtualBox
File Machine View Input Devices Help
kali@kali: ~
Session Actions Edit View Help
For your convenience we have installed a few useful tools which you can find
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* pwndbg (https://github.com/pwndbg/pwndbg) in /opt/pwndbg/
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* pwntools (https://github.com/Gallopsled/pwntools)
* radare2 (http://www.radare.org/)
--[More information]--
For more information regarding individual wargames, visit
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Enjoy your stay!
bandit20@bandit:~$ ls -la
total 36
drwxr-xr-x 2 root root 4096 Sep 26 21:54 .
drwxr-xr-x 150 root root 4096 Sep 26 21:58 ..
-rw-r--r-- 1 root root 220 Feb 13 2026 .bash_logout
-rw-r--r-- 1 root root 3851 Sep 26 21:38 .bashrc
-rw-r--r-- 1 root root 807 Feb 13 2026 .profile
-rwsr-x--- 1 bandit21 bandit20 15604 Sep 26 21:54 suconnect
bandit20@bandit:~$ echo "<bandit20_password>" | nc -lvnp 45454 &
[1] 47
Listening on 0.0.0.0 45454
bandit20@bandit:~$ ./suconnect 45454
Connection received on 127.0.0.1 49762
Read: <bandit20_password>
ERROR: This doesn't match the current password!
Fail!
[1]+ Done echo "<bandit20_password>" | nc -lvnp 45454
bandit20@bandit:~$ cat /etc/bandit_pass/bandit20 | nc -lvnp 45454 &
[1] 68
Listening on 0.0.0.0 45454
bandit20@bandit:~$ ./suconnect 45454
Connection received on 127.0.0.1 48528
Read: kpIJcun20fK2vnp3IwfG8VF7hhxO6p0A
Password matches, sending next password
[1]+ Done cat /etc/bandit_pass/bandit20 | nc -lvnp 45454
bandit20@bandit:~$ exit
logout
Connection to bandit.labs.overthewire.org closed.
┌──(kali㉿kali)-[~]
└─$
Right Ctrl
Search
32°C
Mostly sunny
ENG
IN
72%
14:53
24-10-2026

LEVEL 21 – LEVEL 22

Step 1: Login to Level 21

ssh bandit21@bandit.labs.overthewire.org -p 2220



Step 2 - **ls -la /etc/cron.d/**

Explanation - Lists scheduled task files to find the cron job for Bandit 22.

Step 3 - **cat /etc/cron.d/cronjob_bandit22**

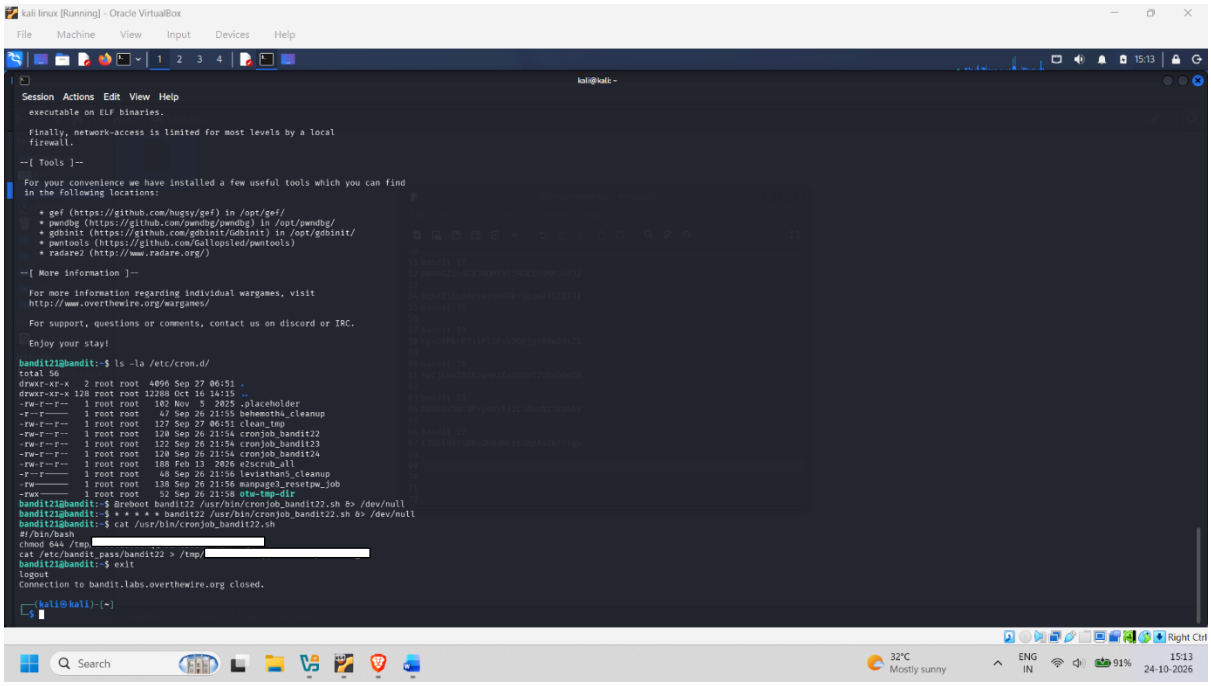
Explanation - Displays the scheduled job and shows which script runs automatically.

Step 4 - **cat /usr/bin/cronjob_bandit22.sh**

Explanation - Displays the script to understand how it copies the Bandit 22 password into a temporary file.

Step 5 - **cat /tmp/<filename>**

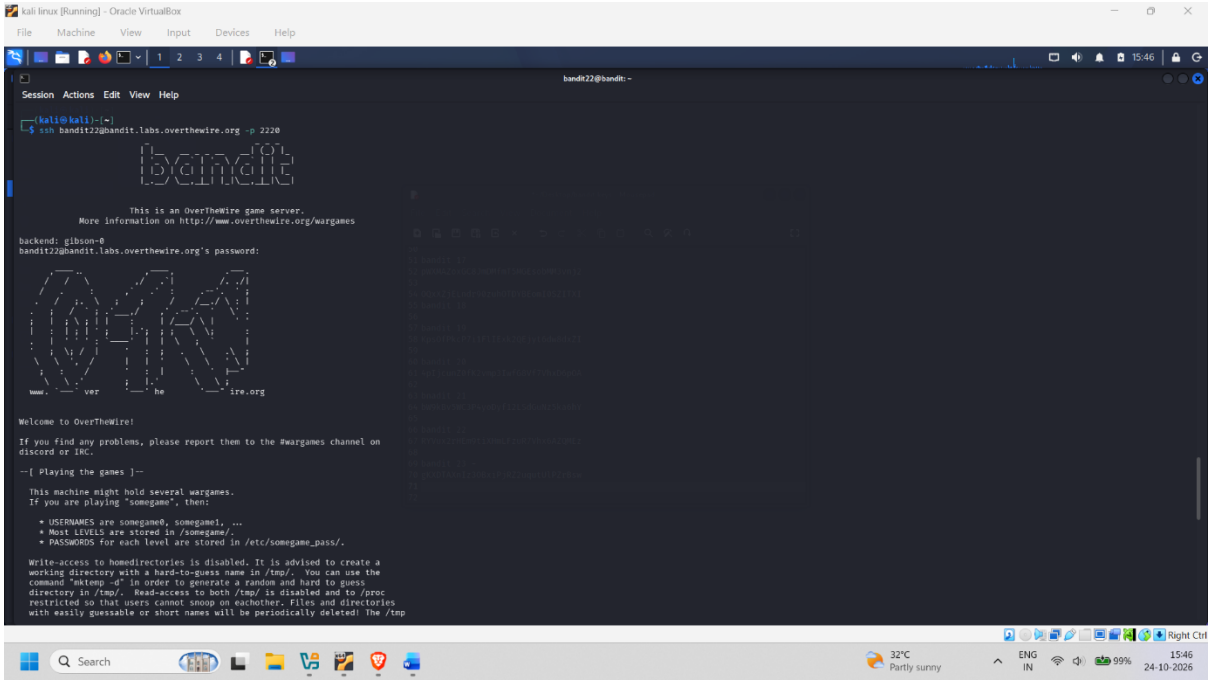
Explanation - Reads the temporary file to retrieve the Bandit 22 password. Replace <filename> with the actual filename found in the script.


kali linux [Running] - Oracle VirtualBox
File Machine View Input Devices Help
kali@kali: ~
Session Actions Edit View Help
executable on ELF binaries.
Finally, network-access is limited for most levels by a local firewall.
--[Tools]--
For your convenience we have installed a few useful tools which you can find in the following locations:
* gef (https://github.com/hugsy/gef) in /opt/gef/
* pwndbg (https://github.com/pwndbg/pwndbg) in /opt/pwndbg/
* gdbinit (https://github.com/gdbinit/Gdbinit) in /opt/gdbinit/
* pwntools (https://github.com/Gallopsled/pwntools)
* radare2 (http://www.radare.org/)
--[More information]--
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For support, questions or comments, contact us on discord or IRC.
Enjoy your stay!
bandit21@bandit:~$ ls -la /etc/cron.d/
total 56
drwxr-xr-x 2 root root 4096 Sep 27 06:51 .
drwxr-xr-x 128 root root 12288 Oct 16 14:15 ..
-rw-r--r-- 1 root root 102 Nov 5 2025 .placeholder
-r--r----- 1 root root 67 Sep 26 21:55 behemoth4_cleanup
-rw-r--r-- 1 root root 127 Sep 27 06:51 clean_tmp
-rw-r--r-- 1 root root 120 Sep 26 21:54 cronjob_bandit22
-rw-r--r-- 1 root root 122 Sep 26 21:54 cronjob_bandit23
-rw-r--r-- 1 root root 120 Sep 26 21:54 cronjob_bandit24
-rw-r--r-- 1 root root 188 Feb 13 2026 e2scrub_all
-r--r----- 1 root root 48 Sep 26 21:56 leviathan5_cleanup
-rw------- 1 root root 138 Sep 26 21:56 manpage3_resetpw_job
-rwx------ 1 root root 52 Sep 26 21:58 otw-tmp-dir
bandit21@bandit:~$ @reboot bandit22 /usr/bin/cronjob_bandit22.sh &> /dev/null
bandit21@bandit:~$ * * * * * bandit22 /usr/bin/cronjob_bandit22.sh &> /dev/null
bandit21@bandit:~$ cat /usr/bin/cronjob_bandit22.sh
#!/bin/bash
chmod 644 /tmp/
cat /etc/bandit_pass/bandit22 > /tmp/
bandit21@bandit:~$ exit
logout
Connection to bandit.labs.overthewire.org closed.
(kali㉿kali)-[~]
$
Right Ctrl
Search
32°C
Mostly sunny
ENG
IN
91%
15:13
24-10-2026

LEVEL 22 -LEVEL 23

Step 1: Login to Level 22

ssh bandit22@bandit.labs.overthewire.org -p 2220



Step 2 - **ls -la /etc/cron.d/**

Explanation: Lists scheduled task files to locate the cron job for Bandit 23.

Step 3 - **cat /etc/cron.d/cronjob_bandit23**

Explanation: Displays the scheduled task that runs the Bandit 23 script automatically.

Step 4 - **cat /usr/bin/cronjob_bandit23.sh**

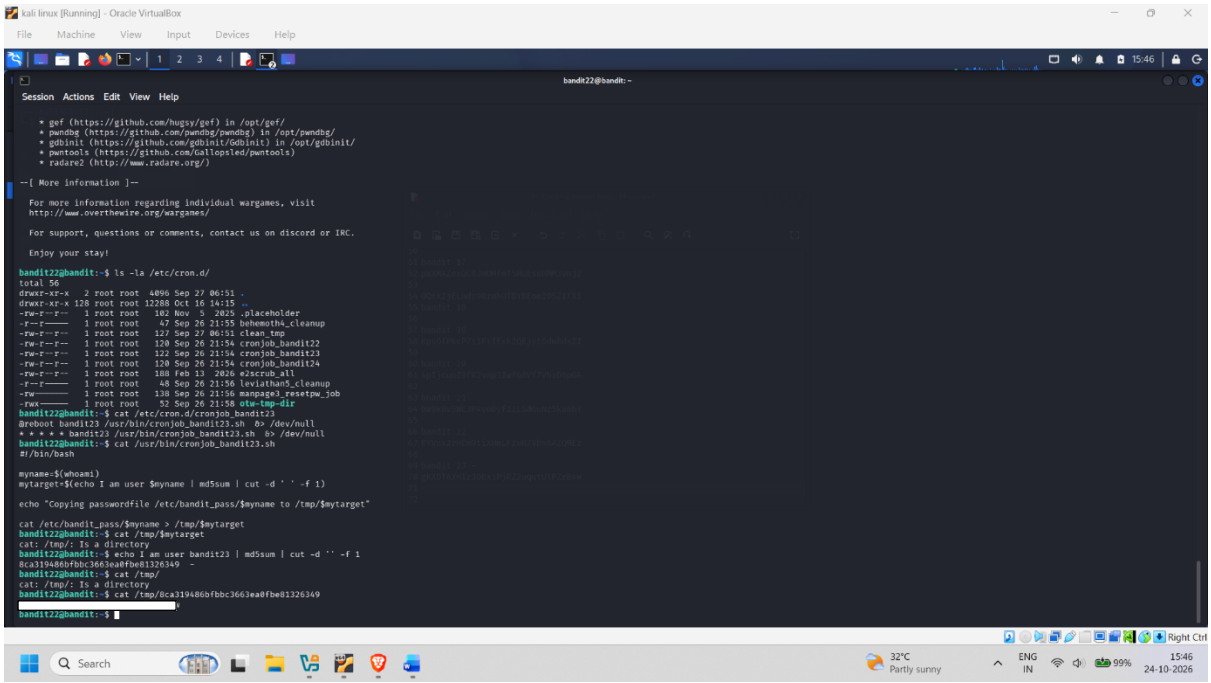
Explanation: Displays the script that generates a filename using an MD5 hash and copies the password into a temporary file.

Step 5 - **echo I am user bandit23 | md5sum | cut -d ' ' -f 1**

Explanation: Generates the MD5 hash used as the temporary filename**.**

Step 6 - **cat /tmp/<generated_hash>**

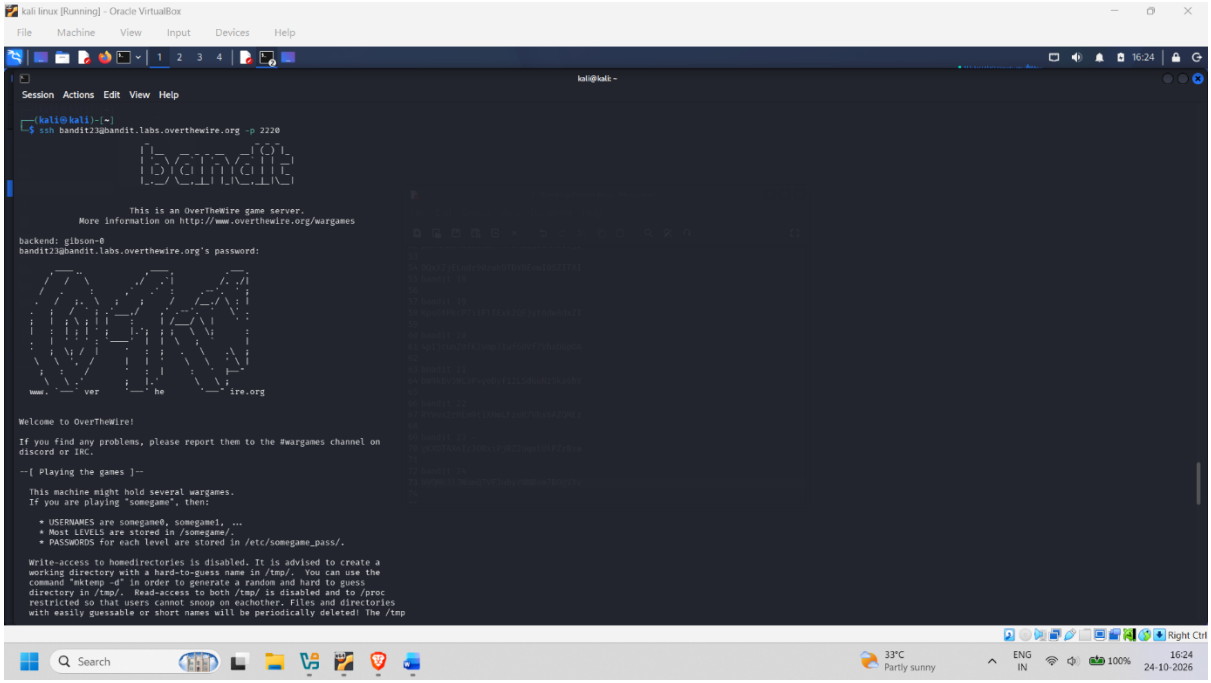
Explanation: Reads the temporary file to retrieve the Bandit 23 password. Replace "<generated_hash>" with the complete hash generated in Step 4.


kali linux [Running] - Oracle VirtualBox
File Machine View Input Devices Help
bandit22@bandit: ~
Session Actions Edit View Help
* gef (https://github.com/hugsy/gef) in /opt/gef/
* pwndbg (https://github.com/pwndbg/pwndbg) in /opt/pwndbg/
* gdbinit (https://github.com/gdbinit/Gdbinit) in /opt/gdbinit/
* pwntools (https://github.com/Gallopsled/pwntools)
* radare2 (http://www.radare.org/)
--[More information]--
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Enjoy your stay!
bandit22@bandit:~$ ls -la /etc/cron.d/
total 56
drwxr-xr-x 2 root root 4096 Sep 27 06:51 .
drwxr-xr-x 128 root root 12288 Oct 16 14:15 ..
-rw-r--r-- 1 root root 102 Nov 5 2025 .placeholder
-r--r----- 1 root root 47 Sep 26 21:55 behemoth4_cleanup
-rw-r--r-- 1 root root 127 Sep 27 06:51 clean_tmp
-rw-r--r-- 1 root root 120 Sep 26 21:54 cronjob_bandit22
-rw-r--r-- 1 root root 122 Sep 26 21:54 cronjob_bandit23
-rw-r--r-- 1 root root 120 Sep 26 21:54 cronjob_bandit24
-rw-r--r-- 1 root root 188 Feb 13 2026 e2scrub_all
-r--r----- 1 root root 48 Sep 26 21:56 leviathan5_cleanup
-rw------- 1 root root 138 Sep 26 21:56 manpage3_resetpw_job
-rwx------ 1 root root 52 Sep 26 21:58 otw-tmp-dir
bandit22@bandit:~$ cat /etc/cron.d/cronjob_bandit23
@reboot bandit23 /usr/bin/cronjob_bandit23.sh &> /dev/null
* * * * * bandit23 /usr/bin/cronjob_bandit23.sh &> /dev/null
bandit22@bandit:~$ cat /usr/bin/cronjob_bandit23.sh
#!/bin/bash
myname=$(whoami)
mytarget=$(echo I am user $myname | md5sum | cut -d ' ' -f 1)
echo "Copying passwordfile /etc/bandit_pass/$myname to /tmp/$mytarget"
cat /etc/bandit_pass/$myname > /tmp/$mytarget
bandit22@bandit:~$ cat /tmp/$mytarget
cat: /tmp/: Is a directory
bandit22@bandit:~$ echo I am user bandit23 | md5sum | cut -d ' ' -f 1
8ca319486bfbbc3663ea0fbe81326349
bandit22@bandit:~$ cat /tmp/
cat: /tmp/: Is a directory
bandit22@bandit:~$ cat /tmp/8ca319486bfbbc3663ea0fbe81326349
bandit22@bandit:~$
Right Ctrl
Search
32°C
Partly sunny
ENG
IN
99%
15:46
24-10-2026

LEVEL 23 – LEVEL 24

Step 1: Login to Level 23

ssh bandit23@bandit.labs.overthewire.org -p 2220



Step 2 - **ls -la /etc/cron.d/**

Explanation: Lists scheduled tasks to identify the cron job for Bandit 24.

Step 3 - **cat /etc/cron.d/cronjob_bandit24**

Explanation: Displays the cron configuration and shows which script runs automatically.

Step 4 - **cat /usr/bin/cronjob_bandit24.sh**

Explanation: Displays the script that executes eligible files in the Bandit 24 directory and deletes them afterward.

Step 5 - **ls -la /var/spool/bandit24/**

Explanation: Checks the directory's contents, ownership, and permissions.

Step 6 - **ls -la /var/spool/bandit24/foo/**

Explanation: Attempts to list the directory contents but returns a permission error because directory access is restricted.

Step 7 - **mkdir -p /tmp/my_pass24**

Explanation: Creates a temporary directory to store the password output.

Step 8 - **chmod 777 /tmp/my_pass24**

Explanation: Gives all users permission to read, write, and access the directory.

Step 9 - **cat << 'EOF' > /tmp/getpass.sh ... EOF**

Explanation: Creates a shell script that copies the Bandit 24 password into a temporary file.

Step 10 - **chmod +x /tmp/getpass.sh**

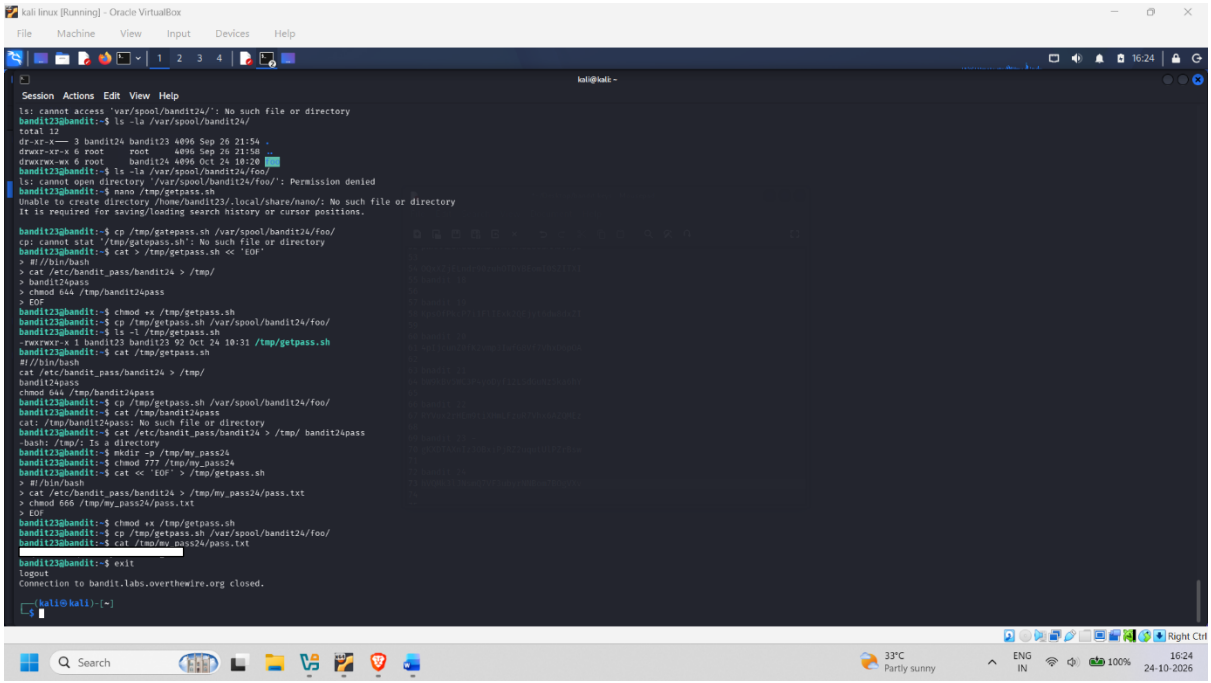
Explanation: Makes the script executable.

Step 11 - **cp /tmp/getpass.sh /var/spool/bandit24/foo/**

Explanation: Copies the script into the directory monitored by the cron job.

Step 12 - **cat /tmp/my_pass24/pass.txt**

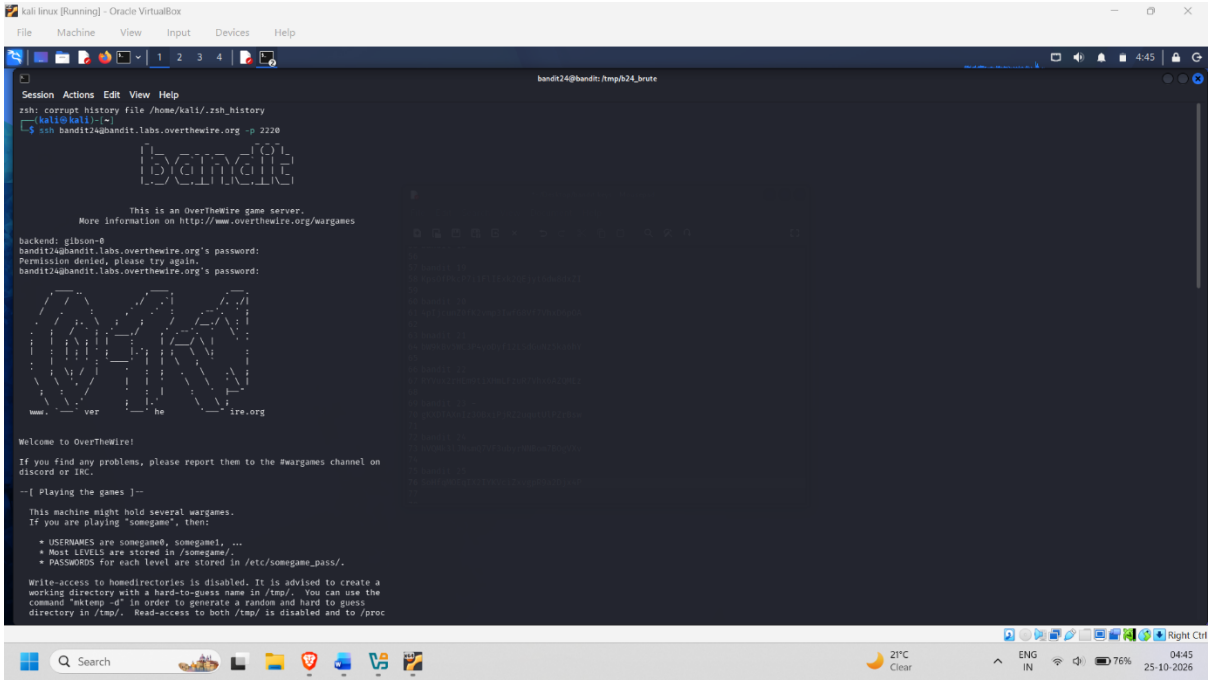
Explanation: Displays the password written by the executed script, provided the cron job ran successfully.



LEVEL 24 – LEVEL 25

Step 1: Login to Level 24

ssh bandit24@bandit.labs.overthewire.org -p 2220



Step 2 - **mkdir -p /tmp/b24_brute`**

Explanation - Creates a temporary working directory to store the generated PIN combinations.

Step 3 - **cd /tmp/b24_brute`**

Explanation - Moves into the working directory.

Step 4 - **PASS=$(cat /etc/bandit_pass/bandit24)**

Explanation - Reads the current bandit24 password and stores it in the `PASS` variable.

Step 5 - **for pin in {0000..9999}; do**

Explanation -Loops through all 10,000 four-digit PIN combinations, from `0000` to `9999`.

Step 6 - **echo "$PASS $pin"**

Explanation - Prints the current password followed by a space and the PIN.

Step 7 - **done > pins.txt**

Explanation - Saves all generated password-and-PIN combinations into `pins.txt`.

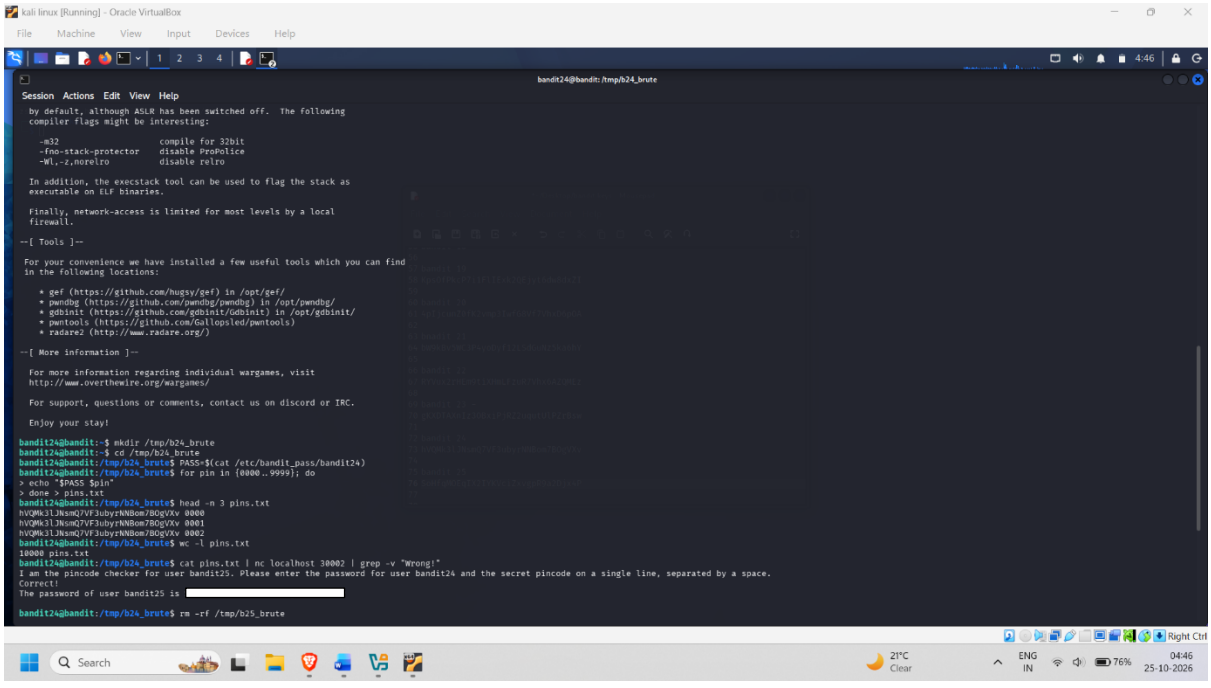
Step 8 - **wc -l pins.txt**

Explanation - Counts the lines in the file to verify that 10,000 combinations were generated.

Step 9 - **cat pins.txt \| nc localhost 30002 \| grep -v "Wrong!"** Explanation - Sends the combinations to the service on port 30002 using Netcat and filters out responses containing Wrong! to help identify the successful response.

Step 10 - **exit**

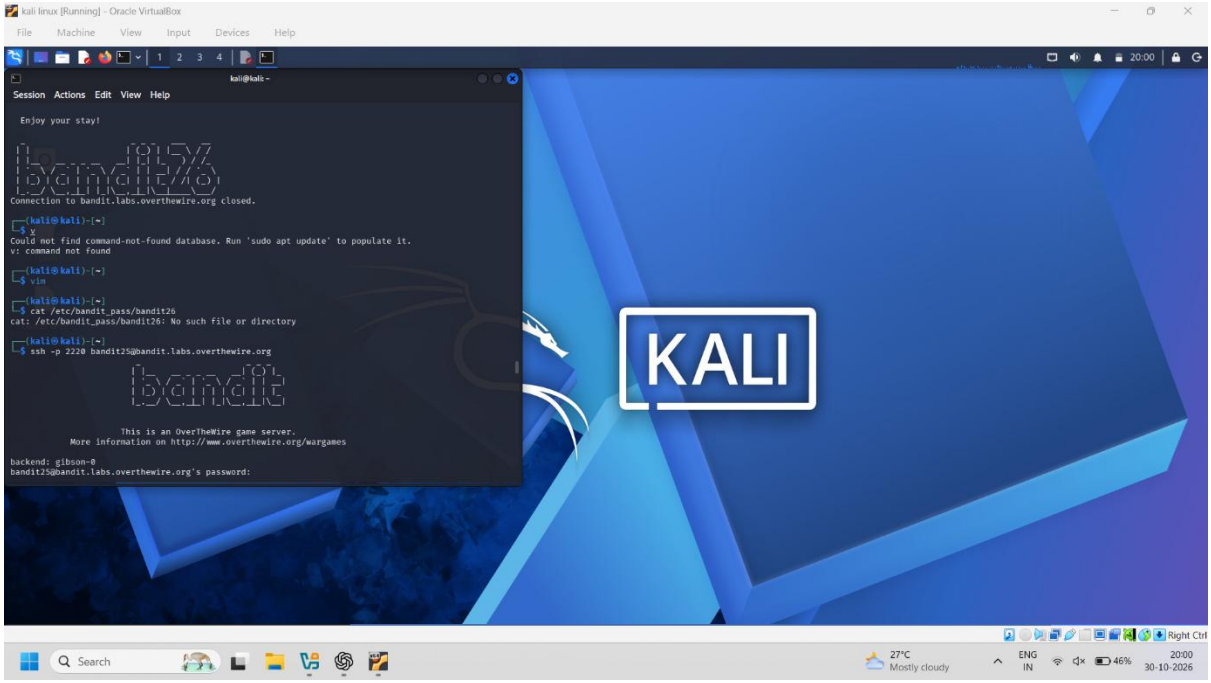
Explanation - Exits the current SSH session.



LEVEL 25 – LEVEL 26

Step 1: Login to Level 25

ssh bandit25@bandit.labs.overthewire.org -p 2220



Step 2 - **ssh -p 2220 bandit25@bandit.labs.overthewire.org**

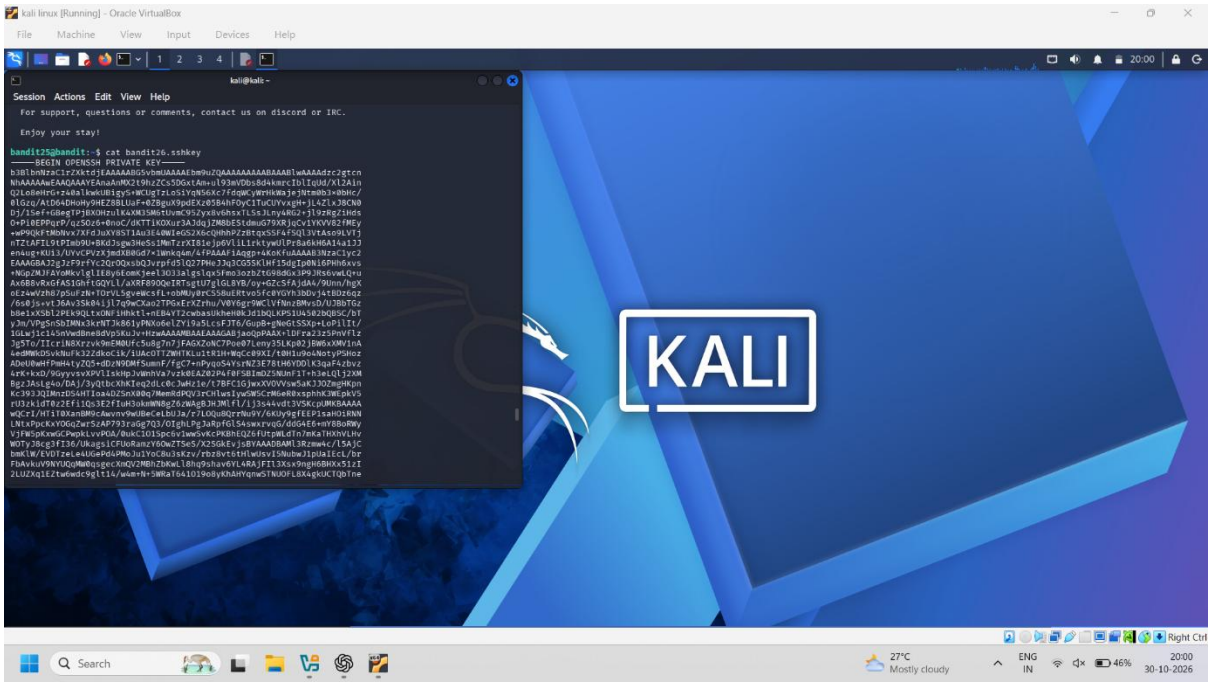
Explanation: Connects to the Bandit25 account.

Step 3 - **ls**

Explanation: Lists files in the home directory.

Step 4 - **cat bandit26.sshkey**

Explanation: Displays the SSH private key.



Step 5 - **exit**

Explanation: Exits the Bandit25 session.

Step 6 - **nano ~/bandit26.sshkey**

Explanation: Creates a file to save the copied SSH private key.

Step 7 - **chmod 600 ~/bandit26.sshkey**

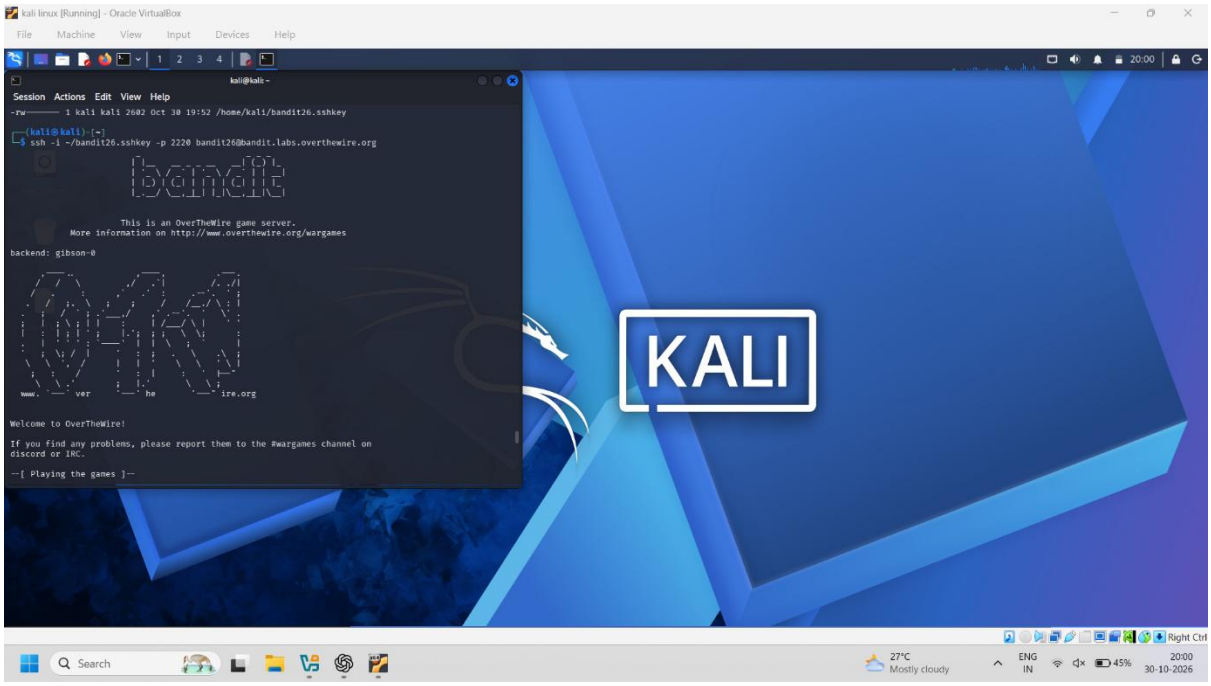
Explanation: Sets secure permissions for the private key.

Step 8 - **ls -l ~/bandit26.sshkey**

Explanation: Checks the private key's permissions.

Step 9 - **ssh -i ~/bandit26.sshkey -p 2220 bandit26@bandit.labs.overthewire.org**

Explanation: Connects to Bandit26 using the SSH private key.



Step 10 - Key press:

v

Explanation: Opens Vim from the `more` pager.

Step 11 - **Vim command:**

vim

:set shell=/bin/bash

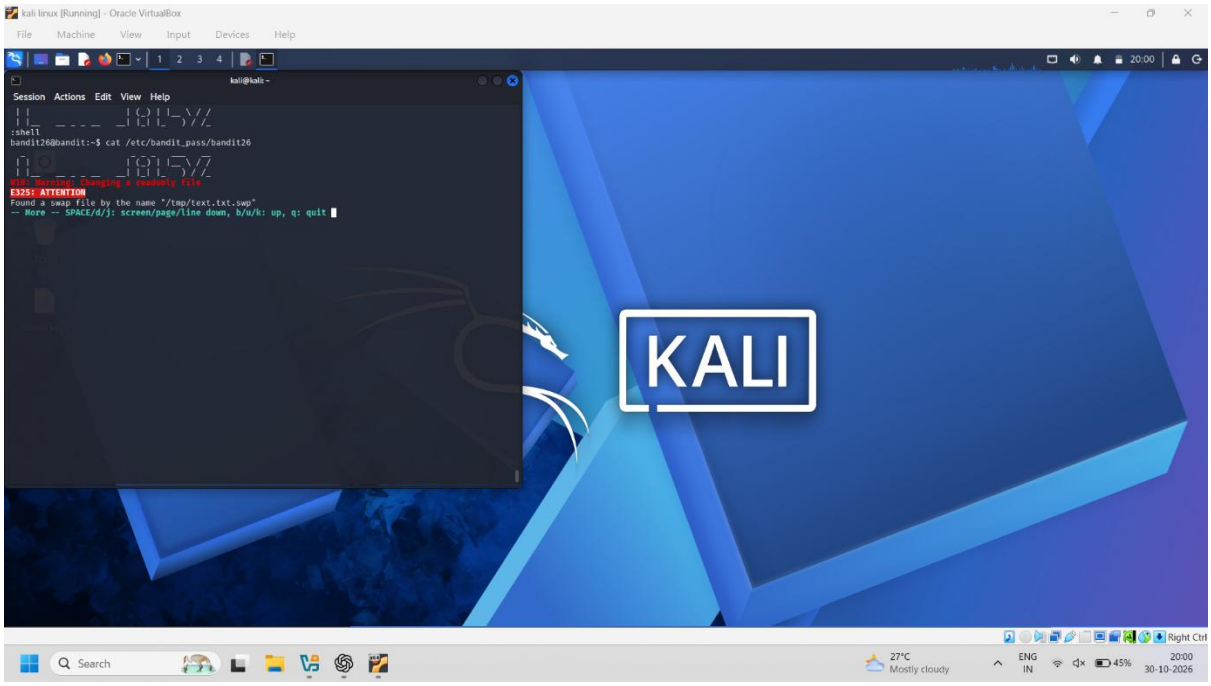
Explanation: Sets Bash as the shell in Vim.

Step 12 - Vim command:

vim

:shell

Explanation: Opens the Bash shell from Vim.



Step 13 - **cat /etc/bandit_pass/bandit26**

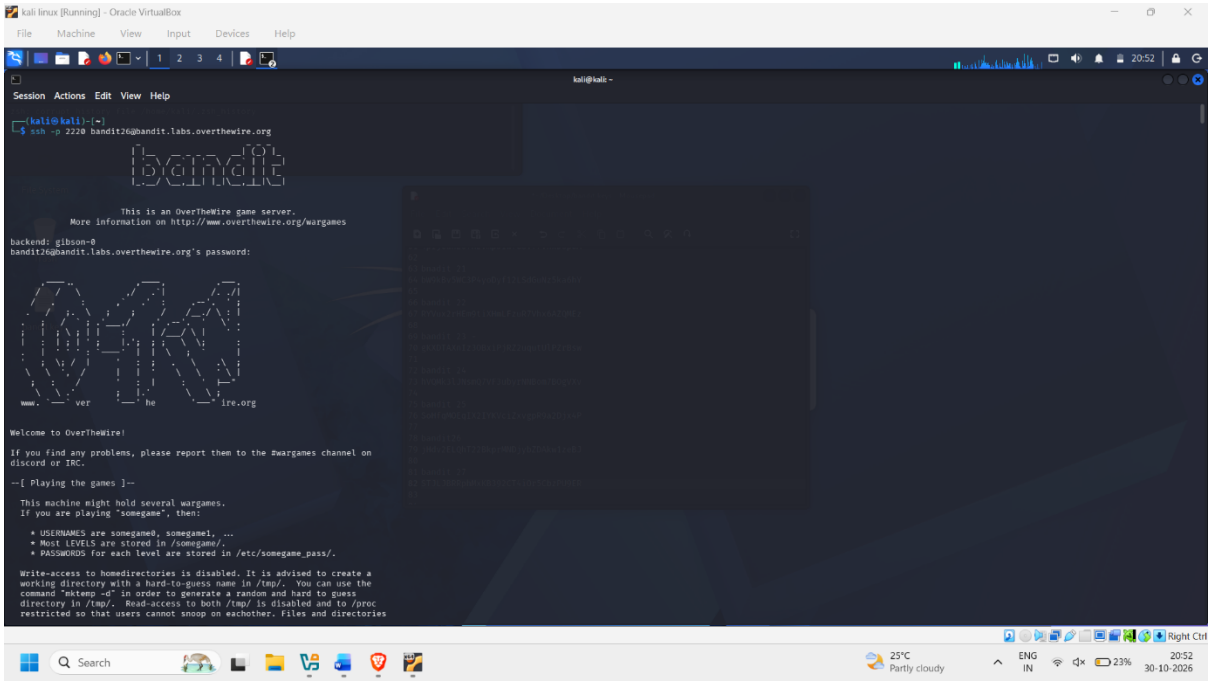
Explanation: Displays the Bandit26 password.



Level 26 – LEVEL 27

Step 1: Login to Level 26

ssh bandit26@bandit.labs.overthewire.org -p 2220



Step 2 - **ssh -p 2220 bandit26@bandit.labs.overthewire.org**

Explanation: Connects to the Bandit26 account.

Step 3 - **Key press:**

v

Explanation: Opens Vim from the `more` pager.

Step 4 - **Vim command:**

vim

:set shell=/bin/bash

Explanation: Sets Bash as the shell in Vim.

Step 5 - **Vim command:**

Vim

:shell

Explanation: Opens the Bash shell from Vim.

Step 6 - **Command:**

ls -la

Explanation: Lists all files in the home directory.

Step 7 - **Command:**

./bandit27-do

Explanation: Runs the special executable provided for this level.

Step 8 - **Command:**

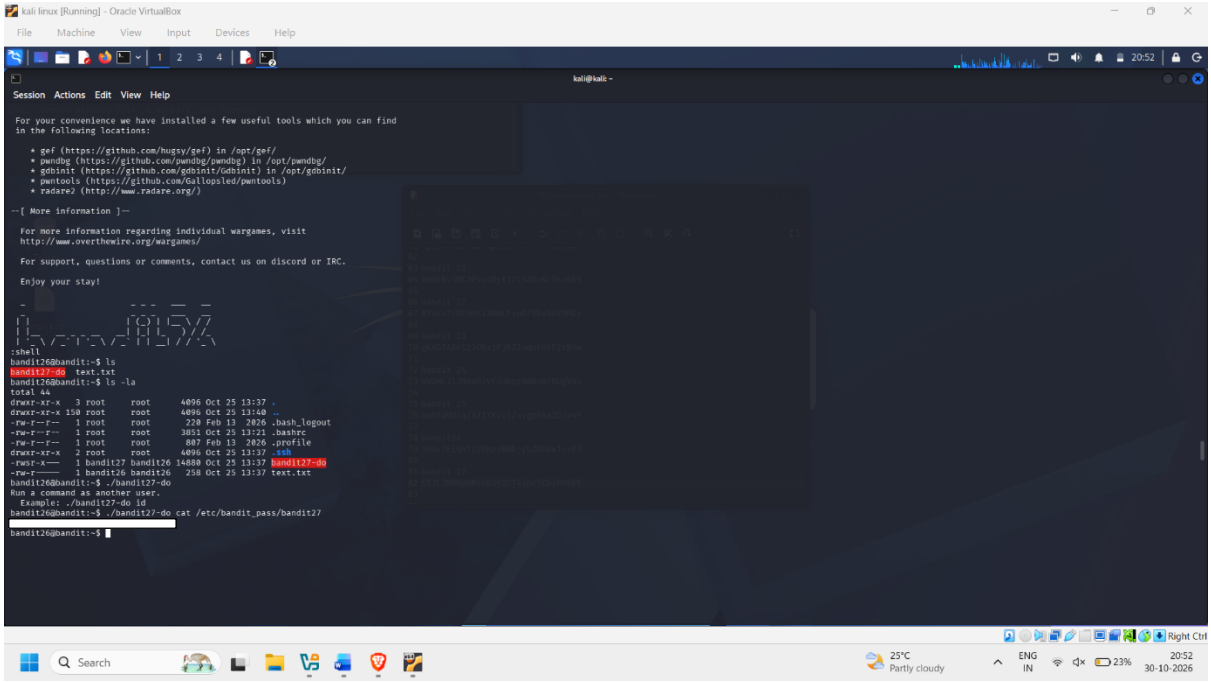
./bandit27-do cat /etc/bandit_pass/bandit27

Explanation: Displays the password for Bandit27 using the special executable.

Step 9 - **Command:**

exit

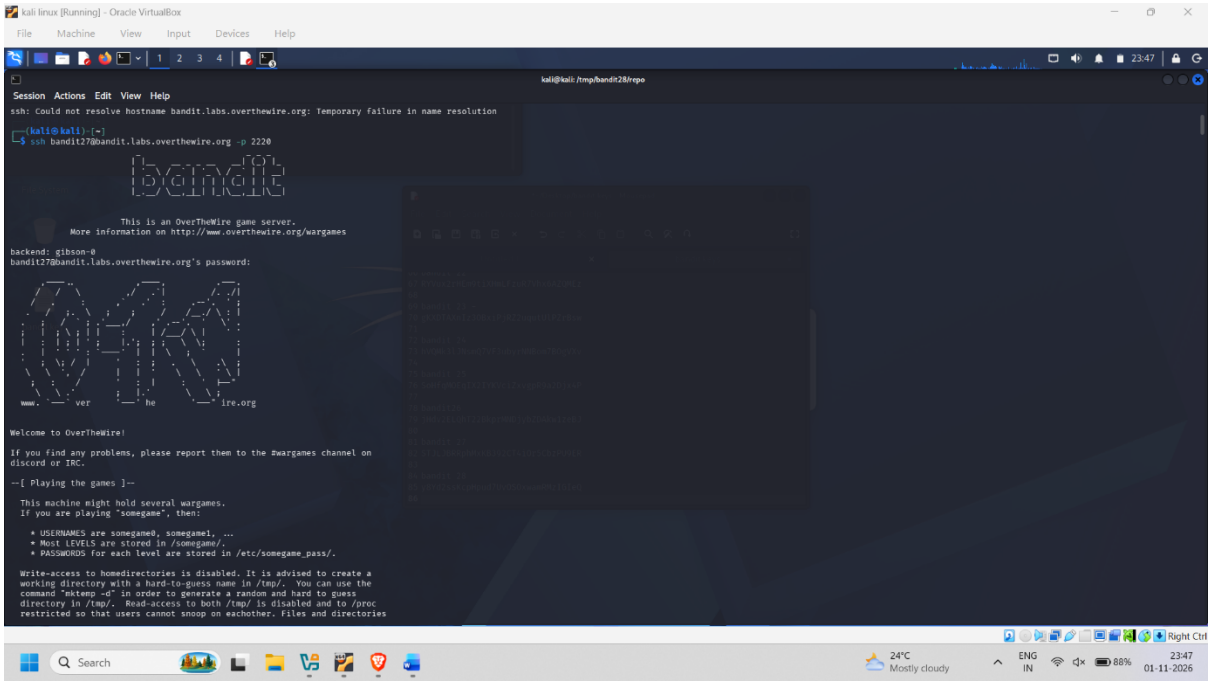
Explanation: Closes the current shell.



Level 27 – LEVEL 28

Step 1: Login to Level 27

ssh bandit27@bandit.labs.overthewire.org -p 2220



Step 2 - **ssh bandit27@bandit.labs.overthewire.org -p 2220**

Explanation - Connects to Bandit Level 27 using SSH.

Step 3 - **exit**

Explanation - Exits the SSH session.

Step 4 - **mkdir -p /tmp/bandit28**

Explanation - Creates a working directory.

Step 5 - **cd /tmp/bandit28`**

Explanation - Moves into the working directory.

Step 6 - **git clone ssh://bandit27-git@bandit.labs.overthewire.org:2220/home/bandit27-git/repo**

Explanation - Clones the Git repository using the Level 27 password.

Step 7 - **cd repo**

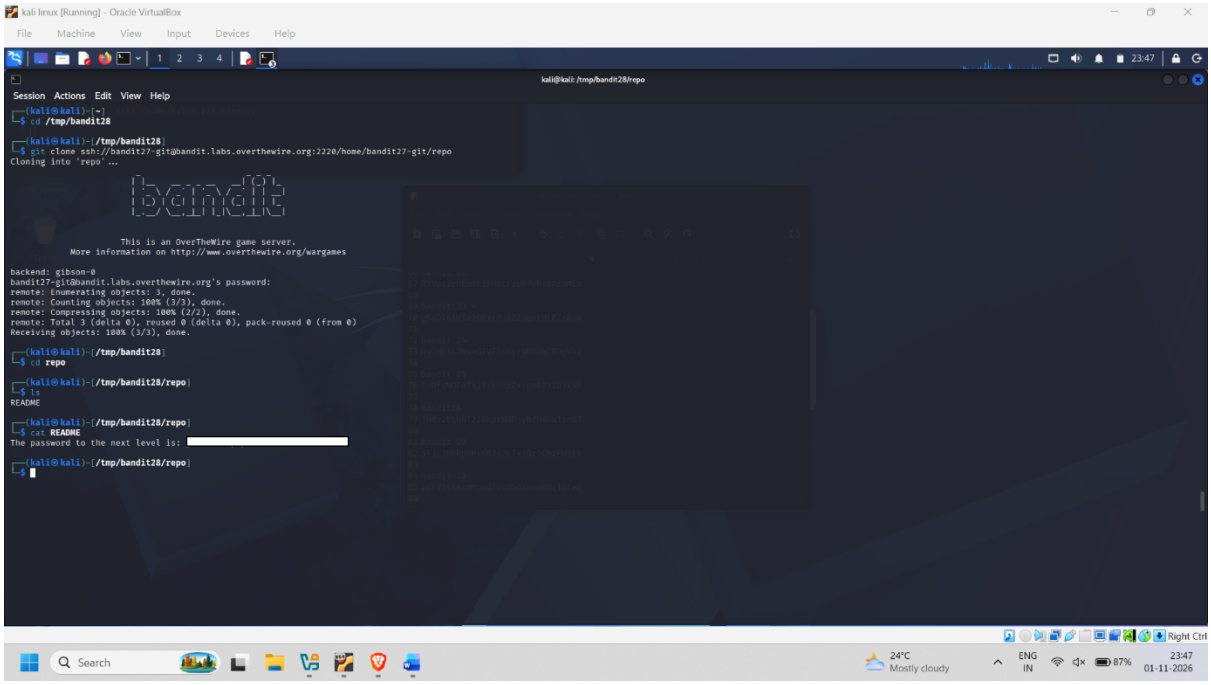
Explanation - Enters the cloned repository.

Step 8 - **ls**

Explanation - Lists the repository files.

Step 9 - **cat README**

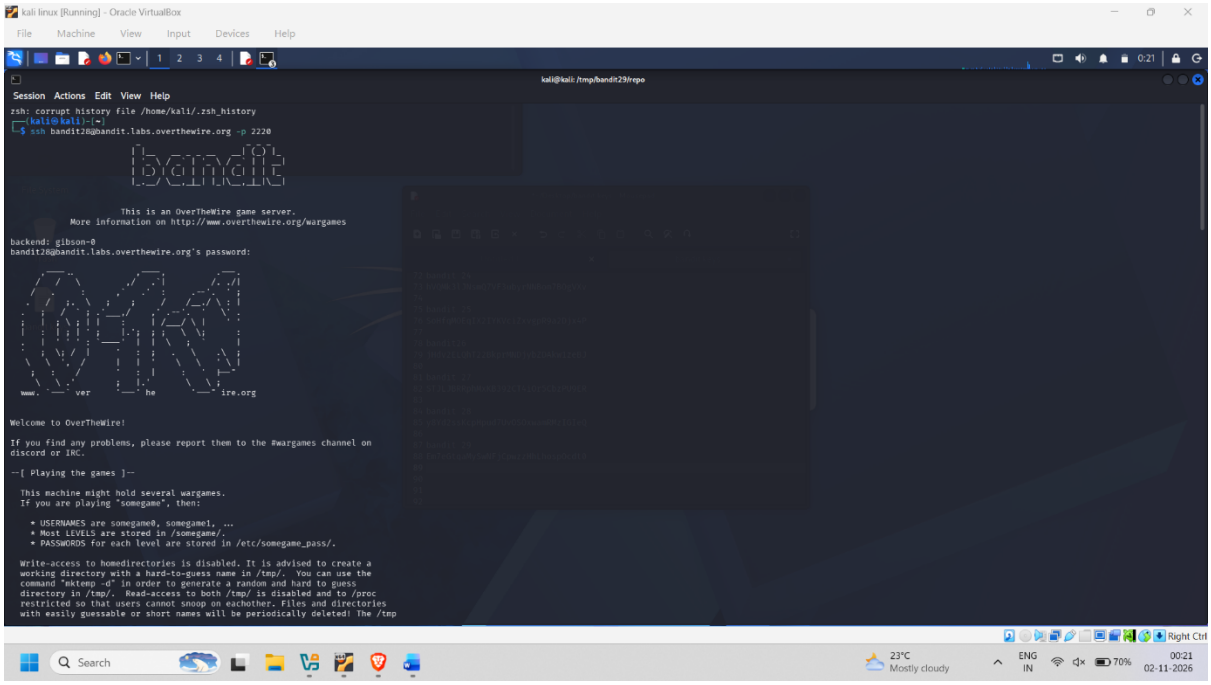
Explanation - Displays the password for Level 28.



LEVEL 28 – LEVEL 29

Step 1: Login to Level 28

ssh bandit28@bandit.labs.overthewire.org -p 2220



Step 2 - **exit**

Explanation - Exits the Level 28 SSH session.

Step 3 - **mkdir -p /tmp/bandit29**

Explanation - Creates a working directory.

Step 4 - **cd /tmp/bandit29**

Explanation - Moves into the working directory.

Step 5 - **git clone ssh://bandit28-git@bandit.labs.overthewire.org:2220/home/bandit28-git/repo**

Explanation - Clones the repository using the Level 28 password.

Step 6 - **cd repo**

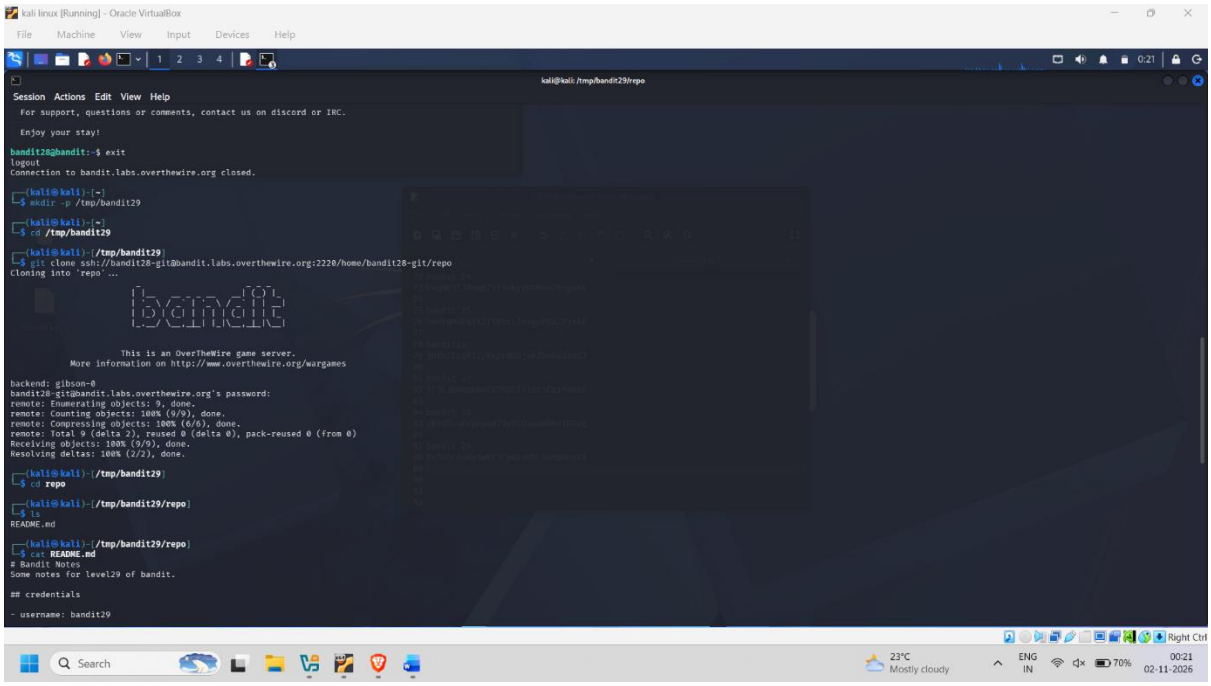
Explanation - Enters the cloned repository.

Step 7 - ls

Explanation - Lists the repository files.

Step 8 - **cat README.md**

Explanation - Displays the README file.

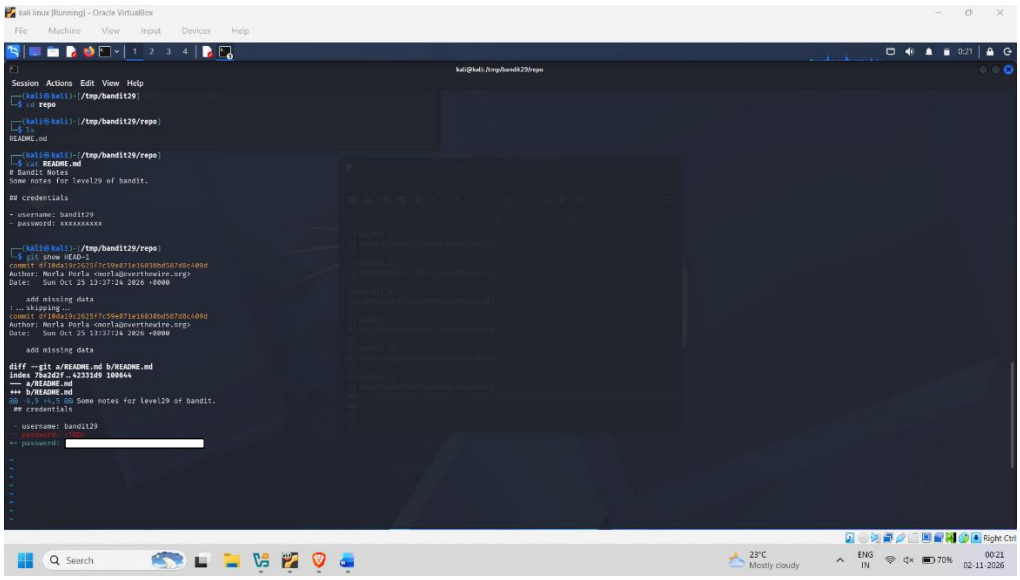


Step 9 - **git log**

Explanation - Shows the repository's commit history.

Step 10 - **git show HEAD~1**

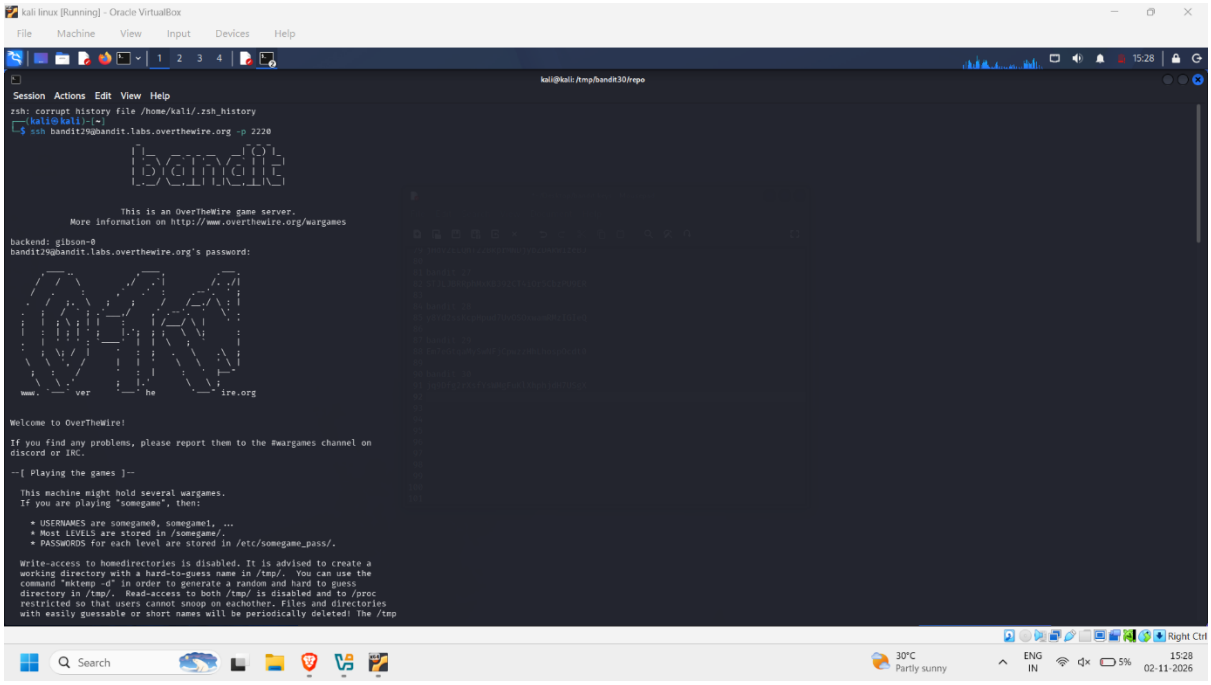
Explanation - Displays the previous commit, which contains the password for Level 29.



LEVEL 29 – LEVEL 30

Step 1: Login to Level 29

ssh bandit29@bandit.labs.overthewire.org -p 2220



Step 2 - exit

Explanation - Exits the current SSH session.

Step 3 - **mkdir -p /tmp/bandit30**

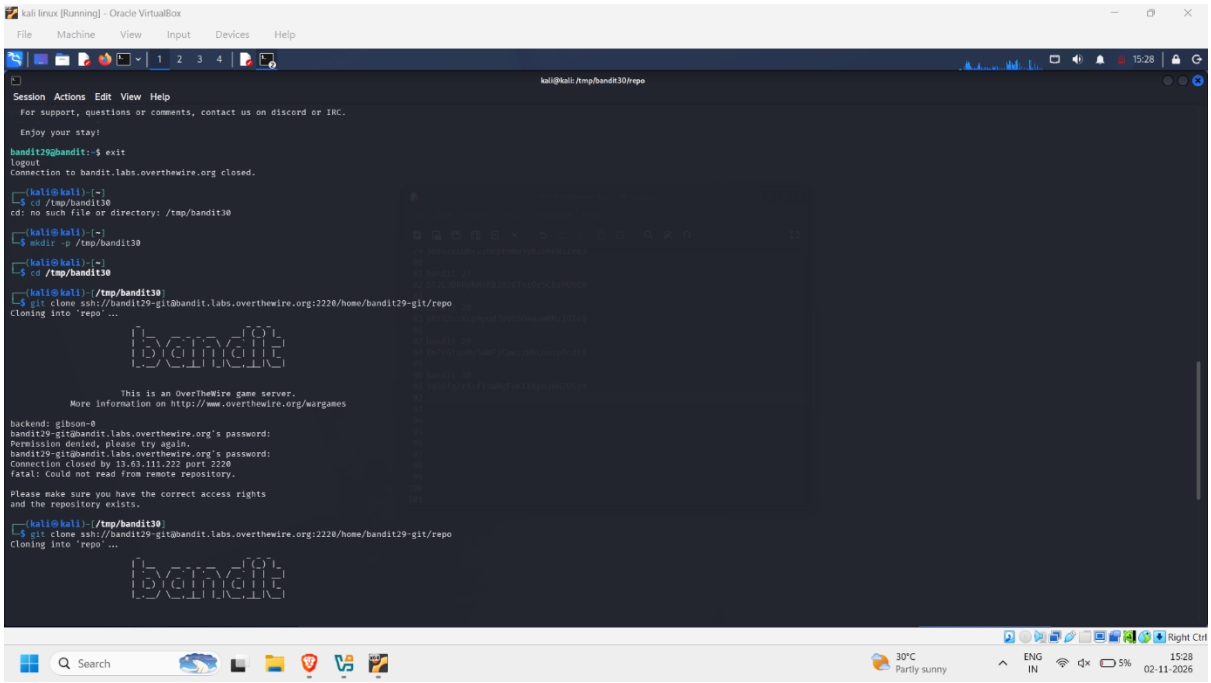
Explanation - Creates a working directory.

Step 4 - **cd /tmp/bandit30`**

Explanation - Opens the working directory**.**

Step 5 - **git clone ssh://bandit29git@bandit.labs.overthewire.org:2220/home/bandit29-git/repo**

Explanation - Clones the Git repository using the Level 29 password.



Step 6 - **cd repo**

Explanation - Enters the cloned repository.

Step 7 - **git branch -a**

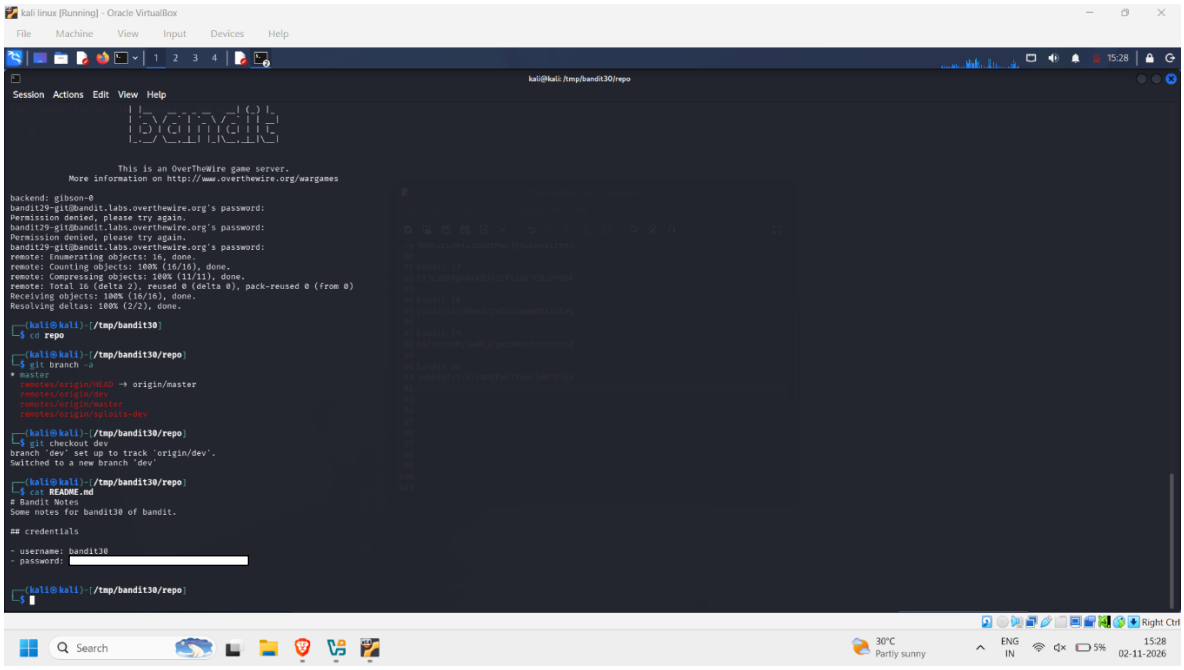
Explanation - Lists all available Git branches.

Step 8 - **git checkout dev**

Explanation - Switches to the development branch.

Step 9 - **cat README.md**

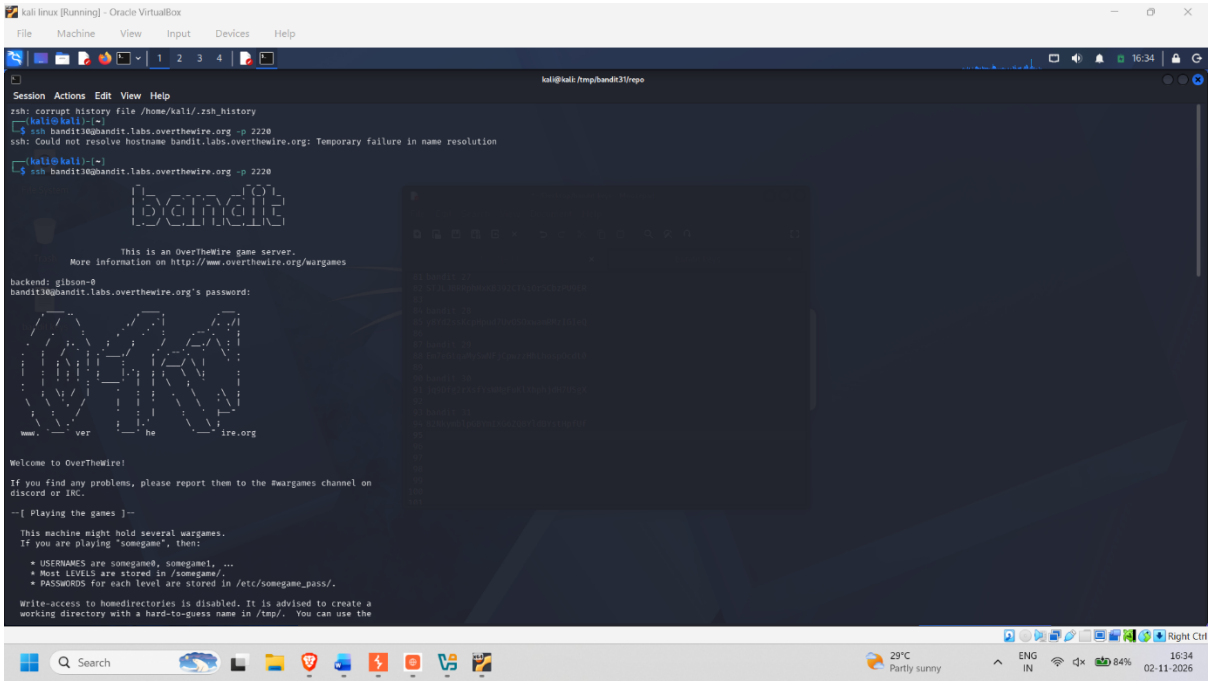
Explanation - Displays the password for Level 30


kali linux [Running] - Oracle VirtualBox
File Machine View Input Devices Help
kali@kali: /tmp/bandit30/repo
Session Actions Edit View Help
This is an OverTheWire game server.
More information on http://www.overthewire.org/wargames
backend: gibson-0
bandit29-git@bandit.labs.overthewire.org's password:
Permission denied, please try again.
bandit29-git@bandit.labs.overthewire.org's password:
Permission denied, please try again.
bandit29-git@bandit.labs.overthewire.org's password:
remote: Enumerating objects: 16, done.
remote: Counting objects: 100% (16/16), done.
remote: Compressing objects: 100% (11/11), done.
remote: Total 16 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (16/16), done.
Resolving deltas: 100% (2/2), done.
(kali㉿kali)-[/tmp/bandit30]
$ cd repo
(kali㉿kali)-[/tmp/bandit30/repo]
$ git branch -a
* master
remotes/origin/HEAD -> origin/master
remotes/origin/dev
remotes/origin/master
remotes/origin/sploits-dev
(kali㉿kali)-[/tmp/bandit30/repo]
$ git checkout dev
branch 'dev' set up to track 'origin/dev'.
Switched to a new branch 'dev'
(kali㉿kali)-[/tmp/bandit30/repo]
$ cat README.md
Bandit Notes
Some notes for bandit30 of bandit.
credentials
- username: bandit30
- password:
(kali㉿kali)-[/tmp/bandit30/repo]
$
Right Ctrl
Search
30°C
Partly sunny
ENG
IN
5%
15:28
02-11-2026

LEVEL 30 – LEVEL 31

Step 1: Login to Level 30

ssh bandit30@bandit.labs.overthewire.org -p 2220



Step 2 - **exit**

Explanation - Exits the current SSH session.

Step 3 - **mkdir -p /tmp/bandit31**

Explanation - Creates a working directory.

Step 4 - **cd /tmp/bandit31**

Explanation - Opens the working directory.

Step 5 - **git clone ssh://bandit30git@bandit.labs.overthewire.org:2220/home/bandit30-git/repo**

Explanation - Clones the Git repository using the Level 30 password.

Step 6 - **cd repo**

Explanation - Enters the cloned repository.

Step 7 - **ls -la**

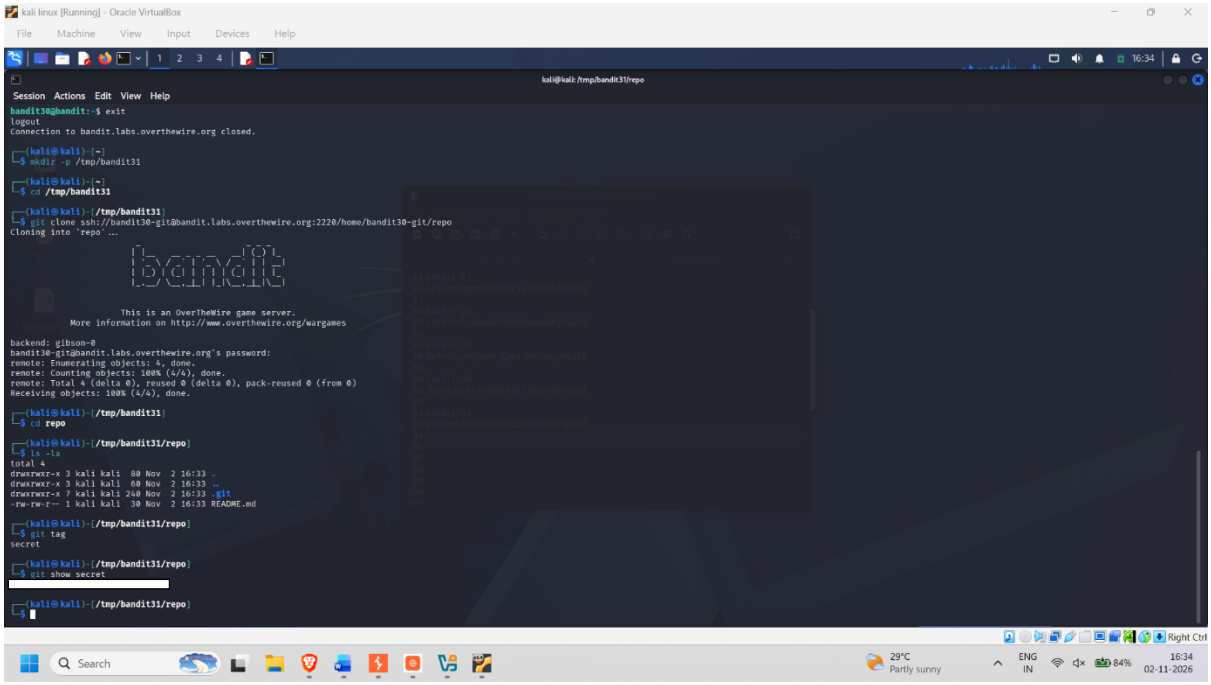
Explanation - Lists all files, including hidden files.

Step 8 - **git tag**

Explanation - Lists the available Git tags.

Step 9 - **git show secret**

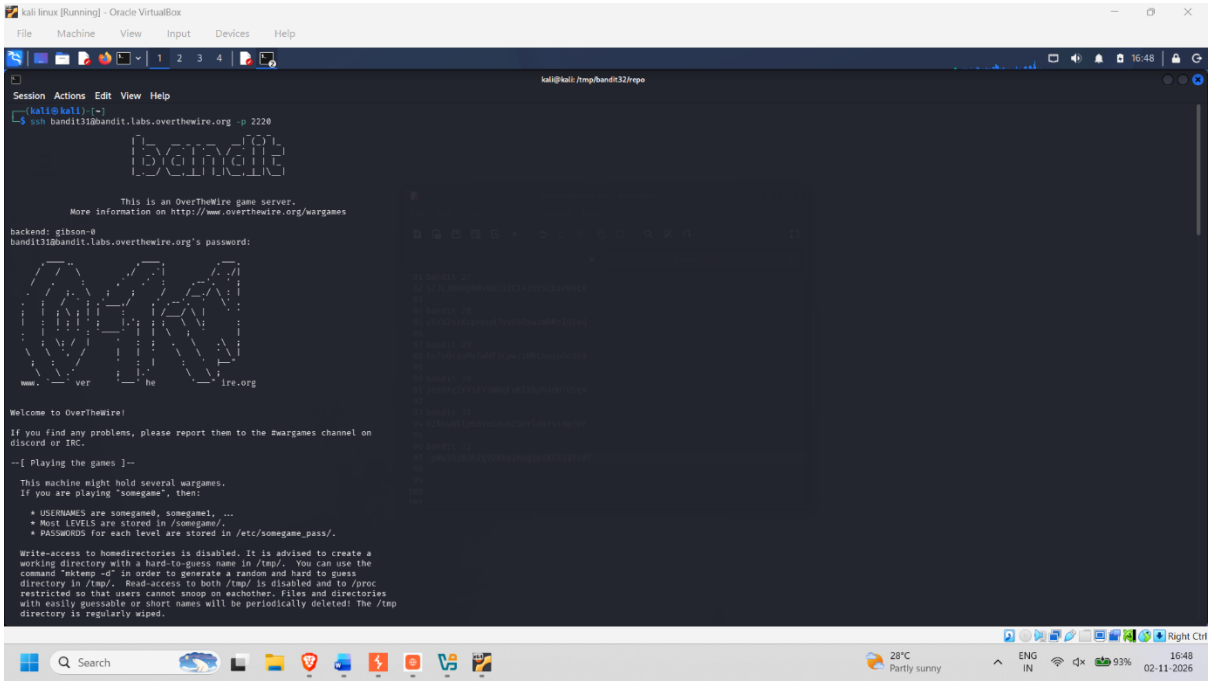
Explanation - Displays the password for Level 31.



LEVEL 31 – LEVEL 32

Step 1: Login to Level 31

ssh bandit31@bandit.labs.overthewire.org -p 2220



Step 2 – **exit**

Explanation - Exits the current SSH session.

Step 3 - **mkdir -p /tmp/bandit32**

Explanation - Creates a working directory.

Step 4 - **cd /tmp/bandit32`**

Explanation - Opens the working directory.

Step 5 - **git clone ssh://bandit31git@bandit.labs.overthewire.org:2220/home/bandit31-git/repo`**

Explanation - Clones the Git repository using the Level 31 password.

Step 6 - **cd repo**

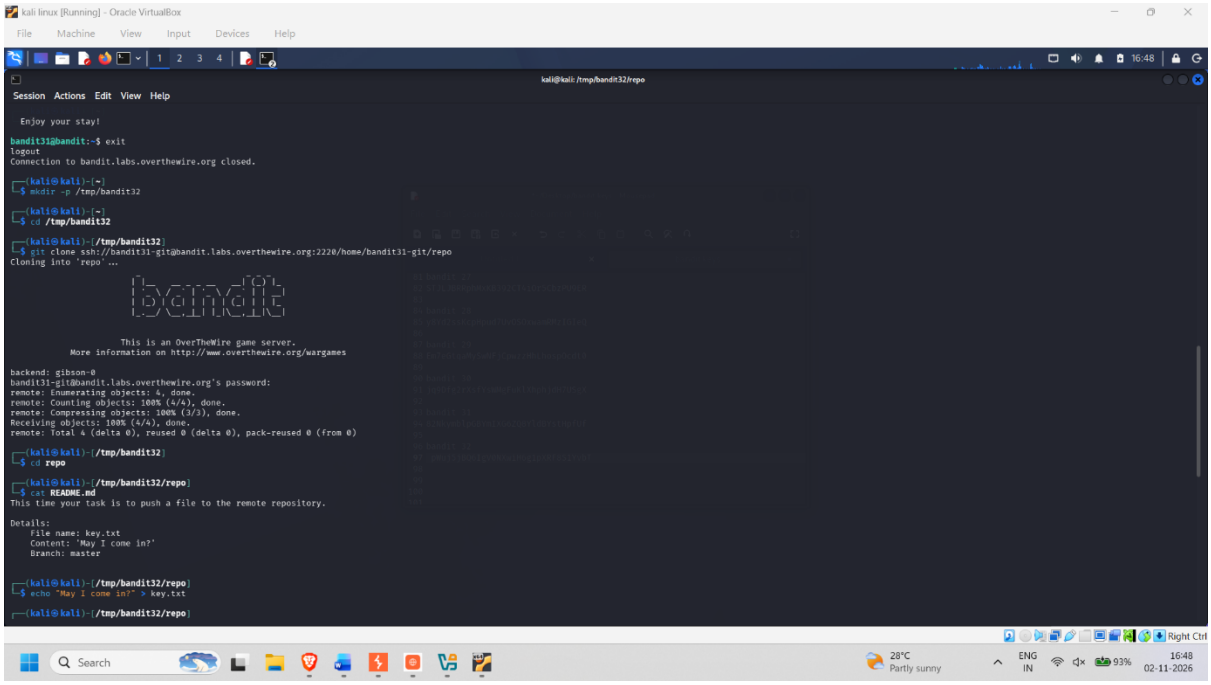
Explanation - Enters the cloned repository.

Step 7 - **cat README.md**

Explanation - Displays the instructions for Level 32.

Step 8 - **echo "May I come in?" > key.txt**

Explanation - Creates a file containing the required password.



Step 9 - **git add -f key.txt**

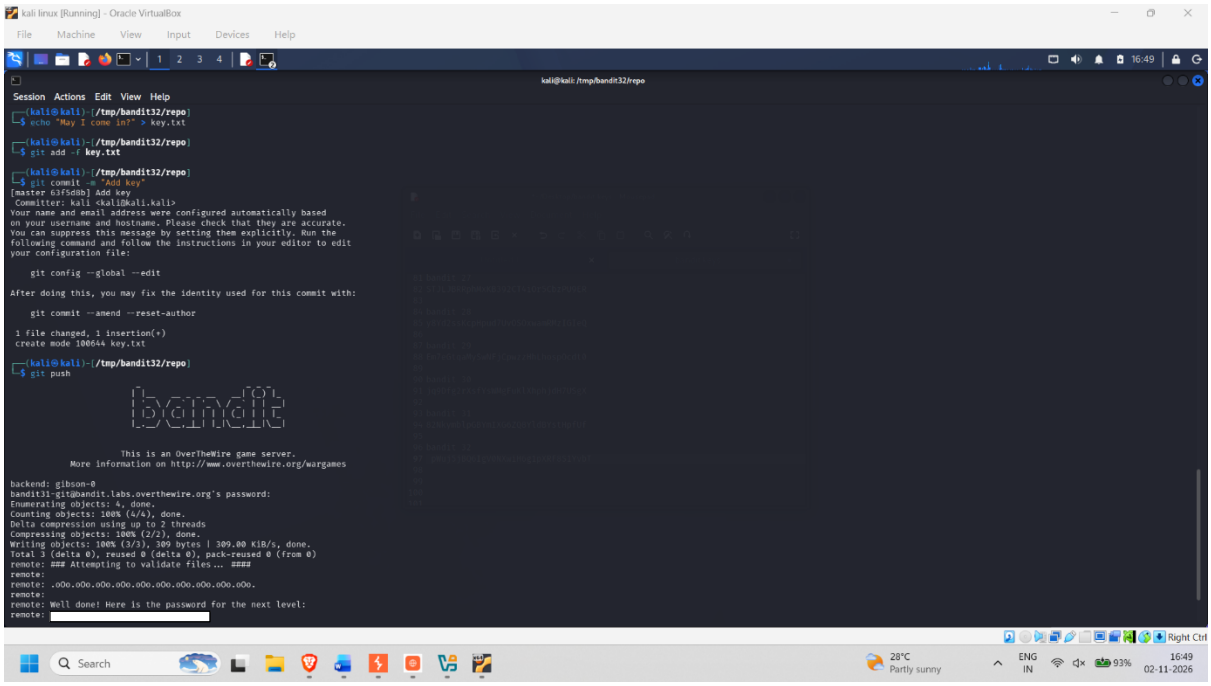
Explanation - Adds the file to Git, even if it is ignored.

Step 10 - **git commit -m "Add key"**

Explanation - Saves the file in a new commit.

Step 11 - **git push**

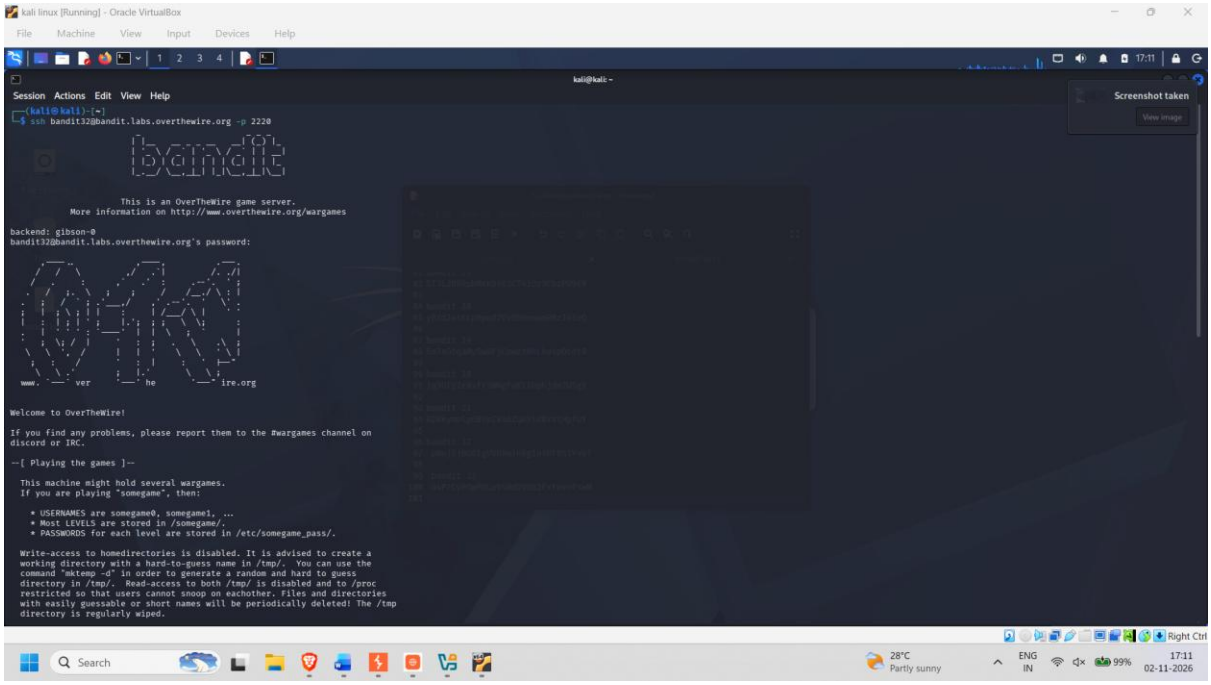
Explanation - Pushes the commit to the remote repository.


kali linux [Running] - Oracle VirtualBox
File Machine View Input Devices Help
kali@kali: /tmp/bandit32/repo
Session Actions Edit View Help
(kali㉿kali)-[/tmp/bandit32/repo]
$ echo "May I come in?" > key.txt
(kali㉿kali)-[/tmp/bandit32/repo]
$ git add -f key.txt
(kali㉿kali)-[/tmp/bandit32/repo]
$ git commit -m "Add key"
[master 63f5d8b] Add key
Committer: kali <kali@kali.kali>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly. Run the
following command and follow the instructions in your editor to edit
your configuration file:
git config --global --edit
After doing this, you may fix the identity used for this commit with:
git commit --amend --reset-author
1 file changed, 1 insertion(+)
create mode 100644 key.txt
(kali㉿kali)-[/tmp/bandit32/repo]
$ git push
This is an OverTheWire game server.
More information on http://www.overthewire.org/wargames
backend: gibson-0
bandit31-git@bandit.labs.overthewire.org's password:
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 309 bytes | 309.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote: ### Attempting to validate files... ####
remote:
remote: .oOo.oOo.oOo.oOo.oOo.oOo.oOo.oOo.oOo.oOo.
remote:
remote: Well done! Here is the password for the next level:
remote:
Right Ctrl
Search
28°C
Partly sunny
ENG
IN
93%
16:49
02-11-2026

LEVEL 32 – LEVEL 33

Step 1: Login to Level 32

ssh bandit32@bandit.labs.overthewire.org -p 2220



Step 2 - **whoami**

Explanation: Checks the current username.

Step 3 - **$0**

Explanation: Starts a normal shell to escape the uppercase shell.

Step 4 - **whoami**

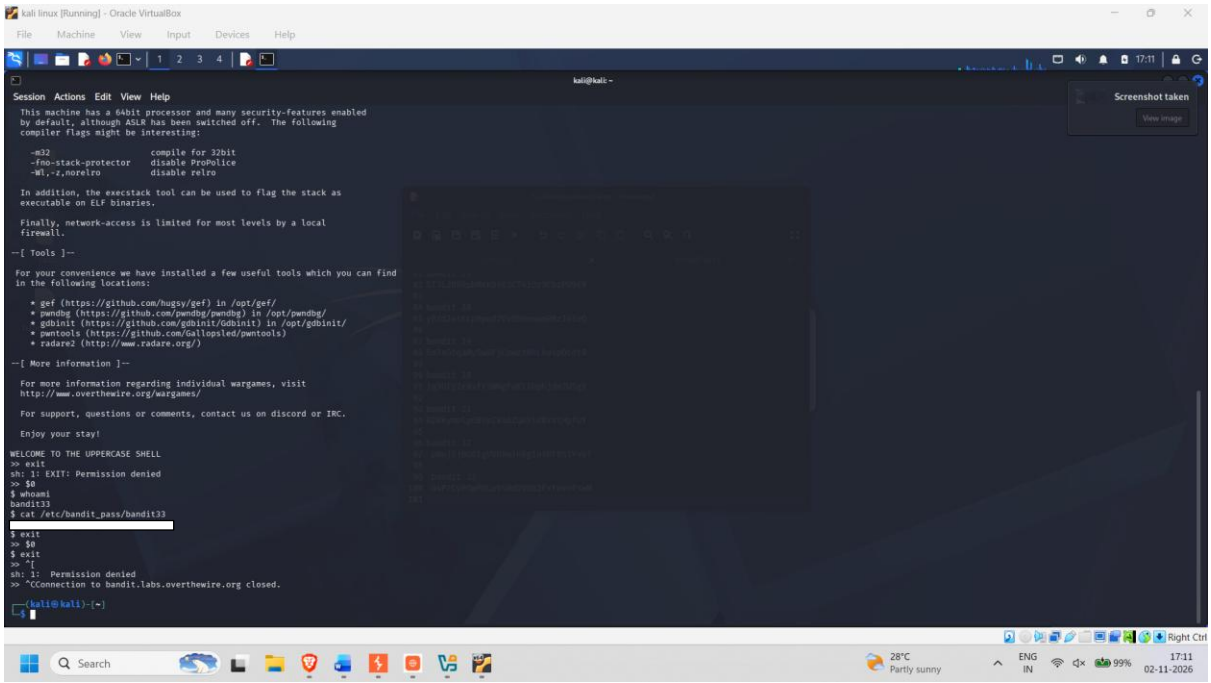
Explanation: Confirms the current username is `bandit33`.

Step 5 - **cat /etc/bandit_pass/bandit33**

Explanation: Displays the password for Level 33.

Step 6 - **exit**

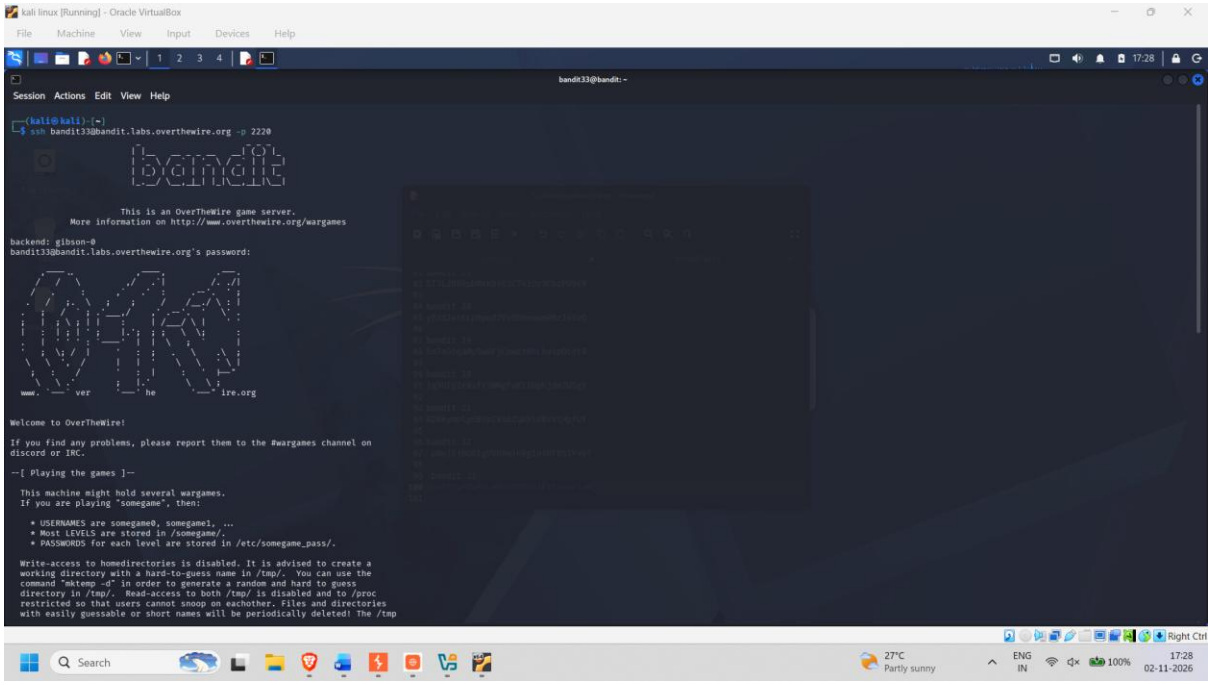
Explanation: Exits the normal shell**.**


kali linux [Running] - Oracle VirtualBox
File Machine View Input Devices Help
kali@kali: ~
Session Actions Edit View Help
This machine has a 64bit processor and many security-features enabled
by default, although ASLR has been switched off. The following
compiler flags might be interesting:
-m32 compile for 32bit
-fno-stack-protector disable ProPolice
-Wl,-z,norelro disable relro
In addition, the execstack tool can be used to flag the stack as
executable on ELF binaries.
Finally, network-access is limited for most levels by a local
firewall.
--[Tools]--
For your convenience we have installed a few useful tools which you can find
in the following locations:
* gef (https://github.com/hugsy/gef) in /opt/gef/
* pwndbg (https://github.com/pwndbg/pwndbg) in /opt/pwndbg/
* gdbinit (https://github.com/gdbinit/Gdbinit) in /opt/gdbinit/
* pwntools (https://github.com/Gallopsled/pwntools)
* radare2 (http://www.radare.org/)
--[More information]--
For more information regarding individual wargames, visit
http://www.overthewire.org/wargames/
For support, questions or comments, contact us on discord or IRC.
Enjoy your stay!
WELCOME TO THE UPPERCASE SHELL
>> exit
sh: 1: EXIT: Permission denied
>> $0
$ whoami
bandit33
$ cat /etc/bandit_pass/bandit33
$ exit
>> $0
$ exit
>> ^[
sh: 1: Permission denied
>> ^CConnection to bandit.labs.overthewire.org closed.
(kali㉿kali)-[~]
Screenshot taken
View image
Right Ctrl
Search
28°C
Partly sunny
ENG
IN
99%
17:11
02-11-2026

LEVEL 33 -LEVEL 34

Step 1: Login to Level 33

ssh bandit32@bandit.labs.overthewire.org -p 2220

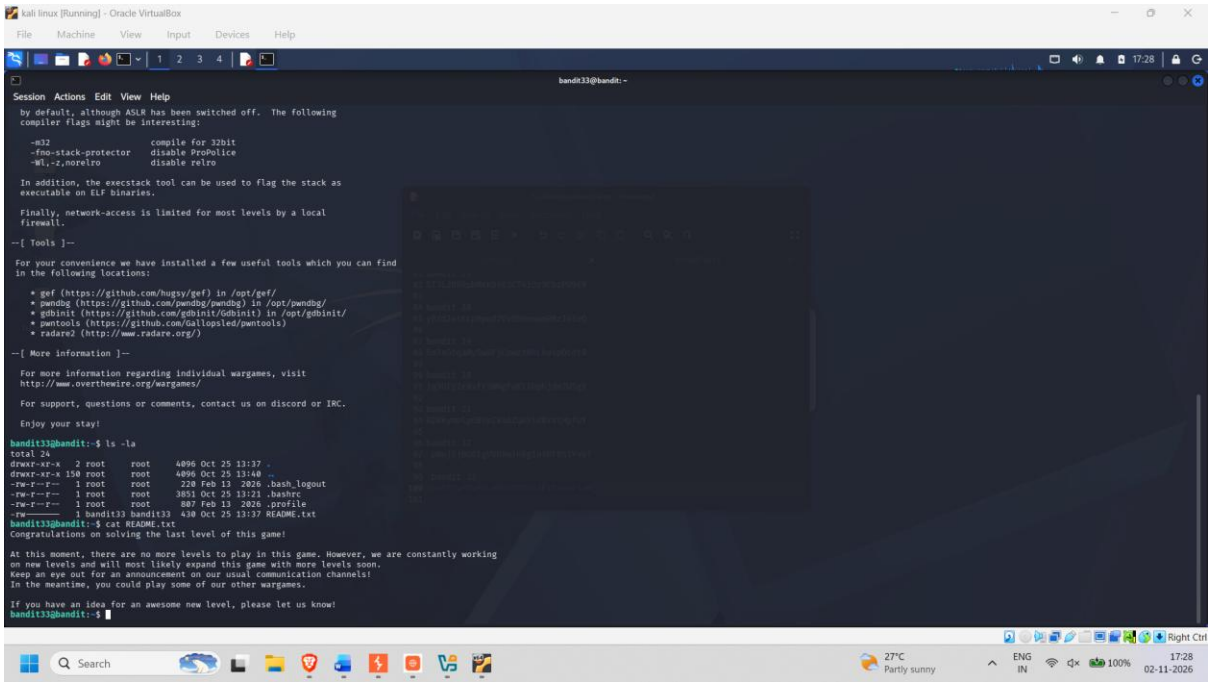


Step 2 - **"ls -la"**

Explanation — Lists all files, including hidden files.

Step 3 - **"cat README.txt"**

Explanation — Displays the final level's completion message.



Result: Successfully completed the Bandit wargame by escaping the uppercase shell and retrieving the final password.

Conclusion: This level demonstrated shell escape techniques and basic Linux commands for accessing files and verifying task completion.